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Foreword

This Technical Specification (TS) has been produced by ETSI Technical Committee Railway Telecommunications (RT).

The present document is part 1 of a multi-part deliverable covering the Future Railway Mobile Communication System (FRMCS); Services over 5G, as identified below:

Part 1: "Application conformance specification for client";

Part 2: "Application conformance specification for server".

Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the ETSI Drafting Rules (Verbal forms for the expression of provisions).

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1 Scope

The present document specifies the protocol conformance testing for testing a Future Railway Mobile Communication System (FRMCS) Client for compliance to railway specific functionality over 5G. The present document addresses only FRMCS Client-Server communication scenarios for:

- Specific Mission Critical Push-To-Talk (MCPTT) functionality.
- Mission Critical Ad Hoc Group Communication (AHGC) and Mission Critical Ad Hoc Group Emergency Alert (AHGEA).
- MCData including SDS, FD, QoS, MP-QUIC and MP-TCP as well IPU-connectivity supporting datagram and streaming mode.
- Enhanced MCPTT functionality including geographical based features.
- Interception.

It does not cover e.g. MCPTT Server-5G Core, MCPTT Server-SIP Core, etc. scenarios which involve interfaces which implementation may widely vary.

The present specification contains:

- the overall test structure.
- the test configurations.
- the conformance requirement and reference to the core specifications.
- the Test purposes; and
- a brief description of the test procedure, the specific test requirements and short message exchange table.

The present document is valid for FRMCS Clients implemented according to 3GPP releases starting from Release 18.

The present document does not specify the protocol conformance testing for the 5G Core bearers which carry the MCPTT data sent or received by the MCPTT Server. The specification of such testing is out of the scope of RAN5.

2 References

2.1 Normative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

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The following referenced documents are necessary for the application of the present document.

[1] ETSI TS 122 179 (V19.3.0): "LTE; 5G; Mission Critical Push to Talk (MCPTT); Stage 1 (3GPP TS 22.179 version 19.3.0 Release 19)".

[2] ETSI TS 122 280 (V19.7.0): "LTE; 5G; Mission Critical Services Common Requirements (MCCoRe); Stage 1 (3GPP TS 22.280 version 19.7.0 Release 19)".

[3] ETSI TS 122 282 (V19.0.0): "LTE; 5G; Mission Critical (MC) data (3GPP TS 22.282 version 19.0.0 Release 19) ".

[4] ETSI TS 123 280 (V19.8.1): "LTE; Common functional architecture to support mission critical services; Stage 2 (3GPP TS 23.280 version 19.8.1 Release 19)".

[5] ETSI TS 123 379 (V18.12.0): "LTE; Functional architecture and information flows to support Mission Critical Push To Talk (MCPTT); Stage 2 (3GPP TS 23.379 version 18.12.0 Release 18)".

[6] ETSI TS 138 508-1 (V18.6.0): "LTE; 5G; 5GS; User Equipment (UE) conformance specification; Part 1: Common test environment (3GPP TS 38.508-1 version 18.6.0 Release 18)".

[7] ETSI TS 137 579-4 (V17.3.0): "LTE; 5G; Mission Critical (MC) services; Part 4: Test Applicability and Implementation Conformance Statement (ICS) proforma specification (3GPP TS 37.579-4 version 17.3.0 Release 17)".

[8] ETSI TS 124 229 (V19.5.0): "Digital cellular telecommunications system (Phase 2+) (GSM); Universal Mobile Telecommunications System (UMTS); LTE; 5G; IP multimedia call control protocol based on Session Initiation Protocol (SIP) and Session Description Protocol (SDP); Stage 3 (3GPP TS 24.229 version 19.5.0 Release 19)".

[9] IETF RFC 3856: "A Presence Event Package for the Session Initiation Protocol (SIP)".

[10] IETF RFC 6665: "SIP-Specific Event Notification".

[11] IETF RFC 1889: "RTP: A Transport Protocol for Real-Time Applications".

[12] ETSI TS 124 380 (V18.6.0): "LTE; Mission Critical Push To Talk (MCPTT) media plane control; Protocol specification (3GPP TS 24.380 version 18.6.0 Release 18)".

[13] ETSI TS 123 289 (V18.8.0): "5G; Mission Critical services over 5G System; Stage 2 (3GPP TS 23.289 version 18.8.0 Release 18)".

[14] ETSI TS 137 579-5 (V18.0.0): "LTE; 5G; Mission Critical (MC) services; Part 5: Abstract test suite (ATS) (3GPP TS 37.579-5 version 18.0.0 Release 18)".

[15] ETSI TS 133 180 (V18.1.0): "LTE; Security of the Mission Critical (MC) service (3GPP TS 33.180 version 18.1.0 Release 18)".

[16] IETF RFC 3261: "SIP - Session Initiation Protocol".

[17] IETF RFC 3903: "Session Initiation Protocol (SIP) Extension for Event State Publication (PUBLISH)".

[18] IETF RFC 3556: "Session Description Protocol (SDP) Bandwidth Modifiers for RTP Control Protocol (RTCP) Bandwidth".

[19] IETF RFC 3550: "RTP: A Transport Protocol for Real-Time Applications".

[20] ETSI TS 136 579-3 (V15.0.0): "LTE; Mission Critical (MC) services over LTE; Part 3: Mission Critical Push To Talk (MCPTT) Server Application conformance specification (3GPP TS 36.579-3 version 15.0.0 Release 15)".

[21] ETSI TS 124 379 (V19.4.0): "LTE; Mission Critical Push To Talk (MCPTT) call control; Protocol specification (3GPP TS 24.379 version 19.4.0 Release 19)".

[22] ETSI TS 133 310 (V19.5.0): "Universal Mobile Telecommunications System (UMTS); LTE; 5G; Network Domain Security (NDS); Authentication Framework (AF) (3GPP TS 33.310 version 19.5.0 Release 19)".

[23] IETF RFC 2408: "Internet Security Association and Key Management Protocol (ISAKMP)".

[24] IETF RFC 8446: "The Transport Layer Security (TLS) Protocol Version 1.3".

[25] ETSI TS 103 765-3 (V1.1.1): "Rail Telecommunications (RT); Future Railway Mobile Communication System (FRMCS); Building Blocks and Functions; Part 3: Train On-Board functions and interfaces".

[26] ETSI TS 129 513 (V19.5.0): "5G; 5G System; Policy and Charging Control signalling flows and QoS parameter mapping; Stage 3 (3GPP TS 29.513 version 19.5.0 Release 19)".

[27] ETSI TS 103 765-1 (V1.1.1): "Rail Telecommunications (RT); Future Railway Mobile Communication System (FRMCS); Building Blocks and Functions; Part 1: Transport Stratum".

[28] ETSI TS 137 579-1 (V17.3.0): "LTE; 5G; Mission Critical (MC) services; Part 1: Common test environment (3GPP TS 37.579-1 version 17.3.0 Release 17)".

[29] ETSI TS 124 282 (V19.5.0): "LTE; Mission Critical Data (MCData) signalling control; Protocol specification (3GPP TS 24.282 version 19.5.0 Release 19)".

[30] IETF RFC 2784: "Generic Routing Encapsulation (GRE)".

[31] IETF RFC 792: "Internet Control Message Protocol".

[32] IETF RFC 9897: "Datagram Congestion Control Protocol (DCCP) Extensions for Multipath Operation with Multiple Addresses".

[33] IETF RFC 9000: "QUIC - A UDP based Multiplexed and Secured Protocol".

[34] IETF RFC 8200: "Internet Protocol, Version 6 (IPv6) Specification".

[35] IETF RFC 4443: "Internet Control Message Protocol (ICMPv6)".

[36] IETF RFC 3706: "A Traffic-Based Method of Detecting Dead Internet Key Exchange (IKE) Peers".

2.2 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

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[i.1] ETSI TR 103 791: "Rail Telecommunications (RT); Future Railway Mobile Communication System (FRMCS); Terminology for FRMCS specifications".

[i.2] ETSI TS 104 070: "Rail Telecommunications (RT); FRMCS conformance specification; Test Applicability and Implementation Conformance Statement (ICS) proforma specification".

[i.3] IETF RFC 3909: "Robots Exclusion Protocol".

[i.4] ETSI TR 121 905 (V18.0.0): "Digital cellular telecommunications system (Phase 2+) (GSM); Universal Mobile Telecommunications System (UMTS); LTE; 5G; Vocabulary for 3GPP Specifications (3GPP TR 21.905 version 18.0.0 Release 18)".

3 Definition of terms, symbols and abbreviations

3.1 Terms

For the present document, the terms given in ETSI TS 122 280 [2] (3GPP TS 22.280), ETSI TS 123 280 [4] (3GPP TS 23.280) and the following apply:

Ad hoc Group Communication: combining of a multiplicity of MC service users into a group for the duration of a communication

NOTE: When the communication is released, the group no longer exists. If the communication is associated with an alert, then the group continues to exist until the alert is also cancelled.

Ad hoc Group Emergency Alert: emergency alert addressed to multiple MC service users which are part of a group

NOTE: When the alert is cancelled, the group no longer exists. If the alert is associated with a communication, then the group continues to exist until the communication is also cancelled.

Functional Alias: user selectable alias that is tied to the assignment or task of the user

NOTE: A MCX User can activate one or multiple functional aliases at the same time. The activation of the functional alias(es) will take place after the user has signed in to the MCX Service using the MCX User ID.

Group Communication: one-to-many or many-to-many communication using an MCX Service

NOTE: The combination of different MC service users into a static group. MC service users have to affiliate for using the group service and need to de-affiliate in case they want to leave. In case of a communication, all affiliated MC service users will be involved.

MCX Service Emergency Alert: notification from the MCX UE to the MCX Service that the MCX User has an emergency condition

3.2 Symbols

Void.

3.3 Abbreviations

For the purposes of the present document, the abbreviations given in ETSI TR 121 905 [i.4] (3GPP TR 21.905) and the following apply:

NOTE: An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in ETSI TR 121 905 [i.4] (3GPP TR 21.905).

5QI	QoS Class Identifier
AHGC	Mission Critical Ad Hoc Group communication
AHGEA	Mission Critical Ad Hoc Group Emergency Alert
FA	Functional Alias
FD	File Distribution
ICS	Implementation Conformance Statement
IPConn	IP Connectivity
IUT	Implementation Under Test
MBMS	Multimedia Broadcast and Multicast Service
MC	Mission Critical
MCData	Mission Critical Data
MCPTT	Mission Critical Push To Talk
MCX	Mission Critical Services
MIME	Multipurpose Internet Mail Extensions
MPF	Multipath Control Function

PLMN	Public Land Mobile Network
RTP	Real-time Transport Protocol
SDP	Session Description Protocol
SDS	Short Data Service
SIP	Session Initiation Protocol
SS	System Configuration
SSRC	Synchronization SouRCe
TP	Transmission Point
TP	Test Purpose
URI	Uniform Resource Identifier

4 General

4.1 Test methodology

4.1.1 Testing of optional functions and procedures

Any function or procedure, which is optional, may be subject to a conformance test if it is implemented in the MCX Server for MCPTT and MCData.

A declaration by the Client supplier (to use the Implementation Conformance Statement (ICS) proforma specified in ETSI TS 104 070 [i.2] is used to determine whether an optional function/procedure has been implemented.

4.1.2 Test interfaces and facilities

Descriptions of the MCPTT Server to MCPTT Server interface is specified in ETSI TS 137 579-1 [28].

The general components required for this conformance test are given in Figure 4.1-1. The MC service client carried by a MC service UE which is capable to connect via 5G NR network to a MCx Service system.

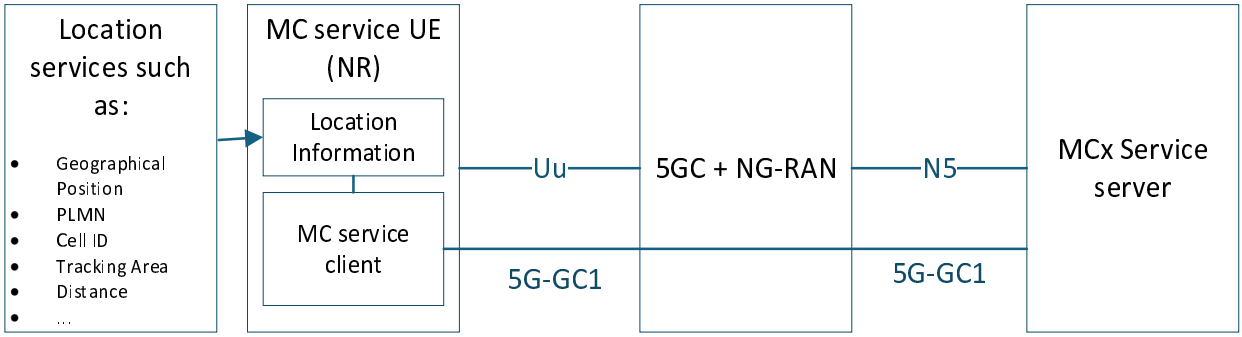
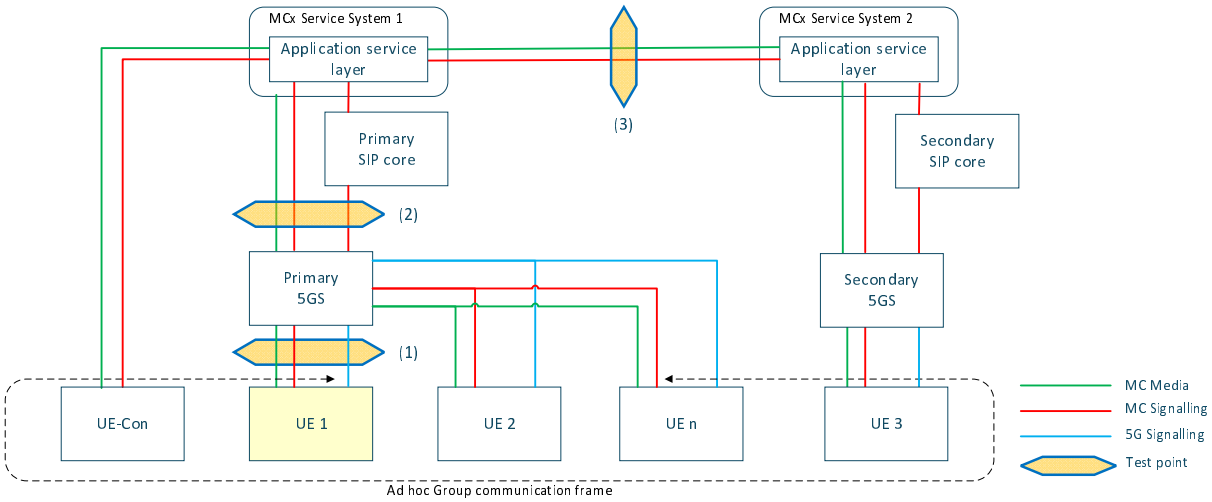


Figure 4.1-1: General configuration and interfaces

4.1.3 MC Conformance testing test points overview

Figure 4.1-2 provides a general overview of all MCx players which may have a role in different conformance testing scenarios together with virtual test points representing the information flow which is intended for conformance testing. The figure is mainly for descriptive purposes and may not necessarily represent a real MCx deployment or implementation.



NOTE 1: For the Ad hoc Group communication multiple UE will be required. They should be connected to the 5G system and provide similar interfaces for MC media plane, MC signalling and 5G signalling. The entity in light yellow will be the Implementation Under Test (IUT). Entities with white fill will be either simulated or IUT or real implementations depending on the test scenario. UE3 is connected to the partner MCx Service system and can be simulated or present as physical device.

NOTE 2: <UE n> identifies a various number of supporting devices. These can be simulated or present as physical system. To prevent any misinterpretations in the test steps because different additional numbers and different configurations could occur, "n" is replaced by a1, b2, c3 and so on, while the character indicates an individual configuration and the digit gives the ongoing number of the device. Same character indicates equivalent parameter configurations (but e.g. different SIP/IMS user accounts for sure).

NOTE 3: The Ad hoc Group communication frame can vary during the tests depending on the number of participants given by the test scope (e.g. pre-configured or list based ad hoc group communication).

NOTE 4: UE-Con is the controller/dispatcher. It is connected directly to the server application function but integrated in the configuration.

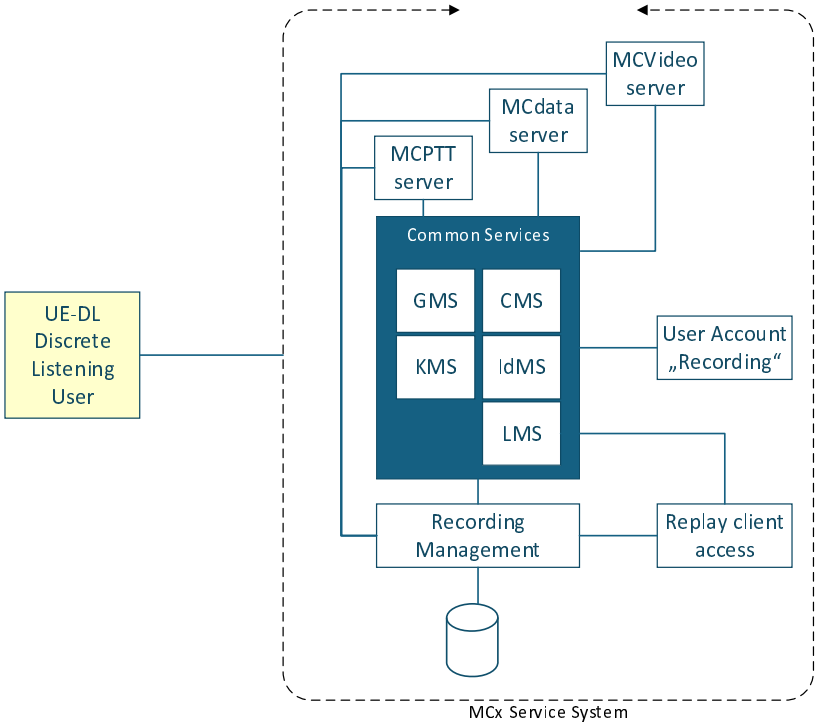
NOTE 5: Number of MC service user UE can vary depending on the test case. For this, entity UE n symbolizes several similar UE with similar interfaces.

NOTE 6: MCx Service system 2 can be homed in a different domain or on a different home server.

Figure 4.1-2: MC Conformance testing test points model

4.1.4 MC Conformance testing - configuration for recording management and discrete listening

To ensure the functionality for recording and discrete listening the conformance test environment shall be configured according to Figure 4.1-3.

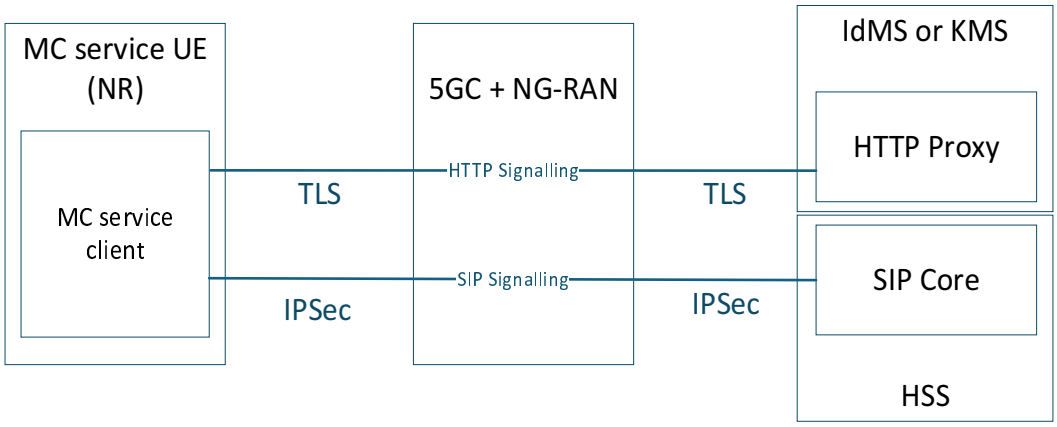


NOTE 1: User account "recording" should be an independent user access which allows only the configuration of the recording function and the allocation of a discrete listening UE. No other functionality should be possible.
NOTE 2: The storage period for the recorded interception data should not violate national data retention rules.
NOTE 3: User account "Replay Client" is only for analysing recorded session data including useful data.
NOTE 4: UE-DL is the discrete listening UE. It can be a physical UE connected to the 5G system according to Figure 4.1-2 or it can be simulated or connected directly to the server application function but integrated in the configuration.

Figure 4.1-3: Configuration for recording and discrete listening functionality

4.1.5 MC Conformance testing - configuration for ad hoc group communication security

To ensure the functionality for recording and discrete listening the conformance test environment shall be configured according to Figure 4.1-4.



NOTE 1: For HTTP usage the HTTP-1 interface is required. The authentication should be performed between HTTP client in the MC UE and the HTTP server (IdMS or KMS).
NOTE 2: The PSK TLS based authorisation methods are given with ETSI TS 133 310 [22] (3GPP TS 33.310).
NOTE 3: The mutual authentication procedure between UE and SIP core should support IMS AKA and mechanisms, described in ETSI TS 133 310 [22] (3GPP TS 33.310), clause 6.2.

Figure 4.1-4: Communication Security

4.1.6 MC Conformance testing - configuration for IP Connectivity (IPConn)

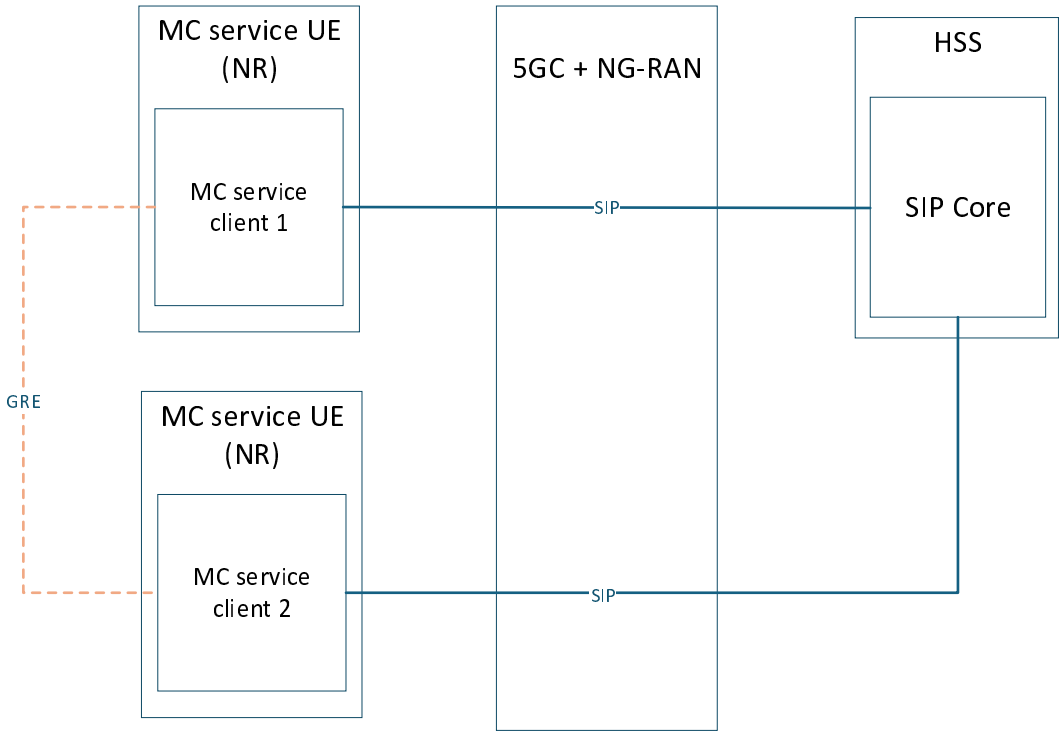


Figure 4.1-5: IPConn communication connections

4.1.7 MC Conformance testing - QUIC datagram or streaming mode

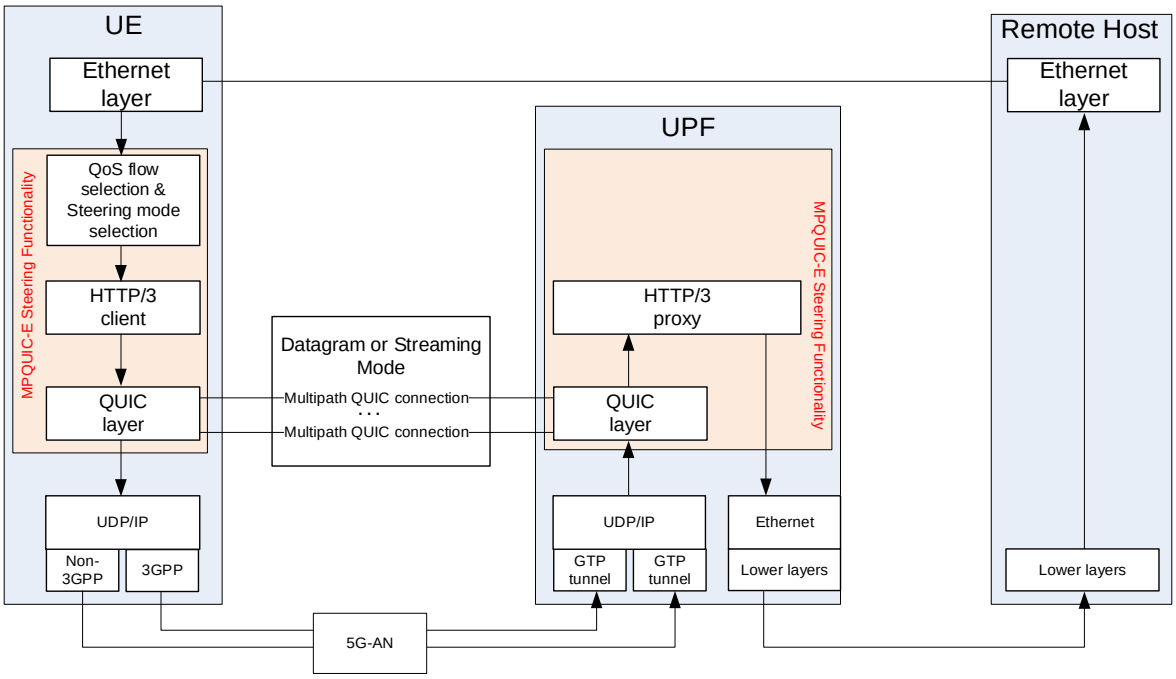


Figure 4.1-6: QUIC connection in datagram or streaming mode

4.2 Implicit testing

For some MCPTT protocol features conformance is not verified explicitly in the present document. This does not imply that correct functioning of these features is not essential, but that these are implicitly tested to a sufficient degree in tests, which are not explicitly dedicated to test the feature.

4.3 Repetition of tests

As a rule, the test cases specified in the present document are highly reproducible and do not need to be repeated unless otherwise stated.

4.4 Handling of differences between conformance requirements in different releases of cores specifications

The conformance requirements, which determine the scope of each test case, are explicitly copy-pasted from relevant core specifications in the especially dedicated for this section of each test with the title 'Conformance requirements'.

NOTE: When in the copy/pasted text there are references to other specifications the reference numbers will not match the reference numbers used in the present specification. This approach has been taken to allow easy copy and then search for conformance requirements in those specifications.

When differences between conformance requirements in different releases of the cores specifications have impact on the Pre-test conditions, Test procedure sequence or/and the Specific message contents, the Conformance requirements related to different releases are specified separately with clear indication of the Release of the spec from which they were copied.

When there is no Release indicated for a conformance requirement text, this should be understood either as the Conformance requirements in the latest version of the spec with release = the TC Applicability release. This can be found in ETSI TS 137 579-4 [7], Table 4-2.

4.5 Reference conditions

The reference environments used by all signalling and protocol tests are specified in ETSI TS 137 579-1 [28]. Where a test requires an environment that is different, this will be specified in the test itself.

4.6 Generic setup procedures

A set of basic generic procedures for MCPTT Client-Server and MCPTT Client-Client communication are given in clause 5.

5 Generic functionality

5.1 Functional aliasing client side

5.1.1 Test Purpose (TP)

(1)

```
with {IUT connected to MCX service system}
ensure that {
when {IUT has no active FA, no FA is in use and IUT has activated the MCX service user}
then {MCX service user activates multiple FA automatically (pre-configured in the MCX service user
setting) using FA support in numerical form (e.g. 12345) as well as organizational arrangement (e.g.
role.mission@departement.company) and containing the additional information to FA (e.g. expiration
timer, priority, location, velocity, etc.) and MCX service user can be reached by only the FA and
identifies itself by the FA
```

}

(2)

with {IUT connected to MCX service system}
ensure that {
    when {IUT has no active FA, no FA is in use and IUT has activated the MCX service user **and** FA needs to be activated manually}
    then {MCX service user activates **five** FA manually and MCX service user activates an unlisted FA which is defined in the MCX service system and MCX service user can be reached by only the FA and identifies itself by the FA and MCX service user can manually deactivate an active FA
            }

(3)

with {FA shared between many MCX users and all of them are notified for this sharing of FA}
ensure that {
    when {the same FA can be used by multiple MCX service user and no active FA for the MCX service user A; second MCX service user B has activated FA1 and maximum allowed number of members for the shared FA not reached **and** activation possible}
    then {MCX service user activates FA which is possible to share}
else when { MCX service user activates FA which is possible to share and maximum allowed number of members for the shared FA already reached - activation not possible **and** other MCX service users are informed about the new activation}
then {FA request rejected}
            }

(4)

with {Automatic activation/deactivation of FAs (e.g., based on location or timer expiration)}
ensure that {
    when {no active FA for the MCX service user on the MCX service system and MCX service system is provided in different areas with attached, different FA}
    then {Automatic FA activation while entering a dedicated location (e.g. based on GPS information, 5G network information or location configuration) on the MCX service system and Automatic FA deactivation while leaving a dedicated location and automatic FA deactivation while FA timer expires}
            }

(5)

with {It shall be possible for a MCX service user to take over a FA which is already activated by another MCX service user}
ensure that {
    when {FA already activated by MCX service user B. MCX service system is configured, that takeover of FA is possible and FA is not configured for multiple use.}
    then {MCX service user A force to take over the FA and FA will be deactivated on MCX service user B and User B gets information about the deactivation and user A receive confirmation of activation and MCX service system provides status and interrogation of FA status including options FA activated **or** FA overtaken by another user **or** FA not active **or** all FA will be interrogated}
    else when {FA already activated by MCX service user B. MCX service system is configured, that takeover of FA is **not** possible and FA is not configured for multiple use}
    then {MCX service user A activates FA which is used by MCX service user B and FA not activated (reject) and present information to user A including options change FA **or** call to responsible or MCX service user requests a list of possible FA}
        }

(6)

with {MCX User shall be reachable by its functional alias from the primary MCX system while migrated to partner MCX Service systems}
ensure that {
    when {MCX service user has activated one or more FA and incoming and outgoing communication to the FA inside MCX service system A is possible and another MCX service system B is connected.}
    then {incoming and outgoing communication from MCX service user via MCX service system A to system B shall be possible}

5.1.2 Conformance requirements

References: The conformance requirements covered by this test case are specified in stage 1 document ETSI TS 122 280 [2] (3GPP TS 22.280), clause 5.9a with detailed requirements:

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.9a-001]]

"The MCX Service shall provide a mechanism for the MCX User to activate one or more functional alias(es) for use by the MCX Users within that MCX Service system (i.e. the primary MCX Service system of the functional alias)."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.9a-001a]]

"The MCX Service system shall be able to use the functional alias as a unique identifier for MCX Users as Participant of an MCX Service Group, e.g. to display a functional alias as speaker identification or for the use in group member lists."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.9a-001b]]

"The MCX Service system shall provide a mechanism for an MCX User to assign a functional alias for use on an MCX Service Group."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.9a-001c]]

"The MCX Service system shall provide a mechanism for an MCX User to assign an activated functional alias for an outgoing MCX Private communication."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.9a-002]]

"The MCX User shall be reachable by its functional alias(es) from within the MCX Service system where the functional alias was activated."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.9a-002a]]

"An MCX User shall be reachable by its functional alias from the primary MCX system while migrated to partner MCX Service systems."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.9a-003]]

"The MCX Service shall provide a mechanism for the MCX User to deactivate a functional alias."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.9a-004]]

"The MCX Service shall upon request provide the MCX User a list of functional aliases from which the user can select for activation/deactivation.

NOTE: The list may contain functional aliases based on a certain context, like location of the MCX User, operational schedule, etc."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.9a-006]]

"If an MCX User attempts to activate a functional alias that is already active for another MCX User and is not allowed to be simultaneously active for multiple MCX Users or the number of simultaneous MCX Users is reached to the upper limit, the MCX Service shall inform the MCX User that the functional alias is already in use."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.9a-007]]

"If an MCX User attempts to activate a functional alias that is already active for at least one other MCX User, and that functional alias is allowed to be simultaneously active for multiple MCX Users and the upper limit of number of simultaneous MCX Users is not reached, the MCX Service shall activate the functional alias for the MCX user and inform all other MCX User(s) with the same functional alias."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.9a-008]]

"If an authorized MCX User attempts to activate a functional alias that is already used by another MCX User, and that functional alias is allowed to be taken over and is not indicated for simultaneous activation to multiple MCX Users, the MCX Service shall offer the MCX User to take over the functional alias from the MCX User using the alias."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.9a-008a]]

"If an authorized MCX User attempts to activate a functional alias that is already active for at least one other MCX User, and the upper limit of number of simultaneous MCX Users is reached, the MCX Service shall reject the MCX User's request."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.9a-009]]

"If an authorized MCX User takes over the functional alias that is already active for another MCX User, the MCX Service shall activate the functional alias to the MCX User and inform the previous MCX User that the alias has been deactivated."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.9a-010]]

"The MCX Service shall allow the MCX User to perform an activation/deactivation of an unlisted functional alias that is defined in the MCX Service system."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.9a-011]]

"An authorized MCX User shall be able to interrogate the MCX Service system of the alias(es) active for a certain MCX User."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.9a-014]]

"The MCX Service shall allow the functional alias to be structured as a combination of organizationally meaningful elements (e.g. user.agency@organization.country, role.mission@department.company, function.equipment.ID@city, label1.label2.label3@firstname.familyname)."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.9a-015]]

"The MCX Service system shall allow an MCX Service Administrator to make use of information (e.g. operational schedules, locations, velocity or direction) from external sources to create or delete a functional alias."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.9a-016]]

"The MCX Service system shall allow the MCX Service Administrator to assign a communication priority for a functional alias."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.9a-017]]

"The MCX Service system shall allow the MCX Service Administrator to assign a time limit to a functional alias after which the functional alias will be deactivated."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.9a-018]]

"The MCX Service shall support automatic activation and de-activation of a functional alias based on the operational criteria (e.g. MCX User ID, login/logoff from the MCX Service system, specific external information supplied by external systems)."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.9a-019]]

"The MCX Service shall provide a mechanism for an MCX Service Administrator to define a functional alias with related geographic areas that can be associated to MCX Users for the purpose of routing Location dependent communications, as part of handling MCX Service Private Communication requests, when the determination of the receiving party is based on the initiating MCX User's current Location."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.9a-022]]

"For private calls to functional aliases the authorizations in clause 5.9a.2 shall replace the authorizations defined in clause 6.7.3."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.9a-023]]

"The MCX Service system shall support private calls and private emergency calls to a functional alias simultaneously activated by multiple MCX Users."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.9a-024]]

"The MCX Service system shall enable an MCX Service Administrator to select the mechanism to be used for handling of private calls to functional aliases that are allowed to be simultaneously active for multiple MCX Users."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.9a-025]]

"The MCX Service system shall be able to provide a mechanism to inform the MCX User about the functional alias activation/deactivation status (e.g. activation/deactivation accepted, activation/deactivation pending, activation/deactivation rejected)."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.9a-027]]

"The MCX Service shall provide a Location information report for every authorized MCX User associated with one functional alias in accordance with sub-clause 6.12."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.9a-029]]

"The MCX Service shall provide, upon request, the list of currently affiliated members of an MCX group which encompasses the MCX Service User ID and the associated functional alias(es) of each member in accordance with sub-clause 6.4.5."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.9a-031]]

"The MCX Service shall provide a mechanism for an authorized MCX User to obtain a list of functional aliases currently activated in the system either periodically or as a onetime request.

NOTE: The mechanism may allow for filtering criteria e.g. geographic area."

5.1.3 Test description

5.1.3.1 Pre-test conditions

System Configuration:

- MCx Service system consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (UE1) also consists of all sub-systems of the MCx user service.
- For the underlying "transport bearer" over which the UE1 (MCx Service system) and the IUT will communicate Parameters are set to the default parameters specific for the basic E-UTRA single simulated cells cell network scenarios, as defined in ETSI TS 138 508-1 [6] (3GPP TS 38.508-1), clause 6.3.4.4. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCx operation in the MCx configuration document).
- MCx Service system is configured to support the following features:
  - A configuration (as file or user settings entry) with multiple FA is available for the UE1. The configuration contains numerical and organizational arrangements of FA and flags for expiration timer, priority and location. MCx Service user accepts the configuration.
  - FA in numerical form (e.g. 12345) and organizational arrangement (e.g. role.mission@departement.company).
  - FA containing additional information (e.g. expiration timer, priority, location, velocity, etc.).
  - No FA is in use.

IUT:

- IUT (UE1).
- The IUT (UE1) consists of all sub-systems of the MCx user service including the ability for floor control.

- The IUT (UE1) is the acting user.
- No FA is in use.

Preamble:

- The MCX Client has been provided either with the address information of the server from which the client can retrieve the MCX UE initial configuration document ETSI TS 137 579-1 [28] (3GPP TS 37.579-1) (steps 1a1-1a2 of procedure 'MCX Initial Configuration and User Authentication', Table 5.3.2.3.1-1) or directly with the Initial UE Configuration Data as specified in Table 5.5.8.1-1.
- The UE User is provided with username/password for user authentication (px_MCX_User_A_username, px_MCX_User_A_password as provided in ETSI TS 137 579-5 [14] (3GPP TS 37.579-5) Table 9.2-1: MCX Client Common PIXIT).
- The UE is provided with the names and values of the Transport Key (TrK) and the Integrity Key (InK)], since the KMS shall encrypt the key material sent to the client with the TrK and sign the response with the TrK or the InK according to ETSI TS 133 180 [15] (3GPP TS 33.180).
- The IUT (UE1) is connected to PLMN of the primary 5GS. UE3 is connected to the PLMN of the secondary 5GS.

5.1.3.2 Test procedure sequence

Table 5.1.3.2-1: Main behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
	EXCEPTION: Procedure 'MCX CO communication' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1), clause 5.4.3	--		-	-
	EXCEPTION: Procedure 'Initial MCX Authentication, Registration, Configuration and Subscription' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1), clause 5.3.2	--		-	-
1a	Execute MCX Group Affiliation Status Change as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1) clause 5.3.34. No FA is active at the MCx Service system. (See note 1). Check: UE1 publish the basic POC settings	-->	SIP PUBLISH according ETSI TS 124 229 [8] (3GPP TS 24.229), IETF RFC 3903 [17]	-	P
1b	MCx Service system confirm the POC settings	<--	STATUS 200 OK (PUBLISH) according ETSI TS 124 229 [8] (3GPP TS 24.229)	-	-
2a	Check: UE1 announce interest for a specific FA to the MCx Service system (Interrogation, entrance to a specific list); UE1 is not allowed to use interrogation	-->	SIP PUBLISH according ETSI TS 124 379 [21] (3GPP TS 24.379) clause 9A.2.1.2, IETF RFC 3856 [9] and IETF RFC 6665 [10]	1	P
3a	Check: MCx Service system responds with a STATUS (see note 2) message	<--	SIP STATUS (see note 2) 403 FORBIDDEN	-	-
2b	Check: UE1 announce interest for a specific FA to the MCx Service system (Interrogation, entrance to a specific list); UE1 is allowed to use interrogation but authorization to the specific FA is not granted	-->	SIP SUBSCRIBE	1	P
3b	Check: MCx Service system responds with a STATUS (see note 2) message	<--	SIP STATUS (see note 2) 401 UNAUTHORIZED	-	-
2c	Check: UE1 announce interest for a specific FA to the MCx Service system (Interrogation, entrance to a specific list)	-->	SIP PUBLISH	1	P
3c	Check: MCx Service system responds with a STATUS (see note 2) message	<--	SIP STATUS (see note 2) 200 OK	-	-

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
4a	Check: PUBLIH message for FA on MCx Service system. The message contains multiple FA as configured in test step 1 including additional information for expiration timer, priority, location, etc. UE is not allowed to use the requested FA e.g. while invalid message content	-->	SIP PUBLISH	1	P
5a	Check: MCx Service system responds with a STATUS (see note 2) message	<--	SIP STATUS (see note 2) 403 FORBIDDEN	-	-
4b	Check: PUBLIH message for FA on MCx Service system. The message contains multiple FA as configured in test step 1 including additional information for expiration timer, priority, location, etc. UE is not authorized to use the requested FA	-->	SIP PUBLISH	1	P
5b	Check: MCx Service system responds with a STATUS (see note 2) message	<--	SIP STATUS (see note 2) 401 UNAUTHORIZED	-	-
4c	Check: PUBLIH message for FA on MCx Service system. The message contains multiple FA as configured in test step 1 including additional information for expiration timer, priority, location, etc.	-->	SIP PUBLISH	1	P
5c	Check: MCx Service system responds with a STATUS (see note 2) message	<--	SIP STATUS (see note 2) 200 OK	-	-
6	In case of STATUS response 401/403 (UNAUTHORIZED/FORBIDDEN) the root cause shall be indicated by the MCx Service system.	--		-	-
	EXCEPTION: Procedure 'MCX CO communication' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1), clause 5.4.3	--		-	-
	EXCEPTION: Procedure 'Initial MCX Authentication, Registration, Configuration and Subscription' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1), clause 5.3.2	--		-	-
	Execute the MCx service registration and affiliation. No FA is active at the MCx Service system.	--		-	-
7	Check: UE1 announce interest for a specific FA to the MCx Service system (Interrogation, entrance to a specific list)	-->	SIP PUBLISH	2	P
8	Check: MCx Service system responds with a STATUS (see note 2) message	<--	SIP STATUS (see note 2) 200 OK	2	-
9a	Check: PUBLIH message for FA on MCx Service system. The message contains multiple FA as configured in test step 1 including additional information for expiration timer, priority, location, etc. UE is not allowed to use the requested FA e.g. while invalid message content	-->	SIP PUBLISH	2	P
10a	Check: MCx Service system responds with a STATUS (see note 2) message	<--	SIP STATUS (see note 2) 403 FORBIDDEN	-	-
9b	Check: PUBLIH message for FA on MCx Service system. The message contains multiple FA as configured in test step 1 including additional information for expiration timer, priority, location, etc. UE is not authorized to activate the requested FA	-->	SIP PUBLISH	2	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
10b	Check: MCx Service system responds with a STATUS (see note 2) message	<--	SIP STATUS (see note 2) 401 UNAUTHORIZED	-	-
9c	Check: PUBLIH message for FA on MCx Service system. The message contains multiple FA as configured in test step 1 including additional information for expiration timer, priority, location, etc.	-->	SIP PUBLISH	2	P
10c	Check: MCx Service system responds with a STATUS (see note 2) message	<--	SIP STATUS (see note 2) 200 OK	-	-
11	Check: INVITE the UE1 into a connection. MCx Service user can be reached only by its FA and use the FA for outgoing identification	<--	SIP INVITE	-	P (server)
12	UE1 send ALERT information in case it is not automatically accepted	-->	SIP STATUS (see note 2) 180 (Ringing)	2	P
13	UE1 send ACTIVE information when call is accepted	-->	SIP ACK	2	P
	UE1 receive NOTIFICATION	<--	SIP NOTIFICATION	-	-
14a	Check: UE1 can manually deactivate the FA with PUBLIH message for FA on MCx Service system. The message contains multiple FA as configured in test step 1 including additional information for expiration timer, priority, location, etc. UE is not allowed to use the requested FA e.g. while invalid message content	-->	SIP PUBLISH	2	P
15a	Check: MCx Service system responds with a STATUS (see note 2) message	<--	SIP STATUS (see note 2) 403 FORBIDDEN	-	-
14b	Check: UE1 can manually deactivate the FA with PUBLIH message for FA on MCx Service system. The message contains multiple FA as configured in test step 1 including additional information for expiration timer, priority, location, etc. UE is not authorized to deactivate the requested FA	-->	SIP PUBLISH	2	P
15b	Check: MCx Service system responds with a STATUS (see note 2) message	<--	SIP STATUS (see note 2) 401 UNAUTHORIZED	-	-
14c	Check: UE1 can manually deactivate the FA with PUBLIH message for FA on MCx Service system. The message contains multiple FA as configured in test step 1 including additional information for expiration timer, priority, location, etc.	-->	SIP PUBLISH	2	P
15c	Check: MCx Service system responds with a STATUS (see note 2) message	<--	SIP STATUS (see note 2) 200 OK	-	-
	EXCEPTION: Procedure 'MCX CT call release' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1) clause 5.3.12	--		-	-
	EXCEPTION: Procedure 'MCX CO communication' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1), clause 5.4.3	--		-	-
	EXCEPTION: Procedure 'Initial MCX Authentication, Registration, Configuration and Subscription' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1), clause 5.3.2 No FA is active at the MCx Service system	--		-	-
16	Check: UE1 announce interest for a specific FA to the MCx Service system (Interrogation, entrance to a specific list)	-->	SIP PUBLISH	3	P
17	Check: MCx Service system responds with a STATUS (see note 2) message	<--	SIP STATUS (see note 2) 200 OK	-	-

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
18a	Check: UE1 activates a FN which is allowed to be shared between many users. Maximum of allowed other MCx Service users have activated the shared FN. PUBLISH message sent to the MCx Service system containing additional information for expiration timer, priority, location, etc.	-->	SIP PUBLISH	3	P
19a	Check: MCx Service system responds with a STATUS (see note 2) message	<--	SIP STATUS (see note 2) 403 FORBIDDEN	-	-
18b	Check: UE1 activates a FN which is allowed to be shared between many users. One or more other MCx Service users have activated the shared FN. PUBLISH message sent to the MCx Service system containing additional information for expiration timer, priority, location, etc.	-->	SIP PUBLISH	3	P
19b	Check: MCx Service system responds with a STATUS (see note 2) message	<--	SIP STATUS (see note 2) 401 UNAUTHORIZED	-	-
18c	Check: UE1 activates a FN which is allowed to be shared between many users. No other MCx Service user has activated the shared FN. PUBLISH message sent to the MCx Service system containing additional information for expiration timer, priority, location, etc.	-->	SIP PUBLISH	3	P
19c	Check: MCx Service system responds with a STATUS (see note 2) message	<--	SIP STATUS (see note 2) 200 OK	-	-
20	Check: UE2 receive a NOTIFICATION message containing the MC Service user ID who performed the query and the MC Service user ID(s) queried (list can be empty) according to ETSI TS 123 280 [4] (3GPP TS 23.280)	<--	NOTIFY	3	-
	EXCEPTION: Procedure 'MCX CT call release' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1) clause 5.3.12	--		-	-
	EXCEPTION: Procedure 'MCX CO communication' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1), clause 5.4.3	--		-	-
	EXCEPTION: Procedure 'Initial MCX Authentication, Registration, Configuration and Subscription' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1), clause 5.3.2 No FA is active at the MCx Service system. MCx Service system provides FA-A for location A and FA-B for location B and expiration time for FA; interrogation allowed for UE1	--		-	-
21	Check: UE1 announce interest for a specific FA to the MCx Service system (Interrogation, entrance to a specific list)	-->	SIP PUBLISH	4	P
22	Check: MCx Service system responds with a STATUS (see note 2) message	<--	SIP STATUS (see note 2) 200 OK	-	-
23	Check: UE1 send PUBLISH message containing an automatic FA information	-->	SIP PUBLISH	4	P
24	Check: MCx Service system responds with a STATUS (see note 2) message	<--	SIP STATUS (see note 2) 200 OK	-	-
25	Check: Wait for expiration of FA. According to IETF RFC 1886 [11] the expiration of a timer leads to a remove of the subscription. FA will automatically be rejected	<--	NOTIFY	4	-
26	Check: UE1 refresh FA activation with PUBLISH message containing the automatic FA information	-->	SIP PUBLISH	4	P
27	Check: MCx Service system responds with a STATUS (see note 2) message	<--	SIP STATUS (see note 2) 200 OK	-	-

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
28a	Check: receipt of an UPDATE message. Changing the location - leave location A, UE1 receive an update and a notification and deactivates FA-A	<--	UPDATE (Geolocation)	4	-
28b	Check: receipt of an NOTIFY message. Changing the location - leave location A, UE1 receive an update and a notification and deactivates FA-A	<--	NOTIFY	4	-
28c	Check: receipt of a n UPDATE message; Changing the location - enter location B, UE1 receive an update and a notification and activates FA-B	<--	UPDATE (Geolocation)	4	-
28d	Check: receipt of an NOTIFY message. Changing the location - leave location B, UE1 receive an update and a notification and deactivates FA-B	<--	NOTIFY	4	-
	EXCEPTION: Procedure 'MCX CT call release' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1) clause 5.3.12	--		-	-
	EXCEPTION: Procedure 'MCX CO communication' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1), clause 5.4.3	--		-	-
	EXCEPTION: Procedure 'Initial MCX Authentication, Registration, Configuration and Subscription' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1), clause 5.3.2 UE2 has active FA. Multiple use is not activated for this FA but take over is activated in the MCx Service system	--		-	-
29	Check: UE1 announce interest for a specific FA to the MCx Service system (Interrogation, entrance to a specific list)	-->	SIP PUBLISH	5	P
30	Check: MCx Service system responds with a STATUS (see note 2) message	<--	SIP STATUS (see note 2) 200 OK	-	-
31	Check: UE1 send PUBLISH message containing FA which is allocated to UE2 (overriding flag not active)	-->	SIP PUBLISH	5	P
32	Check: MCx Service system responds with a STATUS (see note 2) message 403 because FA is currently in use	<--	SIP STATUS (see note 2) 403 FORBIDDEN	5	-
33	Check (see note 3): UE1 send PUBLISH message containing FA which is allocated to UE2 (overriding flag active)	-->	SIP PUBLISH	5	P
34	Check: MCx Service system responds with a STATUS (see note 2) message	<--	SIP STATUS (see note 2) 200 OK	5	-
35a	Check: UE2 receive a notification and deactivates FA	<--	NOTIFY	5	-
35b	Check: UE1 receive a notification for activated FA	<--	NOTIFY	5	-
36	Check: UE1 send PUBLISH message containing FA which is allocated to UE2 (overriding flag not active)	-->	SIP PUBLISH	5	P
37	Check: MCx Service system responds with a STATUS (see note 2) message 403 because FA is currently in use	<--	SIP STATUS (see note 2) 403 FORBIDDEN	5	-
38	Check: UE1 send PUBLISH message containing FA which is allocated to UE2 (interrogation flag active)	-->	SIP PUBLISH	5	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
39	Check: UE1 receive a notification with interrogated content (FA activated/multiple usage/deactivated)	<--	NOTIFY	5	-
	EXCEPTION: Procedure 'MCX CO communication' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1), clause 5.4.3	--		-	-
	EXCEPTION: Procedure 'Initial MCX Authentication, Registration, Configuration and Subscription' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1), clause 5.3.2 UE1 has active FA on MCx Service system 1. UE3 has active FA on MCx Service system 2 and MCx Service systems are connected vice versa	--		-	-
40	Check: Initiate a call from UE3 connected to MCx Service system 2 to FA of UE1 allocated to MCx Service system 1	-->	SIP INVITE	6	P
41	Check: INVITE the UE1 into a connection. MCx Service user can be reached only by its FA and use the FA for outgoing identification	<--	SIP STATUS (see note 2) 100 (Trying)	6	-
42	UE1 send ALERT information in case it is not automatically accepted	<--	SIP STATUS (see note 2) 180 (Ringing)	6	-
43a	UE1 send ACTIVE information when call is accepted	<--	SIP ACK	6	-
43b	Check: UE1 terminates the call by sending BYE	-->	SIP BYE	6	P
43c	MCx Service system responds with STATUS message to the call termination request	<--	SIP STATUS (see note 2) 200 (OK)	6	-
	EXCEPTION: Procedure 'MCX CT call release' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1), clause 5.3.12	--		-	-
44	Check: Initiate a call from UE1 attached to MCx Service system 1 to FA of UE3 attached to MCx Service system 2	-->	SIP INVITE	6	P
45	Check: INVITE the UE2 into a connection. MCx Service user can be reached only by its FA and use the FA for outgoing identification	<--	SIP STATUS (see note 2) 100 (Trying)	6	-
46	UE1 send ALERT information in case it is not automatically accepted	<--	SIP STATUS (see note 2) 180 (Ringing)	6	-
47a	UE1 send ACTIVE information when call is accepted	<--	ACK	6	-
47b	Check: UE1 terminates the call by sending BYE	-->	SIP BYE	6	P
47c	MCx Service system responds with STATUS message to the call termination request	<--	SIP STATUS (see note 2) 200 (OK)	6	-
	EXCEPTION: Procedure 'MCX CT call release' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1), clause 5.3.12	--		-	-

NOTE 1: Trigger for the initialisation of the affiliation can be manual action by the user (e.g. pushing group affiliation button on the GUI) or automized via external interface or configuration setting in the MCx Service user configuration.

NOTE 2: According to ETSI TS 124 229 [8] (3GPP TS 24.229) a request should be rejected by sending a suitable 4xx response.

NOTE 3: MCx Service user application shall indicate information to user A including options change FA or call to responsible or MCX service user requests a list of possible FA. Via this menu the user can activate the FA overriding flag and/or interrogation flag.

5.1.3.3 Specific message contents

Table 5.1.3.3-1: SIP PUBLISH from the UE
(Table 5.1.3.2-1; step 1a ETSI TS 137 579-4 [7] (3GPP TS 37.579-4) Table 5.5.2.11-1)

Derivation Path: ETSI TS 137 579-4 [7] (3GPP TS 37.579-4), clause 5.5.2.11; condition POC-SETTINGS-EVENT, MCPTT-INFO-EVENT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Method	**"PUBLISH"**			
Request-URI	MCPTT_PublicServiceID	MCPTT originating user public ID		MCPTT
Message-body			IETF RFC 3261 [16] IETF RFC 3909 [i.3]	
MIME Multipart Media Encapsulation				
application/poc-settings+xml	<answer-mode>	Call answer mode		
	<user-profile-index>	User profile		
application/mcptt-Params+xml	<mcptt-access-token>	Authentication token		
	<mcptt-client-id>	MCPTT client ID		

Table 5.1.3.3-2: SIP PUBLISH from the UE
(Table 5.1.3.2-1; step 2a, ETSI TS 137 579-4 [7] (3GPP TS 37.579-4) Table 5.5.2.11-1)

Derivation Path: ETSI TS 137 579-4 [7] (3GPP TS 37.579-4), clause 5.5.2.11; condition MCPTT-INFO-EVENT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Method	**"SUBSCRIBE"**			
Request-URI	MCPTT_PublicServiceID	MCPTT originating user public ID		MCPTT
Message-body			IETF RFC 3261 [16]	
Extensible Markup Language				
application/mcpttinfo+xml	<mcpttURI>	MCPTT user identity		MCPTT
<entry>				
<anyExt>	<manual-deactivation-not-allowed-if-location-criteria-met>	FA is location depending		True/False
	<FunctionalAliasList>	Functional alias subscription list (PIDF (FA-ID))		FUNCTIONAL ALIASING

Table 5.1.3.3-3: SIP INVITE from the UE
(Table 5.1.3.2-1; step 11, ETSI TS 137 579-4 [7] (3GPP TS 37.579-4) Table 5.5.2.5.1-1)

Derivation Path: ETSI TS 137 579-4 [7] (3GPP TS 37.579-4) clause 5.5.2.5.1; condition MCPTT-INFO-EVENT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Method	**"INVITE"**			
Request-URI	MCPTT_PublicServiceID	MCPTT originating user public ID		MCPTT
Message-header				
P-asserted identity/SIP PAI Address	<SIP PAI User part>			
	<SIP PAI Host part>			
Route				
addr-spec[1]	SIP URI			
user-info and host	P-CSCF address of the SS	P-CSCF address as assigned to the UE via NAS signalling or P-CSCF discovery		
port	protected server port of the SS	as assigned during registration		
uri-parameters	List of parameters			
Message-body				
Application/mcpttinfo+xml				
<mcptt-Params>	<session-type>	Type of call, e.g. adhoc, prearranged		MCPTT
	<mcptt-request-uri>			
	<mcptt-calling-user-id>			
	<mcptt-calling-group-id>			
	<anyExt>			

Table 5.1.3.3-4: SIP STATUS from the MCx Service system
(step 11, Table 5.1.3.2-1; ETSI TS 137 579-4 [7] (3GPP TS 37.579-4) Table 5.5.2.16-1)

Derivation Path: Derivation Path: IETF RFC 3261 [16]; condition STATUS-CODE				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Status-Line				
Status-Code	100		ETSI TS 124 229 [8] (3GPP TS 24.229)	
Reason-Phrase	(Trying)			

Table 5.1.3.3-5: SIP STATUS from the UE
(step 12, Table 5.1.3.2-1; step x, ETSI TS 137 579-4 [7] (3GPP TS 37.579-4) Table 5.5.2.16-1)

Derivation Path: Derivation Path: IETF RFC 3261 [16]; condition STATUS-CODE				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Status-Line				
Status-Code	180		ETSI TS 124 229 [8] (3GPP TS 24.229)	
Reason-Phrase	(Ringing)			

Table 5.1.3.3-6: SIP STATUS
(step 1b, Table 5.1.3.2-1; step x, ETSI TS 137 579-4 [7] (3GPP TS 37.579-4) Table 5.5.2.17-1)

Derivation Path: Derivation Path: IETF RFC 3261 [16]; condition STATUS-CODE				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Status-Line				
Status-Code	200		ETSI TS 124 229 [8] (3GPP TS 24.229)	
Reason-Phrase	(OK)			

Table 5.1.3.3-7: SIP STATUS from the MCx Service system
(step 3b, Table 5.1.3.2-1; ETSI TS 137 579-4 [7] (3GPP TS 37.579-4) Table 5.5.2.18-1)

Derivation Path: Derivation Path: IETF RFC 3261 [16]; condition STATUS-CODE				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Status-Line				
Status-Code	401		ETSI TS 124 229 [8] (3GPP TS 24.229)	
Reason-Phrase	(Unauthorized)			

Table 5.1.3.3-8: SIP STATUS from the MCx Service system
(step 3a, Table 5.1.3.2-1; ETSI TS 137 579-4 [7] (3GPP TS 37.579-4) Table 5.5.2.18-1)

Derivation Path: Derivation Path: IETF RFC 3261 [16]; condition STATUS-CODE				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Status-Line				
Status-Code	403		ETSI TS 124 229 [8] (3GPP TS 24.229)	
Reason-Phrase	(Forbidden)			

Table 5.1.3.3-9: SIP NOTIFY
(step 20, Table 5.1.3.2-1; ETSI TS 137 579-4 [7] (3GPP TS 37.579-4) Table 5.5.2.8-1)

Derivation Path: ETSI TS 137 579-4 [7] (3GPP TS 37.579-4), clause 5.5.2.8; condition PRESENCE-EVENT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Method	**"NOTIFY"**			
Request-URI	MCPTT_PublicServiceID	MCPTT originating user public ID		MCPTT
Message-body			IETF RFC 3261 [16]	
Extensible Markup Language	<mcpttURI>	MCPTT user identity		
application/event-type+xml	<presence>	PIDF	PIDF according to ETSI TS 137 579-4 [7] Table 5.5.3.5.1-1	PRESENCE-EVENT
	<xcap-diff>	xcap-div	xcap-diff as described in ETSI TS 137 579-4 [7] Table 5.5.3.12-1	CONFIG. GROUP-CONFIG
	<poc-settings>		PoC-Settings as described in ETSI TS 137 579-4 [7] Table 5.5.3.11.2-1	POC-SETTINGS-EVENT

Table 5.1.3.3-10: SIP UPDATE
(step 28a, Table 5.1.3.2-1; ETSI TS 137 579-4 [7] (3GPP TS 37.579-4) Table 5.5.2.15-1))

Derivation Path: ETSI TS 137 579-4 [7] (3GPP TS 37.579-4), clause 5.5.2.15); condition SDS-MESSAGE-PART				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Method	**"UPDATE"**			
Request-URI	MCPTT_PublicServiceID	MCPTT originating user public ID		MCPTT
Message-body			IETF RFC 3261 [16]	
Extensible Markup Language	<mcpttURI>	MCPTT user identity		

Table 5.1.3.3-11: SIP ACK from the MCx Service system
(step 28a, Table 5.1.3.2-1; ETSI TS 137 579-4 [7] (3GPP TS 37.579-4) Table 5.5.2.1.2-1))

Derivation Path: ETSI TS 137 579-4 [7] (3GPP TS 37.579-4), clause 5.5.2.1.2); condition ROUTE-INFORMATION-PART				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Method	**"ACK"**			
Request-URI	MCPTT_PublicServiceID	MCPTT originating user public ID		MCPTT
Route			IETF RFC 3261 [16]	
Route-param list	<destination>	MCPTT user endpoint		
	<port>	Port information		
	<protocol>	Used protocol type		

Table 5.1.3.3-12: SIP BYE from the UE
(step 28a, Table 5.1.3.2-1; ETSI TS 137 579-4 [7] (3GPP TS 37.579-4) Table 5.5.2.2.1-1))

Derivation Path: ETSI TS 137 579-4 [7] (3GPP TS 37.579-4), clause 5.5.2.2.1); condition ROUTE-INFORMATION-PART				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Method	**"ACK"**			
Request-URI	MCPTT_PublicServiceID	MCPTT originating user public ID		MCPTT
Route			IETF RFC 3261 [16]	
Route-param list	<destination>	MCPTT user endpoint		
	<port>	Port information		
	<protocol>	Used protocol type		
P-asserted identity	<SIP PAI user part>			
	<SIP PAI host part>			

5.2 Floor Control in general

5.2.1 Test purpose (TP)

(1)

```
with {Floor control request/granted/release}
ensure that {
    when {MCX service user can handle the floor control and MCX service user is in an active
communication with floor control and ongoing incoming or outgoing communication with floor control}
    then {MCX service user can detect correctly the activation of floor control and MCX service user
requests the floor control the MCX service system response result "granted" and "release" after
successful usage and MCX service user indicates the returned status and MCX service user indicates
talker status according to notification audio-visual}
            }
```

(2)

with {MCX service user can handle the floor control}
ensure that {
    when {MCX service user A and MCX service user B are in an active communication with floor control and MCX service user requests the floor control}
    then {MCX service system reject the request with status "busy, overriding not allowed" (floor used by another MCX service user) **or** override the floor control (without revoke of floor control)}
            }

(3)

with {Floor control revoke floor control}
ensure that {
    when {MCX service user can revoke the floor control and MCX service user A and MCX service user B are in an active communication with floor control and MCX service user A has active floor control with continuous voice stream (see [5] Figure 10.9.1.6.1-1) and MCX service user B has the authorization to revoke the floor}
    then {Floor control is released after floor control timer expiry **or** revoke by MCX service user B **and** information for idle floor to all the participants and in multi-talker case idle floor will be noticed by all participants}
            }

(4)

with {MCX system can queue the floor control requests by multiple users}
ensure that {
    when {MCX service user A and MCX service user B are in an active communication with floor control, and other MCX service user have requested the floor and multi-taker floor control is not activated by the MCX service system}
    then {MCX service system detects the several floor requests and queue them and MCX service user request the own position in the floor control queue}
    else when {MCX service user A and MCX service user B are in an active communication with floor control, and other MCX service user have requested the floor and multi-taker floor control is not activated by the MCX service system and floor control queue contains entries and talker releases the floor control}
    then {MCS service system provides notification to all MCX service users and next MCX service user **or** MCX service user with the highest priority in the floor control queue becomes active talker}
            }

(5)

with {multi-Talker floor control}
ensure that {
    when {Multiple MCX service users can activate the floor control at the same time, and multiple MCX service users are in an active communication with floor control, and other MCX service user have requested the floor and multi-talker floor control is activated by the MCX service system}
    then {Multiple MCX service users (minimum two) activate the floor control and voice transmission shall be handled correctly by the MCX service system}
            }

(6)

with {Floor priorities with migrating MCX user (multiple MC systems)}
ensure that {
    when {Multiple MCX service users are configured with different floor control priorities and MCX service user with active communication and floor control queuing during the migration}
    then {Active floor control migrates from MCX service system A to another MCX service system B and the priority of the floor control and the position in the floor control queue shall not be changed during migration}
            }

(7)

with {Active ad hoc group communication involving multiple participants with the authorization to use multi-talker control}
ensure that {
    when {Number of MCPTT Users requesting the permission to talk exceeds}
    then {The MCPTT Service provides a mechanism to override and MCPTT User will receive a notification e.g. override}
            }

(8)

```
with {Active ad hoc group communication involving multiple participants with the authorization to
use multi-talker control and maximum number of simultaneous talkers can be changed by an authorized
MCPTT User}
ensure that {
    when {Number of MCPTT Users requesting the permission to talk exceeds and an authorized MCPTT
User change the maximum number of simultaneous talkers during active ad hoc group communication}
    then {Additional MCPTT Users authorized for multi-talker service can talk and further MCPTT
Users can request the permission to talk until the new maximum number of simultaneous talkers
exceed}
            }
```

(9)

```
with {An MCPTT system configured for a pre-arranged group where a designated controller has the
privilege for implicit floor establishment upon call initiation}
ensure that
    when {The authorized controller initiates a pre-arranged group call with implicit floor control}
    then {The group call is successfully established, and the floor is automatically and immediately
granted to the initiating controller without the need for an explicit and implicit floor request
(PTT) procedure and controller's audio is successfully streamed to all called participants}
```

5.2.2 Conformance requirements

[ETSI TS 122 179 [1] (3GPP TS 22.179), clause [R-6.2.3.7.4.1-001]]

"If the number of MCPTT Users requesting the permission to talk exceeds the maximum number of simultaneous talkers in an MCPTT Group configured for multi-talker control, the MCPTT Service shall apply the override mechanism."

[ETSI TS 122 179 [1] (3GPP TS 22.179), clause [R-6.2.3.7.2-006]]

"The MCPTT Service shall allow an authorized MCPTT User to change the maximum number of simultaneous talkers at any time during a group call in an MCPTT Group configured for multi-talker control."

5.2.3 Test description

5.2.3.1 Pre-test conditions

System Configuration:

- MCx Service system consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (UE1) also consists of all sub-systems of the MCx user service.
- For the underlying "transport bearer" over which the UE1 (MCx Service system) and the IUT will communicate Parameters are set to the default parameters specific for the basic E-UTRA single simulated cells cell network scenarios, as defined in ETSI TS 138 508-1 [6] (3GPP TS 38.508-1), clause 6.3.4.4. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCx operation in the MCx configuration document).
- MCx Service system is configured to support the following features:
  - Implicit and explicit floor control.
  - Multiple user floor control.
  - Multi-talker floor control.
  - Prioritisation of floor control requests.
  - Acknowledgement not required according to ETSI TS 124 380 [12] (3GPP TS 24.380), Table 8.2.2.1-1.

IUT:

- IUT (UE1).
- The IUT (UE1) consists of all sub-systems of the MCx user service including the ability for floor control.
- The IUT (UE1) is the acting user.

Preamble:

- The IUT (UE1) is connected to PLMN of the primary 5GS. UE3 is connected to the PLMN of the secondary 5GS.
- The MCx Service system (MCx_S1) is connected to the IUT (UE1, MCx client) and to the supporting devices UE2 (MCx client) and UE-Con (MCx client with controller/dispatcher functionality).
- MCx service SIP registration and affiliation are executed on MCx_S1.
- Partner MCx Service system (MCx_S2) is connected to MCx_S1.

5.2.3.2 Test procedure sequence

Table 5.2.3.2-1: Main behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
	EXCEPTION: Procedure 'MCX CO communication' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1), clause 5.4.3	--		-	-
	EXCEPTION: Procedure 'Initial MCX Authentication, Registration, Configuration and Subscription' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1), clause 5.3.2	--		-	-
1a	EXCEPTION: UE originating case. UE1start INVITE	-->	SIP INVITE according ETSI TS 124 229 [8] (3GPP TS 24.229)	-	-
2a	Incoming response to the INVITE; MCx_S1 responds to the INVITE request (see note 1)	<--	100 (TRYING)	-	-
3a	OPTIONAL: incoming response to TRYING if no automatic answering mode is activated	<--	180 (RINGING)	-	-
4a	Call established confirmation from MCx_S1	<--	200 (OK)	-	-
1b	UE terminating case. UE receive INVITE	<--	SIP INVITE	-	-
2b	UE1 responds to the INVITE request (see note 1)	-->	100 (TRYING)	-	P
3b	OPTIONAL: outgoing response to TRYING if no automatic answering mode is activated	-->	180 (RINGING)	-	-
4b	Call established confirmation towards MCx_S1	-->	200 (OK)	-	P
5	Check: UE1 opens encapsulated multipart session description protocol in the INVITE message. It contains following information: Media attribute (a): MCPTT implicit request - Receiver expects the information about (MCPTT) FLOOR control messages in the real time control protocol part	-->	Media format specific parameters: fmtp: mc_implicit_request	1	P
6	UE1 send RTCP (MCPT) Floor. MCx client on UE1 indicate the occupation status (see note 2) Check: UE1 REQUEST the floor; following information included: Subtype 0 Floor Request {Floor priority | user ID | track info | floor indicator | FA | location}	-->	RTCP (MCPT) Floor REQUEST according to ETSI TS 124 380 [12] (3GPP TS 24.380), clause 8.2.4 and IETF RFC 1889 [11]	1	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
6a	Check: UE1 receive GRANTED; following information included: Subtype 1 Floor Granted {duration | audio SSRC of granted participant | floor priority | user ID | queue size | SSRC of queued user | queued user ID | queue ID | floor indicator | track info | floor indicator}	<--	RTCP (MCPT) Floor GRANTED according to ETSI TS 124 380 [12] (3GPP TS 24.380), clause 8.2.5	-	-
6b	Check: UE1 receives TAKEN message by MCx_S1; the following information included: Subtype 2 Floor Taken {RTCP SSRC of sending floor | granted party ID | permission to request the floor | user ID | message sequence | track info | floor indicator | audio SSRC of granted user | FA | list of granted users | list of audio SSRC | list of FA | location info | list of locations of granted users}	<--	RTCP (MCPT) Floor TAKEN According to ETSI TS 124 380 [12] (3GPP TS 24.380), clause 8.2.9	-	-
6c	Check: UE1 receives DENY by MCx_S1; the following information included: Subtype 3 Floor Deny {Reject Cause (see note 3) | User granted the floor | floor indicator}	<--	RTCP (MCPT) Floor DENY Reject causes according to ETSI TS 124 380 [12] (3GPP TS 24.380) clause 8.2.6.2	-	-
6d	Check: UE1 receives RELEASE message by MCx_S1; the following information included: Subtype 4 Floor Release {user ID | track info | floor indicator}	<--	RTCP (MCPT) Floor RELEASE according to ETSI TS 124 380 [12] (3GPP TS 24.380) clause 8.2.7	-	-
6e	Check: UE1 receives IDLE message by MCx_S1; the following information included: Subtype 5 Floor Idle {SSRC of floor control server | message sequence | track info | floor indicator}	<--	RTCP (MCPT) Floor IDLE according to ETSI TS 124 380 [12] (3GPP TS 24.380) clause 8.2.8	-	-
6f	Check: UE1 receives REVOKE message by MCx_S1; the following information included: Subtype 6 Floor Revoke {SSRC of floor control server | reject cause (see note 3) | track info | floor indicator}	<--	RTCP (MCPT) Floor REVOKE according to ETSI TS 124 380 [12] (3GPP TS 24.380) clause 8.2.10	-	-
6g	Check: UE1 receives ACK message by MCx_S1; the following information included: Subtype 10 Floor Ack {SSRC of sender | source field | message type | track info | location indicator}	<--	RTCP (MCPT) Floor ACK according to ETSI TS 124 380 [12] (3GPP TS 24.380) clause 8.2.13	-	-
7	Check: UE1 terminates the call by sending BYE	-->	SIP BYE	-	P
8	MCx Service system responds with STATUS message to the call termination request	<--	SIP STATUS (see note 1) 200 (OK)	-	-
	EXCEPTION: Procedure 'MCX CT call release' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1) clause 5.3.12	--		-	-
	EXCEPTION: Procedure 'MCX CO communication' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1) clause 5.4.3	--		-	-
	EXCEPTION: Procedure 'Initial MCX Authentication, Registration, Configuration and Subscription' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1) clause 5.3.2	--		-	-
9a	EXCEPTION: UE originating case. UE1start INVITE	-->	SIP INVITE	-	-
10a	Incoming response to the INVITE; MCx_S1 responds to the INVITE request (see note 1)	<--	100 (TRYING)	-	-
11a	OPTIONAL: incoming response to TRYING if no automatic answering mode is activated	<--	180 (RINGING)	-	-
12a	Call established confirmation from MCx_S1	<--	200 (OK)	-	-
9b	UE terminating case. UE receive INVITE	<--	SIP INVITE	-	-

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
10b	UE1 responds to the INVITE request (see note 1)	-->	100 (TRYING)	-	P
11b	OPTIONAL: outgoing response to TRYING if no automatic answering mode is activated	-->	180 (RINGING)	-	-
12b	Call established confirmation towards MCx_S1	-->	200 (OK)	-	P
13	Check: UE1 open encapsulated multipart session description protocol in the INVITE message. It contains following information: Media attribute (a): MCPTT implicit request - Receiver expects the information about (MCPTT) FLOOR control messages in the real time control protocol part	-->	Media format specific parameters: fmtp: mc_implicit_request	3	P
14	Floor revoke request by UE1 according to ETSI TS 123 379 [5] (3GPP TS 23.379) clause 10.9.1.6.1 Check: UE1 send REVOKE message to MCx Service system; the following information included: Subtype 6 Floor Revoke {SSRC of floor control server | reject cause (see note 3) | track info | floor indicator}	-->	RTCP (MCPT) Floor REVOKE	3	P
14a	Check: MCx_S1 check the authorization for the revoke request; Not Authorized; UE1 receives DENY message by a remote participant; the following information included: Subtype 3 Floor Deny {Reject Cause | User granted the floor | floor indicator}	<--	RTCP (MCPT) Floor DENY	-	-
14b	Check: MCx_S1 check the authorization for the revoke request; Authorized; MCx_S1 provides the REVOKE message to all involved participants; the following information included: Subtype 6 Floor Revoke {SSRC of floor control server | reject cause (see note 3) | track info | floor indicator}	<--	RTCP (MCPT) Floor REVOKE	-	-
14c	Check: MCx_S1 provides IDLE message to all remote participants; the following information included: Subtype 5 Floor Idle {SSRC of floor control server | message sequence | track info | floor indicator}	<--	RTCP (MCPT) Floor IDLE	-	-
15	Check: UE1 terminates the call by sending BYE	-->	SIP BYE	-	P
16	MCx Service system responds with STATUS message to the call termination request	<--	SIP STATUS (see note 1) 200 (OK)	-	-
	EXCEPTION: Procedure 'MCX CT call release' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1) clause 5.3.12	--		-	-
	EXCEPTION: Procedure 'MCX CO communication' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1) clause 5.4.3	--		-	-
	EXCEPTION: Procedure 'Initial MCX Authentication, Registration, Configuration and Subscription' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1) clause 5.3.2	--		-	-
17a	EXCEPTION: UE originating case. UE1start INVITE	-->	SIP INVITE	-	-
18a	Incoming response to the INVITE; MCx_S1 responds to the INVITE request (see note 1)	<--	100 (TRYING)	-	-
19a	OPTIONAL: incoming response to TRYING if no automatic answering mode is activated	<--	180 (RINGING)	-	-
20a	Call established confirmation from MCx_S1	<--	200 (OK)	-	-

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
17b	UE terminating case. UE receive INVITE	<--	SIP INVITE	-	-
18b	UE1 responds to the INVITE request (see note 1)	-->	100 (TRYING)	-	P
19b	OPTIONAL: outgoing response to TRYING if no automatic answering mode is activated	-->	180 (RINGING)	-	-
20b	Call established confirmation towards MCx_S1	-->	200 (OK)	-	P
21	Minimum two user, UE2 & UE3 are involved into the active communication and activate FLOOR control; Multiple access granted Check: UE1 opens encapsulated multipart session description protocol in the INVITE message. It contains following information: Media attribute (a): MCPTT implicit request - Receiver expects the information about (MCPTT) FLOOR control messages in the real time control protocol part	-->	Media format specific parameters: fmtp: mc_implicit_request	-	P
22a	UE1 request the floor	-->	RTCP (MCPT) Floor REQUEST	-	P
22b	UE1 granted for multi-talker FLOOR control. Check: UE1 GRANTED the floor; following information included: Subtype 1 Floor Granted {duration | audio SSRC of granted participant | floor priority | user ID | queue size | SSRC of queued user | queued user ID | queue ID | floor indicator | track info | floor indicator}	<--	RTCP (MCPT) Floor GRANTED	-	-
23a	Check: UE1 receives RTP data (voice) from both, UE2 & UE3; RTP stream contains payload type information	<--	RTP stream	5	-
23b	Check: UE1 send RTP data (voice) from both, UE2 & UE3; RTP stream contains payload type information	-->	RTP stream	5	P
24	UE1 with no permission for multi-talker FLOOR control. FLOOR control activated. Queued by the MCx_S1 Check: UE1 sends floor request message	-->	RTCP (MCPT) Floor REQUEST	4	P
24a	Check marked elements: UE1 receives TAKEN message by MCx_S1; the following information included: Subtype 2 Floor Taken {RTCP SSRC of sending floor | granted party ID | permission to request the floor | user ID | message sequence | track info | floor indicator | audio SSRC of granted user | FA | list of granted users | list of audio SSRC | list of FA | location info | list of locations of granted users}	<--	RTCP (MCPT) Floor TAKEN	-	-
	Check: UE1 send floor ACK message to confirm the received Granted message	-->	RTCP (MCPT) Floor ACK		
24b	Check: UE1 transmit no RTP data	-->	No outgoing RTP stream	4	P
25	UE1 with no permission for multi-talker FLOOR control. FLOOR control activated. All UE have released FLOOR control Check: UE1 sends floor request message	-->	RTCP (MCPT) Floor REQUEST	4	P
25a	Check: UE1 GRANTED the floor; following information included: Subtype 1 Floor Granted {duration | audio SSRC of granted participant | floor priority | user ID | queue size | SSRC of queued user | queued user ID | queue ID | floor indicator | track info | floor indicator}	<--	RTCP (MCPT) Floor GRANTED	-	-
25b	Check: RTP data (voice) can be received from UE1; RTP stream contains payload type information	-->	RTP stream	5	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
26	Check: UE1 terminates the call by sending BYE	-->	SIP BYE	-	P
27	MCx Service system responds with STATUS message to the call termination request	<--	SIP STATUS (see note 1) 200 (OK)	-	-
	EXCEPTION: Procedure 'MCX CT call release' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1) clause 5.3.12	--		-	-
	EXCEPTION: Procedure 'MCX CO communication' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1) clause 5.4.3	--		-	-
	EXCEPTION: Procedure 'Initial MCX Authentication, Registration, Configuration and Subscription' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1), clause 5.3.2	--		-	-
28a	EXCEPTION: UE originating case. UE1start INVITE	-->	SIP INVITE	-	-
29a	Incoming response to the INVITE; MCx_S1 responds to the INVITE request (see note 1)	<--	100 (TRYING)	-	-
30a	OPTIONAL: incoming response to TRYING if no automatic answering mode is activated	<--	180 (RINGING)	-	-
31a	Call established confirmation from MCx_S1	<--	200 (OK)	-	-
28b	UE terminating case. UE receive INVITE	<--	SIP INVITE	-	-
29b	UE1 responds to the INVITE request (see note 1)	-->	100 (TRYING)	-	P
30b	OPTIONAL: outgoing response to TRYING if no automatic answering mode is activated	-->	180 (RINGING)	-	-
31b	Call established confirmation towards MCx_S1	-->	200 (OK)	-	P
32	UE2 & UE3 are granted for multi-talker FLOOR control Check: UE1 additionally request the floor	-->	RTCP (MCPT) Floor REQUEST	6	P
33	UE1 receive floor GRANTED; following information included: Subtype 1 Floor Granted {duration | audio SSRC of granted participant | floor priority | user ID | queue size | SSRC of queued user | queued user ID | queue ID | floor indicator | track info | floor indicator}	<--	RTCP (MCPT) Floor GRANTED	-	-
34	Check: UE1 send RTP data (voice); RTP stream contains payload type information	-->	RTP stream	6	P
35	UE1 with no permission for multi-talker FLOOR control. FLOOR control activated. Queued by the MCx_S1	-->	RTCP (MCPT) Floor REQUEST	6	P
36a	Check marked elements: UE1 receives TAKEN message by MCx_S1; the following information included: Subtype 2 Floor Taken {RTCP SSRC of sending floor | granted party ID | permission to request the floor | user ID | message sequence | track info | floor indicator | audio SSRC of granted user | FA | list of granted users | list of audio SSRC | list of FA | location info | list of locations of granted users}	<--	RTCP (MCPT) Floor TAKEN	-	-
36b	UE1 migrate from MCx_S1 to MCx_S2 while FLOOR control is activated Check: UE1 receives TAKEN message by MCx_S2; the following information included: Subtype 2 Floor Taken {RTCP SSRC of sending floor | granted party ID | permission to request the floor | user ID | message sequence | track info | floor indicator | audio SSRC of granted user | FA | list of granted users | list of audio SSRC | list of FA | location info | list of locations of granted users}	<--	RTCP (MCPT) Floor TAKEN	-	-

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
36c	Compare elements in list of granted users. Position of UE1 shall not be changed	--	No differences	6	P
37a	UE2 & UE3 are granted for multi-talker FLOOR control Check: UE1 sends floor request message	-->	RTCP (MCPT) Floor REQUEST	6	P
37b	Check: both UE GRANTED the floor; following information included: Subtype 1 Floor Granted {duration | audio SSRC of granted participant | floor priority | user ID | queue size | SSRC of queued user | queued user ID | queue ID | floor indicator | track info | floor indicator}	<--	RTCP (MCPT) Floor GRANTED	-	-
38a	UE1 with permission for multi-talker FLOOR control. FLOOR control activated. Handled by the MCx_S1 Check: UE1 sends floor request message	-->	RTCP (MCPT) Floor REQUEST	6	P
38b	Check marked elements: UE1 receives GRANTED the floor from MCx_S1; following information included: Subtype 1 Floor Granted {duration | audio SSRC of granted participant | floor priority | user ID | queue size | SSRC of queued user | queued user ID | queue ID | floor indicator | track info | floor indicator}	<--	RTCP (MCPT) Floor GRANTED	-	-
38c	UE1 migrate from MCx_S1 to MCx_S2 while FLOOR control is activated. Check marked elements: UE1 receives GRANTED the floor from MCx_S2; following information included: Subtype 1 Floor Granted {duration | audio SSRC of granted participant | floor priority | user ID | queue size | SSRC of queued user | queued user ID | queue ID | floor indicator | track info | floor indicator}	<--	RTCP (MCPT) Floor GRANTED	-	-
38d	Compare elements in floor priority. Position of UE1 shall not be changed.	--	No differences	6	P
39	Check: UE1 terminates the call by sending BYE	-->	SIP BYE	-	P
40	MCx Service system responds with STATUS message to the call termination request	<--	SIP STATUS (see note 1) 200 (OK)	-	-
	EXCEPTION: Procedure 'MCX CT call release' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1) clause 5.3.12	--		-	-
	EXCEPTION: Procedure 'MCX CO communication' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1) clause 5.4.3	--		-	-
	EXCEPTION: Procedure 'Initial MCX Authentication, Registration, Configuration and Subscription' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1) clause 5.3.2	--		-	-
41a	EXCEPTION: UE originating case. UE1start INVITE	-->	SIP INVITE	-	-
42a	Incoming response to the INVITE; MCx_S1 responds to the INVITE request (see note 1)	<--	100 (TRYING)	-	-
43a	OPTIONAL: incoming response to TRYING if no automatic answering mode is activated	<--	180 (RINGING)	-	-
44a	Call established confirmation from MCx_S1	<--	200 (OK)	-	-
41b	UE terminating case. UE receive INVITE	<--	SIP INVITE	-	-
42b	UE1 responds to the INVITE request (see note 1)	-->	100 (TRYING)	-	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
43b	OPTIONAL: outgoing response to TRYING if no automatic answering mode is activated	-->	180 (RINGING)	-	-
44b	Call established confirmation towards MCx_S1	-->	200 (OK)	-	P
45	UE2 & UE3 are granted for multi-talker FLOOR control. MCx_S1 allows max. two multi-talker at the same time and override is activated Check: both UE GRANTED the floor; following information included: Subtype 1 Floor Granted	--	RTCP (MCPT) Floor GRANTED	7	-
46	UE1 with permission for multi-talker send FLOOR request message. Floor priority is set equal to UE2 & UE3. FLOOR control activated Check: UE1 sends floor request message	-->	RTCP (MCPT) Floor REQUEST	7	P
47	Check: UE1 receives TAKEN message by MCx_S1; the following information included: Subtype 2 Floor Taken	<--	RTCP (MCPT) Floor TAKEN	-	-
48	UE1 with permission for multi-talker send FLOOR request message. Floor priority is set higher than UE2 or UE3 (override). FLOOR control activated	-->	RTCP (MCPT) Floor REQUEST	7	P
49	Check: UE1 GRANTED the floor; following information included: Subtype 1 Floor Granted	<--	RTCP (MCPT) Floor GRANTED	-	-
50	UE2 or UE3 override floor granted override Check: UE2 or UE3 receives TAKEN message by MCx_S1; the following information included: Subtype 2 Floor Taken	<--	RTCP (MCPT) Floor TAKEN	-	-
51	Check: UE1 terminates the call by sending BYE	-->	SIP BYE	-	P
52	MCx Service system responds with STATUS message to the call termination request	<--	SIP STATUS (see note 1) 200 (OK)	-	-
	EXCEPTION: Procedure 'MCX CT call release' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1) clause 5.3.12	--		-	-
	EXCEPTION: Procedure 'MCX CO communication' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1) clause 5.4.3	--		-	-
	EXCEPTION: Procedure 'Initial MCX Authentication, Registration, Configuration and Subscription' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1) clause 5.3.2	--		-	-
53a	EXCEPTION: UE originating case. UE1 send INVITE for ad hoc group communication	-->	SIP INVITE	-	P
54a	Incoming response to the INVITE; MCx_S1 responds to the INVITE request (see note 1)	<--	100 (TRYING)	-	-
55a	OPTIONAL: incoming response to TRYING if no automatic answering mode is activated	<--	180 (RINGING)	-	-
56a	Call established confirmation from MCx_S1	<--	200 (OK)	-	-
53b	UE terminating case. UE1 send INVITE for ad hoc group communication	<--	SIP INVITE	-	-
54b	UE1 responds to the INVITE request (see note 1)	-->	100 (TRYING)	-	P
55b	OPTIONAL: outgoing response to TRYING if no automatic answering mode is activated	-->	180 (RINGING)	-	-
56b	Call established confirmation towards MCx_S1	-->	200 (OK)	-	P
57	UE1 with permission for multi-talker send FLOOR request message. FLOOR control activated Check: UE1 sends floor request message	-->	RTCP (MCPT) Floor REQUEST	8	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
58	Check: UE1 receive floor GRANTED message; following information included: Subtype 1 Floor Granted	<--	RTCP (MCPT) Floor GRANTED	-	-
59	UE2 with permission for multi-talker send FLOOR request message while UE1 is still GRANTED. FLOOR control activated Check: UE2 sends floor request message	-->	RTCP (MCPT) Floor REQUEST	8	P
60	Check: UE2 receive floor GRANTED message; following information included: Subtype 1 Floor Granted	<--	RTCP (MCPT) Floor GRANTED	-	-
61	UE3 with permission for multi-talker send FLOOR request message while UE1 and UE2 still GRANTED. FLOOR control activated Check: Check: UE3 sends floor request message	-->	RTCP (MCPT) Floor REQUEST	8	P
62	Check: UE3 receive floor GRANTED message; following information included: Subtype 1 Floor Granted	<--	RTCP (MCPT) Floor GRANTED	-	-
63	Check: UE1 terminates the call by sending BYE	-->	SIP BYE	-	P
64	MCx Service system responds with STATUS message to the call termination request	<--	SIP STATUS (see note 1) 200 (OK)	-	-
	EXCEPTION: Procedure 'MCX CT call release' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1) clause 5.3.12	--		-	-
	EXCEPTION: Procedure 'MCX CO communication' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1) clause 5.4.3	--		-	-
	EXCEPTION: Procedure 'Initial MCX Authentication, Registration, Configuration and Subscription' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1) clause 5.3.2	--		-	-
65a	EXCEPTION: UE originating case. UE1 send INVITE for ad hoc group communication	-->	SIP INVITE	-	P
66a	Incoming response to the INVITE; MCx_S1 responds to the INVITE request (see note 1)	<--	100 (TRYING)	-	-
67a	OPTIONAL: incoming response to TRYING if no automatic answering mode is activated	<--	180 (RINGING)	-	-
68a	Call established confirmation from MCx_S1	<--	200 (OK)	-	-
65b	EXCEPTION: UE terminating case. UE1 receive INVITE for ad hoc group communication	<--	SIP INVITE	-	P
66b	Incoming response to the INVITE; MCx_S1 responds to the INVITE request (see note 1)	-->	100 (TRYING)	-	-
67b	OPTIONAL: incoming response to TRYING if no automatic answering mode is activated	-->	180 (RINGING)	-	-
68b	Call established confirmation from UE1	-->	200 (OK)	-	-
69	UE2 and UE3 have requested the floor, MCx_S1 has granted for both. UE1 with permission for multi-talker send FLOOR request message. Maximum numbers for simultaneous talkers exceed. FLOOR control activated Check: UE1 sends floor request message	-->	RTCP (MCPT) Floor REQUEST	-	P
70	Check: UE1 receives TAKEN message by MCx_S1; the following information included: Subtype 2 Floor Taken	<--	RTCP (MCPT) Floor TAKEN	-	-

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
71	MCx_S1 has permission to change the maximum number of simultaneous talkers. Number increased	--		-	-
72	Check: UE1 sends floor request message	-->	RTCP (MCPT) Floor REQUEST	-	P
73	UE1 with permission for multi-talker send FLOOR request message. FLOOR control activated Check: UE1 GRANTED the floor; following information included: Subtype 1 Floor Granted	<--	RTCP (MCPT) Floor GRANTED	-	-
74	Check: UE1 terminates the call by sending BYE	-->	SIP BYE	-	P
75	MCx Service system responds with STATUS message to the call termination request	<--	SIP STATUS (see note 1) 200 (OK)	-	-
	EXCEPTION: Procedure 'MCX CO communication' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1) clause 5.4.3	--		-	-
	EXCEPTION: Procedure 'Initial MCX Authentication, Registration, Configuration and Subscription' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1) clause 5.3.2	--		-	-
76a	Pre-arranged group communication with implicit floor control EXCEPTION: UE-Con send INVITE for pre-arranged group communication	-->	SIP INVITE	-	P
77a	Incoming response to the INVITE; MCx_S1 responds to the INVITE request (see note 1)	<--	100 (TRYING)	-	-
78a	OPTIONAL: incoming response to TRYING if no automatic answering mode is activated	<--	180 (RINGING)	-	-
79a	Call established confirmation from MCx_S1	<--	200 (OK)	-	-
76b	UE1 receive INVITE for pre-arranged group communication	<--	SIP INVITE	-	-
77b	Incoming response to the INVITE; MCx_S1 responds to the INVITE request (see note 1)	-->	100 (TRYING)	-	-
78b	OPTIONAL: incoming response to TRYING if no automatic answering mode is activated	-->	180 (RINGING)	-	-
79b	Call established confirmation from UE1	-->	200 (OK)	-	-
80	FLOOR control activated. UE-Con receive immediately floor granted message without explicit request Check: UE-Con receive GRANTED; following information included: Subtype 1 Floor Granted	<--	RTCP (MCPT) Floor GRANTED	-	-
81	Check: Encapsulated multipart session description protocol in the INVITE message. It contains following information: Media attribute (a): MCPTT implicit request - Receiver expects the information about (MCPTT) FLOOR control messages in the real time control protocol part	--	Media format specific parameters: fmtp: mc_implicit_request	9	P
82	Check: RTP data (voice) from UE-Con can be received at all involved UE; RTP stream contains payload type information	-->	RTP stream	9	P
83	Check: UE1 terminates the call by sending BYE	-->	SIP BYE	-	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
84	MCx Service system responds with STATUS message to the call termination request	<--	SIP STATUS (see note 1) 200 (OK)	-	-

NOTE 1: When the UE is confirming the successful resource reservation using an UPDATE request (or a PRACK request) and the UE receives a 180 (Ringing) response or a 200 (OK) response to the initial INVITE request before receiving a 200 (OK) response to the UPDATE request (or a 200 (OK) response to the PRACK request)], the UE does not treat this as an error case and does not release the session.

NOTE 2: The application shall indicate the detected status of the floor parameter on the graphical user interface as icon and text:

- Status is IDLE, REVOKE or RELEASED: green; text: e.g. "push PTT to talk!"
- Status REQUESTED, DENY or TAKEN by other user, no multi-talker: red; text: none
- Status REQUESTED, DENY or TAKEN by other user, multi-talker: blue; text: e.g. "push PTT to talk!"
- Status is GRANTED to own UE: yellow; text: e.g. "please talk!"

NOTE 3: Reject Causes are given in ETSI TS 124 380 [12] (3GPP TS 24.380), clause 8.2.6.2.

5.2.3.3 Specific message contents

Table 5.2.3.3-1: SIP INVITE from the UE
(Table 5.2.3.2-1; step 1a, ETSI TS 137 579-4 [7] (3GPP TS 37.579-4) Table 5.5.2.5.1-1)

Derivation Path: Table 5.1.3.3-1: SIP INVITE from the UE

Table 5.2.3.3-2: SIP STATUS 100 from the MCx Service system
(step 2a, Table 5.1.3.2-1; ETSI TS 137 579-4 [7] (3GPP TS 37.579-4) Table 5.5.2.16-1)

Derivation Path: Table 5.1.3.3-4: SIP STATUS from the MCx Service system

Table 5.2.3.3-3: SIP STATUS 180 from the UE
(step 3a, Table 5.2.3.2-1; step x, ETSI TS 137 579-4 [7] (3GPP TS 37.579-4) Table 5.5.2.16-1)

Derivation Path: Derivation Path: Table 5.1.3.3-5: SIP STATUS from the UE

Table 5.2.3.3-4: SIP STATUS 200
(step 4a, Table 5.2.3.2-1; step x, ETSI TS 137 579-4 [7] (3GPP TS 37.579-4) Table 5.5.2.17-1)

Derivation Path: Table 5.1.3.3-6: SIP STATUS

Table 5.2.3.3-5: SDP message
(step 7, Table 5.2.3.2-1; ETSI TS 137 579-4 [7] (3GPP TS 37.579-4) Table 5.5.3.1.0-1)

Derivation Path: Derivation Path: IETF RFC 3556 [18]; condition SDP-ANSWER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Subtype	0 - Floor Request	UE request the floor		SDP
	1 - Floor Granted	MCx Service system grants		
	2 - Floor Taken	Other UE has priority to speak		
	3 - Floor Deny			
	4 - Floor Release	UE or MCx Service system releases the floor usage		
	5 - Floor Idle	No party has active floor		
	6 - Floor Revoke			
	10 - Floor Ack			
Media attribute			ETSI TS 124 229 [8] (3GPP TS 24.229)	MCPTT
Mission Critical Push To Talk (MCPTT)	<list of parameter>	Relevant parameter given in the test step		

Table 5.2.3.3-6: RTP
(step 5, Table 5.2.3.2-1; IETF RFC 3550 [19])

Derivation Path: Derivation Path: IETF RFC 3550 [19]; condition STREAM-INFORMATION				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Stream setup				
Method	SDP			
Payload-type	Used codec e.g. ITU-T G.711 PCMU			

Table 5.2.3.3-7: SIP BYE from the UE
(step 9, Table 5.2.3.2-1; ETSI TS 137 579-4 [7] (3GPP TS 37.579-4) Table 5.5.2.2.1-1))

Derivation Path: Table 5.1.3.3-12: SIP BYE from the UE

5.3 MCPTT Groups configured for multi-talker control

5.3.1 Test purpose

(1)

```
with {multi-Talker configuration by MCPTT Administrator}
ensure that {
    when {MCPTT Group with activated multi-talker floor control is configured on the MCx Service
system}
    then {multiple MCPTT service users can activate the floor control simultaneously and transmit
voice streams and all affiliated MCPTT service users receive the transmitted audio}
}
```

(2)

```
with {maximum number of simultaneous talker configuration}
ensure that {
    when {an authorized MCPTT service user changes the maximum number of simultaneous talkers during
an ongoing MCPTT Group call configured for multi-talker}
    then {the MCx Service system enforces the new limit, revokes talk permission until the maximum
number of simultaneous talkers is reached and MCx Service system provides notification of the
revocation to all affected participants}
}
```

(3)

with {floor control request with priority hierarchy}
ensure that {
    when {several MCPTT service users request permission to transmit simultaneously in an MCPTT Group configured for multi-talker floor control}
    then {the MCx Service system determines the transmitting participants based on the configured priority hierarchy, grants permission to the higher-priority MCPTT users and rejects lower-priority requests with notification of rejection}
}

(4)

with {override of active transmission in multi-Talker floor control}
ensure that {
    when {an authorized MCPTT service user with higher priority request to override an ongoing MCPTT transmission in MCPTT Group communication which is configured for multi-talker}
    then {the MCx Service system revokes the active transmission, grants transmit permission to the overriding MCPTT user **and** overridden MCPTT user(s) receive a notification that their transmission has been terminated}
}

(5)

with {notification of denied transmission request}
ensure that {
    when {MCPTT service user request permission to transmit in a multi-talker configured MCPTT Group and the request is denied}
    then {the MCPTT Service provide explicit notification to the requesting MCPTT service user including the reason for denial (e.g., insufficient priority or limit exceeded)}
}

(6)

with {queue handling of floor control requests}
ensure that {
    when {multiple MCPTT service users request permission to transmit in a multi-talker group **and** the requests are queued}
    then {the MCx Service system maintains the queue, provide the queued MCPTT service user with their position at the time of request **and** MCx Service system allows them to query their position on demand until the floor is granted, or request is withdrawn}
}

(7)

with {call termination on hang time expiry}
ensure that {
    when {MCPTT Group communication is ongoing and no further activity occurs until the configured hang time expires}
    then {the MCx Service system terminates the call, release all resources and provide indication of termination due to hang time expiry to all participants}
}

(8)

with {call time limit expiry handling}
ensure that {
    when {MCPTT Group communication approach the configured call time limit}
    then {the MCx Service system provides a pre-expiry indication to all participants before the time limit is reached **and** upon reaching the limit, the MCx Service system release the call **and** all participants receive notification, that the call was terminated due to time limit}
}

(9)

with {participant pre-emption handling due to insufficient capacity}
ensure that {
    when {one or more MCPTT service users are pre-empted from an ongoing MCPTT Group communication due to insufficient capacity}
    then {MCx Service system ensures that the pre-empted MCPTT users receive a notification that they were removed from the call due to capacity limitations **and** the remaining participants continue the call with the updated participant list}
}

5.3.2 Conformance requirements

[ETSI TS 122 179 [1] (3GPP TS 22.179), clause [R-6.2.3.7.2-001]]

"An MCPTT Group shall be configurable to allow multi-talker control."

[ETSI TS 122 179 [1] (3GPP TS 22.179), clause [R-6.2.3.7.2-002]]

"The MCPTT Service shall provide a mechanism for multiple MCPTT Users to talk simultaneously in an MCPTT Group configured for multi-talker control."

[ETSI TS 122 179 [1] (3GPP TS 22.179), clause [R-6.2.3.7.2-003]]

"The MCPTT Service shall determine which Participant(s) are allowed to transmit to all other Participant(s) in an MCPTT Group configured for multi-talker control."

[ETSI TS 122 179 [1] (3GPP TS 22.179), clause [R-6.2.3.7.2-004]]

"The MCPTT Service shall support all Participant(s) to receive audio from all other Participant(s) that are transmitting in an MCPTT Group configured for multi-talker control."

[ETSI TS 122 179 [1] (3GPP TS 22.179), clause [R-6.2.3.7.2-005]]

"The MCPTT Service shall provide a mechanism for the MCPTT Administrator to configure the maximum number of simultaneous talkers in an MCPTT Group configured for multi-talker control."

[ETSI TS 122 179 [1] (3GPP TS 22.179), clause [R-6.2.3.7.2-006]]

"The MCPTT Service shall allow an authorized MCPTT User to change the maximum number of simultaneous talkers at any time during a group call in an MCPTT Group configured for multi-talker control."

[ETSI TS 122 179 [1] (3GPP TS 22.179), clause [R-6.2.3.7.2-007]]

"The MCPTT Service shall allow an authorized MCPTT User to change the priority hierarchy of simultaneous talkers at any time during a group call in an MCPTT Group configured for multi-talker control."

[ETSI TS 122 179 [1] (3GPP TS 22.179), clause [R-6.2.3.7.2-008]]

"In case the maximum number of simultaneous talkers is reduced by an authorized MCPTT User, the permission to talk will be revoked, until the new maximum number of simultaneous talkers is met.

NOTE: The selection of the participants for revoking the permission to talk is left for implementation."

[ETSI TS 122 179 [1] (3GPP TS 22.179), clause [R-6.2.3.7.3-001]]

"The MCPTT Service shall enable authorized Participants to request to transmit to an MCPTT Group configured for multi-talker control."

[ETSI TS 122 179 [1] (3GPP TS 22.179), clause [R-6.2.3.7.3-002]]

"At call setup the MCPTT Service shall provide a notification, for example audio and/or visual, to the MCPTT Group Member attempting to transmit that there are no other Group Members who have affiliated to the MCPTT Group configured for multi-talker control."

[ETSI TS 122 179 [1] (3GPP TS 22.179), clause [R-6.2.3.7.3-003]]

"The MCPTT Service shall determine the transmitting Participant(s) when there are simultaneous requests for permission to transmit within the same call for an MCPTT Group configured for multi-talker control."

[ETSI TS 122 179 [1] (3GPP TS 22.179), clause [R-6.2.3.7.3-004]]

"Following an MCPTT Request for permission to transmit on the Selected MCPTT Group configured for multi-talker control the MCPTT Service shall provide an Affiliated MCPTT Group Member that made and was granted the request an indication of being allowed to transmit."

[ETSI TS 122 179 [1] (3GPP TS 22.179), clause [R-6.2.3.7.3-004a]]

"Following an MCPTT Request for permission to transmit on the Selected MCPTT Group configured for multi-talker control the MCPTT Service shall provide an Affiliated MCPTT Group Member that made and was denied the request, the reason for denial."

[ETSI TS 122 179 [1] (3GPP TS 22.179), clause [R-6.2.3.7.3-004b]]

"Following an MCPTT Request for permission to transmit on the Selected MCPTT Group configured for multi-talker control the MCPTT Service shall provide an Affiliated MCPTT Group Member that made and was queued the request, the position in the queue.

NOTE: The queueing mechanism is left for implementation."

[ETSI TS 122 179 [1] (3GPP TS 22.179), clause [R-6.2.3.7.3-005]]

"Once in the queue, the MCPTT service shall provide to an Affiliated MCPTT Group Member, the position in the queue on demand."

[ETSI TS 122 179 [1] (3GPP TS 22.179), clause [R-6.2.3.7.4.1-001]]

"If the number of MCPTT Users requesting the permission to talk exceeds the maximum number of simultaneous talkers in an MCPTT Group configured for multi-talker control, the MCPTT Service shall apply the override mechanism."

[ETSI TS 122 179 [1] (3GPP TS 22.179), clause [R-6.2.3.7.4.1-002]]

"The MCPTT Service shall enable MCPTT Administrators to create a priority hierarchy for determining what Participants, Participant types (e.g., dispatcher) and urgent transmission types shall be granted a request to override an active MCPTT transmission on an MCPTT Group configured for multi-talker control."

[ETSI TS 122 179 [1] (3GPP TS 22.179), clause [R-6.2.3.7.4.1-003]]

"The priority hierarchy used for granting a request to override an active MCPTT transmission on a group configured for multi-talker control shall contain at least four (4) levels."

[ETSI TS 122 179 [1] (3GPP TS 22.179), clause [R-6.2.3.7.4.1-004]]

"The transmitting Participant on an MCPTT Group a group configured for multi-talker control shall be determined by the relative priorities of the Participants and Call type based on priority (e.g., MCPTT Emergency)."

[ETSI TS 122 179 [1] (3GPP TS 22.179), clause [R-6.2.3.7.4.1-005]]

"Transmission requests of Participants with insufficient relative priority shall be rejected."

[ETSI TS 122 179 [1] (3GPP TS 22.179), clause [R-6.2.3.7.4.1-006]]

"The MCPTT Service shall provide a mechanism for Participants, to override an active MCPTT transmission of a transmitting Participant when the priority level of the overriding Participant or Call type based on priority (e.g., MCPTT Emergency) are ranked higher than the priority level of the transmitting Participant or Call type based on priority for an MCPTT Group configured for multi-talker control."

[ETSI TS 122 179 [1] (3GPP TS 22.179), clause [R-6.2.3.7.4.1-007]]

"If an authorized Participant overrides an MCPTT transmission, the MCPTT Service shall provide a means of notifying the overridden Participant(s) that the transmission has been overridden for an MCPTT Group configured for multi-talker control."

[ETSI TS 122 179 [1] (3GPP TS 22.179), clause [R-6.2.3.7.4.1-008]]

"The MCPTT Service shall revoke the transmit permission of the overridden transmitting Participant on an MCPTT Group configured for multi-talker control."

[ETSI TS 122 179 [1] (3GPP TS 22.179), clause [R-6.2.4-001]]

"If a Participant of an MCPTT Group call is pre-empted, the MCPTT Service shall terminate the call or continue the call with an indication to the transmitting Participant that one or more receiving Participants was pre-empted."

[ETSI TS 122 179 [1] (3GPP TS 22.179), clause [R-6.2.4-002]]

"If MCPTT User(s) are pre-empted from an ongoing MCPTT call as there is insufficient capacity to support their ongoing participation, the MCPTT Service shall ensure that the MCPTT User(s) receive a notification that they have been removed from the call for reasons of lack of capacity."

[ETSI TS 122 179 [1] (3GPP TS 22.179), clause [R-6.2.4-003]]

"The MCPTT Service shall terminate a call after the Hang Time expires."

[ETSI TS 122 179 [1] (3GPP TS 22.179), clause [R-6.2.4-005]]

"The MCPTT Service shall provide an indication to the Participants that the call is within a configurable amount of time before the call time limit is reached."

[ETSI TS 122 179 [1] (3GPP TS 22.179), clause [R-6.2.4-006]]

"The MCPTT Service shall release the call when the call time limit has been reached."

[ETSI TS 122 179 [1] (3GPP TS 22.179), clause [R-6.2.4-007]]

"The MCPTT Service shall provide an indication to the Participants that the call time limit has been reached."

[ETSI TS 122 179 [1] (3GPP TS 22.179), clause [R-6.2.4-008]]

"The MCPTT Service shall release an MCPTT Group call if any of the termination conditions are met (e.g., last Participant leaving, second last Participant leaving, initiator leaving) or the minimum number of Affiliated MCPTT Group Members are not present."

5.3.3 Test description

5.3.3.1 Pre-test conditions

System Configuration:

- MCx Service system consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (UE1) also consists of all sub-systems of the MCx user service.
- For the underlying "transport bearer" over which the UE1 (MCx Service system) and the IUT will communicate Parameters are set to the default parameters specific for the basic E-UTRA single simulated cells cell network scenarios, as defined in ETSI TS 138 508-1 [6] (3GPP TS 38.508-1), clause 6.3.4.4. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCx operation in the MCx configuration document).
- MCx Service system is configured to support the following features:
  - Pre-arranged groups containing minimum UE2 and UE-Con.
  - Ad-hoc Group Communication functionality.
  - Implicit and explicit floor control.
  - Multiple user floor control.
  - Multi-talker floor control.
  - Prioritisation of floor control requests.
  - Acknowledgement required according to ETSI TS 124 380 [12] (3GPP TS 24.380), Table 8.2.2.1-1.
  - In case the MCx Service system detects floor inactivity for a configurated period, the call will be terminated to unblock resources. The inactivity timer for any group communication type is configured to 240 seconds.

- In case floor control will be used by the participants, the MCX service system provide an overall duration for an ongoing communication. This call time limit timer is configured to 500 seconds.

- Multimedia Broadcast Multicast Service (MBMS) is supported by MCx_S1 and UE1, UE2 and UE-Con. According to ETSI TS 124 380 [12] (3GPP TS 24.380) clause 8.2.6.1 the floor DENY message is only supported in unicast bearer which is specified in the MBMS context. Unicast figures an end-to-end connection.

IUT:

- IUT (UE1).

- The IUT (UE1) consists of all sub-systems of the MCx user service including the ability for floor control, pre-arranged group communication functionality and ad-hoc group communication functionality. Ensure the correct configuration.

- The IUT (UE1) is the acting user.

Preamble:

- The IUT (UE1) and UE2 are connected to PLMN of the primary 5GS.

- The MCx Service system (MCx_S1) is connected to the IUT (UE1, MCx client)], UE2 (MCx client) and UE-Con (MCx client with controller/dispatcher functionality).

- MCx service SIP registration and affiliation are executed on MCx_S1.

5.3.3.2 Test procedure sequence

Table 5.3.3.2-1: Main behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: MCPTT Administrator configures MCPTT Group for multi-talker floor control on the MCx Service system; access via HTML web interface; maximum number of simultaneous talker set to "MaxTalkers"=2	-->	HTTPS link to configuration interface	-	-
2	Check: MCPTT Administrator configures MCPTT Group for multi-talker floor control on the MCx Service system; access via XML configuration file interface; maximum number of simultaneous talker set to "MaxTalkers"=2	-->	XML access (e.g. via SCP or SFTP connection)	-	-
	EXCEPTION: Procedure 'MCX CO communication' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1) clause 5.4.3	--		-	-
	EXCEPTION: Procedure 'Initial MCX Authentication, Registration, Configuration and Subscription' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1) clause 5.3.2	--		-	-
3a	Pre-arranged group communication case Check: UE1 initiates pre-arranged group communication call	-->	SIP INVITE	1	P
3b	Ad-hoc Group Communication case Check: UE1 initiates ad-hoc group communication call	-->	SIP INVITE	1	P
4	UE1 receives STATUS information from MCx_S1 regarding the establishment procedure	<--	SIP STATUS 100 (Trying)	-	-
	EXCEPTION: UE2 and UE-Con receive INVITE to the group communication in step 3a/3b provided by MCx_S1	<--	SIP INVITE	-	-

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
5	UE2 and UE-Con return STATUS information: group communication established	-->	SIP STATUS 200 (OK)	1	P
6	UE1 receives STATUS information: group communication established	<--	SIP STATUS 200 (OK)	-	-
7	Check: UE1 and UE2 send floor REQUEST messages at the same time	-->	RTCP (MCPT) Floor REQUEST	1	P
8a	Check: UE1 receive floor GRANTED message without sending explicit floor request message; following information included: Subtype 1 Floor Granted	<--	RTCP (MCPT) Floor Granted	-	-
8a1	Check: UE1 send floor ACK message to confirm the received Granted message	-->	RTCP (MCPT) Floor ACK	1	P
8b	Check: UE2 receive floor GRANTED message; following information included: Subtype 1 Floor Granted	<--	RTCP (MCPT) Floor Granted	-	-
8b1	Check: UE2 send floor ACK message to confirm the received Granted message	-->	RTCP (MCPT) Floor ACK	1	P
9a	Check: RTP data (voice) from UE1 can be received at all involved UE; RTP stream contains payload type information	-->	RTP stream	1	P
9b	Check: RTP data (voice) from UE2 can be received at all involved UE; RTP stream contains payload type information	-->	RTP stream	-	-
10	Check: UE1 and UE-Con sends floor request message at the same time	-->	RTCP (MCPT) Floor REQUEST	1	P
11a	Check: UE1 receive floor GRANTED message without sending explicit floor request message; following information included: Subtype 1 Floor Granted	<--	RTCP (MCPT) Floor Granted	-	-
11a1	Check: UE1 send floor ACK message to confirm the received Granted message	-->	RTCP (MCPT) Floor ACK	1	P
11b	Check: UE-Con receive floor GRANTED message; following information included: Subtype 1 Floor Granted	<--	RTCP (MCPT) Floor Granted	-	-
11b1	Check: UE-Con send floor ACK message to confirm the received Granted message	-->	RTCP (MCPT) Floor ACK	1	P
12a	Check: RTP data (voice) from UE1 can be received at all involved UE; RTP stream contains payload type information	-->	RTP stream	1	P
12b	Check: RTP data (voice) from UE-Con can be received at all involved UE; RTP stream contains payload type information	-->	RTP stream	-	-
13	UE1 release the floor	-->	RTCP (MCPT) Floor RELEASE	1	P
14	Check: MCx_S1 send floor ACK	<--	RTCP (MCPT) Floor ACK	-	-
15	UE2 send floor request message	-->	RTCP (MCPT) Floor REQUEST	-	-
16	Check: UE-Con receive floor GRANTED message; following information included: Subtype 1 Floor Granted (see note 1)	<--	RTCP (MCPT) Floor GRANTED	-	-
17	Check: UE-Con send floor ACK message to confirm the received Granted message	-->	RTCP (MCPT) Floor ACK	1	P
18	Check: UE1 send floor request message	-->	RTCP (MCPT) Floor REQUEST	2	P
19	Check: UE1 receives TAKEN message by MCx_S1; the following information included: Subtype 2 Floor Taken {RTCP SSRC of sending floor | granted party ID | permission to request the floor | user ID | message sequence | track info | floor indicator | audio SSRC of granted user | FA | list of granted users | list of audio SSRC | list of FA | location info | list of locations of granted users}	<--	RTCP (MCPT) Floor TAKEN with {floor indicator= 1 for multi-talker | list of granted users} According to ETSI TS 124 380 [12] (3GPP TS 24.380) Figure 6.2.4.1-1	-	-
20	Check: UE1 send floor ACK message to confirm the received Taken message	-->	RTCP (MCPT) Floor ACK	2	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
21a	Check: MCPTT Administrator configures MCPTT Group for multi-talker floor control on the MCx Service system; access via HTML web interface; maximum number of simultaneous talker set to "MaxTalkers"=3	-->	HTTPS link to configuration interface	-	-
21b	Check: MCPTT Administrator configures MCPTT Group for multi-talker floor control on the MCx Service system; access via XML configuration file interface; maximum number of simultaneous talker set to "MaxTalkers"=3	-->	XML access (e.g. via SCP or SFTP connection)	-	-
	EXCEPTION: group communication establishment shall not terminate while changing MaxTalkers parameter	--		-	-
22	MCx_S1 send REVOKE information to all involved participants; the following information included: Subtype 6 Floor Revoke {SSRC of floor control server | reject cause (see note 1) | track info | floor indicator	<--	RTCP (MCPT) Floor REVOKE message containing Cause #6 - No resources available	-	-
23	Check: involved UE send floor ACK message to confirm the received Granted message	-->	RTCP (MCPT) Floor ACK	2	P
24a	Check: UE1 sends floor request message	-->	RTCP (MCPT) Floor REQUEST	2	P
25a	Check: UE1 receive floor GRANTED message; following information included: Subtype 1 Floor Granted (see note 1)	<--	RTCP (MCPT) Floor GRANTED	-	-
25a1	Check: UE1 send floor ACK message to confirm the received Granted message	-->	RTCP (MCPT) Floor ACK	2	P
24b	Check: UE2 sends floor request message	-->	RTCP (MCPT) Floor REQUEST	-	-
25b	Check: UE2 receive floor GRANTED message; following information included: Subtype 1 Floor Granted (see note 1)	<--	RTCP (MCPT) Floor GRANTED	-	-
25b1	Check: UE2 send floor ACK message to confirm the received Granted message	-->	RTCP (MCPT) Floor ACK	2	P
24c	Check: UE-Con sends floor request message	-->	RTCP (MCPT) Floor REQUEST	-	-
25c	Check: UE-Con receive floor GRANTED message; following information included: Subtype 1 Floor Granted (see note 1)	<--	RTCP (MCPT) Floor GRANTED	-	-
25c1	Check: UE-Con send floor ACK message to confirm the received Granted message	-->	RTCP (MCPT) Floor ACK	2	P
26a	Check: UE1 send floor RELEASE message; the following information included: Subtype 4 Floor Release {user ID | track info | floor indicator}	-->	RTCP (MCPT) Floor RELEASE	2	P
26a1	Check: MCx_S1 send floor ACK	<--	RTCP (MCPT) Floor ACK	-	-
26b	UE2 send floor RELEASE message; the following information included: Subtype 4 Floor Release {user ID | track info | floor indicator}	-->	RTCP (MCPT) Floor RELEASE	2	-
26b1	Check: MCx_S1 send floor ACK	<--	RTCP (MCPT) Floor ACK	-	-
26c	UE-Con send floor RELEASE message; the following information included: Subtype 4 Floor Release {user ID | track info | floor indicator}	-->	RTCP (MCPT) Floor RELEASE	2	-
26c1	Check: MCx_S1 send floor ACK	<--	RTCP (MCPT) Floor ACK	-	-
27a	Check: MCPTT Administrator configures MCPTT Group for multi-talker floor control on the MCx Service system; access via HTML web interface; maximum number of simultaneous talker set to "MaxTalkers"= 1 and the following priorities (see note 3): UE1=2, UE2=4, UE-Con=0	-->	HTTPS link to configuration interface	-	-

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
27b	Check: MCPTT Administrator configures MCPTT Group for multi-talker floor control on the MCx Service system; access via XML configuration file interface; ; maximum number of simultaneous talker set to "MaxTalkers"=1 and the following priorities (see note 3): UE1=2, UE2=4, UE-Con=0	-->	XML access (e.g. via SCP or SFTP connection)	-	-
	EXCEPTION: group communication establishment shall not terminate while changing MaxTalkers parameter	--		-	-
28a	Check: UE1 send floor REQUEST message	-->	RTCP (MCPT) Floor REQUEST; message containing field {Floor priority=2}	3	P
29a	Check: UE1 receive floor GRANTED message; following information included: Subtype 1 Floor Granted (see note 1)	<--	RTCP (MCPT) Floor GRANTED	-	-
29a1	Check: UE1 send floor ACK message to confirm the received Granted message	-->	RTCP (MCPT) Floor ACK	2	P
28b	Check: UE2 send floor REQUEST message	-->	RTCP (MCPT) Floor REQUEST; message containing field {Floor priority=4}	3	P
29b	Check: UE2 receive floor DENY message; following information included: Subtype 3 Floor Deny {Reject Cause (see note 2) | User granted the floor | floor indicator}	<--	RTCP (MCPT) Floor DENY; message containing field {reject cause=#1 (Another MCPTT client has permission)}	-	-
29b1	Check: UE2 send floor ACK message to confirm the received Deny message	-->	RTCP (MCPT) Floor ACK	3	P
28c	Check: UE-Con send floor REQUEST message	-->	RTCP (MCPT) Floor REQUEST; message containing field {Floor priority=0}	3	P
29c	Check: UE1 receive floor DENY message; following information included: Subtype 3 Floor Deny {Reject Cause (see note 2) | User granted the floor | floor indicator}	<--	RTCP (MCPT) Floor DENY; message containing field {reject cause=#4 (Media Burst pre-empted)}	-	-
29c1	Check: UE1 send floor ACK message to confirm the received Deny message	-->	RTCP (MCPT) Floor ACK	3	P
30	UE-Con send floor RELEASE message; the following information included: Subtype 4 Floor Release {user ID | track info | floor indicator}	-->	RTCP (MCPT) Floor RELEASE	3	-
31	Check: MCx_S1 send floor ACK	<--	RTCP (MCPT) Floor ACK	-	-
32a	Check: MCPTT Administrator configures MCPTT Group for multi-talker floor control on the MCx Service system; access via HTML web interface; maximum number of simultaneous talker set to "MaxTalkers"= 2 and the following priorities (see note 3): UE1=2, UE2=2 UE-Con=0	-->	HTTPS link to configuration interface	-	-
32b	Check: MCPTT Administrator configures MCPTT Group for multi-talker floor control on the MCx Service system; access via XML configuration file interface; ; maximum number of simultaneous talker set to "MaxTalkers"=2 and the following priorities (see note 3): UE1=2, UE2=2, UE-Con=0	-->	XML access (e.g. via SCP or SFTP connection)	-	-
	EXCEPTION: group communication establishment shall not terminate while changing MaxTalkers parameter	--		-	-
33a	Check: UE1 send floor REQUEST message	-->	RTCP (MCPT) Floor REQUEST; message containing field {Floor priority=2}	4	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
34a	Check: UE1 receive floor GRANTED message; following information included: Subtype 1 Floor Granted (see note 1)	<--	RTCP (MCPT) Floor GRANTED	-	-
34a1	Check: UE1 send floor ACK message to confirm the received Granted message	-->	RTCP (MCPT) Floor ACK	4	P
33b	Check: UE2 send floor REQUEST message	-->	RTCP (MCPT) Floor REQUEST; message containing field {Floor priority=2}	4	P
34b	Check: UE2 receive floor GRANTED message; following information included: Subtype 1 Floor Granted (see note 1)	<--	RTCP (MCPT) Floor GRANTED	-	-
34b1	Check: UE2 send floor ACK message to confirm the received Granted message	-->	RTCP (MCPT) Floor ACK	4	P
35	Check: UE-Con send floor REQUEST message	-->	RTCP (MCPT) Floor REQUEST; message containing field {Floor priority=0}	4	P
36a	Check: UE1 receive floor REVOKE message to the active transmission. The following information included: Subtype 6 Floor Revoke {SSRC of floor control server | reject cause (see note 2) | track info | floor indicator}	<--	RTCP (MCPT) Floor REVOKE (see note 4) with Reject Cause #1 - Another MCPTT client has permission	-	-
36a1	Check: UE1 send floor ACK message to confirm the received Revoke message	-->	RTCP (MCPT) Floor ACK	4	P
36b	Check: UE2 receive floor REVOKE message to the active transmission. The following information included: Subtype 6 Floor Revoke {SSRC of floor control server | reject cause (see note 2) | track info | floor indicator}	<--	RTCP (MCPT) Floor REVOKE (see note 4) with Reject Cause #1 - Another MCPTT client has permission	-	-
36b1	Check: UE2 send floor ACK message to confirm the received Revoke message	-->	RTCP (MCPT) Floor ACK	4	P
37	Check: UE-Con receives floor GRANTED	<--	RTCP (MCPT) Floor GRANTED	-	-
38	Check: UE-Con send floor ACK message to confirm the received Granted message	-->	RTCP (MCPT) Floor ACK	4	P
39a	Check: RTP data (voice) from UE-Con can be received at all involved UE; no other RTP source; outgoing condition RTP stream contains payload type information	-->	RTP stream	4	P
39b	Check: RTP data (voice) from UE-Con can be received at all involved UE; incoming condition RTP stream contains payload type information	<--	RTP stream	4	-
40	UE-Con send floor RELEASE message; the following information included: Subtype 4 Floor Release {user ID | track info | floor indicator}	-->	RTCP (MCPT) Floor RELEASE	4	-
41	Check: MCx_S1 send floor ACK	<--	RTCP (MCPT) Floor ACK	-	-
41a	Check: MCPTT Administrator configures MCPTT Group for multi-talker floor control on the MCx Service system; access via HTML web interface; maximum number of simultaneous talker set to "MaxTalkers"= 2 and the following priorities (see note 3): UE1=4, UE2=2 UE-Con=0	-->	HTTPS link to configuration interface	-	-
41b	Check: MCPTT Administrator configures MCPTT Group for multi-talker floor control on the MCx Service system; access via XML configuration file interface; maximum number of simultaneous talker set to "MaxTalkers"=2 and the following priorities (see note 3): UE1=4, UE2=2, UE-Con=0	-->	XML access (e.g. via SCP or SFTP connection)	-	-
	EXCEPTION: group communication establishment shall not terminate while changing MaxTalkers parameter	--		-	-

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
42a	Check: UE2 send floor REQUEST message	-->	RTCP (MCPT) Floor REQUEST; message containing field {Floor priority=2}	5	P
43a	Check: UE2 receive floor GRANTED message; following information included: Subtype 1 Floor Granted	<--	RTCP (MCPT) Floor GRANTED	-	-
43a1	Check: UE2 send floor ACK message to confirm the received Granted message	-->	RTCP (MCPT) Floor ACK	5	P
42b	Check: UE-Con send floor REQUEST message	-->	RTCP (MCPT) Floor REQUEST; message containing field {Floor priority=0}	5	P
43b	Check: UE-Con receive floor GRANTED message; following information included: Subtype 1 Floor Granted	<--	RTCP (MCPT) Floor GRANTED	-	-
43b1	Check: UE-Con send floor ACK message to confirm the received Granted message	-->	RTCP (MCPT) Floor ACK	5	P
42c	Check: UE1 send floor REQUEST message	-->	RTCP (MCPT) Floor REQUEST; message containing field {Floor priority=4}	5	P
43c	Check: UE1 receive floor DENY message; following information included: Subtype 3 Floor Deny {Reject Cause (see note 2) | User granted the floor | floor indicator}	<--	RTCP (MCPT) Floor DENY; message containing field {reject cause #7 - Queue full	-	-
43c1	Check: UE1 send floor ACK message to confirm the received Deny message	-->	RTCP (MCPT) Floor ACK	5	P
44a	Check: MCPTT Administrator configures MCPTT Group for multi-talker floor control on the MCx Service system; access via HTML web interface; maximum number of simultaneous talker set to "MaxTalkers"= 1, "MaxQueueCount"=2 and the following priorities (see note 3): UE1=2, UE2=2 UE-Con=2	-->	HTTPS link to configuration interface	-	-
44b	Check: MCPTT Administrator configures MCPTT Group for multi-talker floor control on the MCx Service system; access via XML configuration file interface; ; maximum number of simultaneous talker set to "MaxTalkers"= 1, "MaxQueueCount"=2 and the following priorities (see note 4): UE1=2, UE2=2, UE-Con=0	-->	XML access (e.g. via SCP or SFTP connection)	-	-
	EXCEPTION: group communication establishment shall not terminate while changing MaxTalkers parameter	--		-	-
45a	Check: UE-Con send floor REQUEST message	-->	RTCP (MCPT) Floor REQUEST; message containing field {Floor priority=0}	6	P
46a	Check: UE-Con receive floor GRANTED message; following information included: Subtype 1 Floor Granted	<--	RTCP (MCPT) Floor GRANTED	-	-
46a1	Check: UE-Con send floor ACK message to confirm the received Granted message	-->	RTCP (MCPT) Floor ACK	6	P
45b	Check: UE2 send floor REQUEST message	-->	RTCP (MCPT) Floor REQUEST; message containing field {Floor priority=2}	6	P
46b	Check: UE2 receive floor QUEUE POSITION INFO message; following information included: Subtype 8 Floor Queue Position Info	<--	RTCP (MCPT) Floor QUEUE POSITION INFO containing the queue info field with the position index 1 and the ID of the queued user	-	-
46b1	Check: UE2 send floor ACK message to confirm the received Floor Queue Position Info message	-->	RTCP (MCPT) Floor ACK	6	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
45c	Check: UE1 send floor REQUEST message	-->	RTCP (MCPT) Floor REQUEST; message containing field {Floor priority=2}	6	P
46c	Check: UE1 receive floor QUEUE POSITION INFO message; following information included: Subtype 8 Floor Queue Position Info	<--	RTCP (MCPT) Floor QUEUE POSITION INFO containing the queue info field with the position index 2 and the ID of the queued user	-	-
46c1	Check: UE1 send floor ACK message to confirm the received Floor Queue Position Info message	-->	RTCP (MCPT) Floor ACK	6	P
47	Check: UE1 send periodically floor QUEUE POSITION REQUEST message; following information included: Subtype 9 Floor Queue Position Info Request	-->	RTCP (MCPT) Floor QUEUE POSITION REQUEST	6	P
48	Check: UE1 receive floor QUEUE POSITION INFO message; following information included: Subtype 8 Floor Queue Position Info	<--	RTCP (MCPT) Floor QUEUE POSITION INFO containing the queue info field with the position index 2 and the ID of the queued user	-	-
49	Check: UE1 send floor ACK message to confirm the received Floor Queue Position Info message	-->	RTCP (MCPT) Floor ACK	6	P
50	UE2 send floor RELEASE message; the following information included: Subtype 4 Floor Release {user ID | track info | floor indicator}	-->	RTCP (MCPT) Floor RELEASE	6	-
51	Check: MCx_S1 send floor ACK	<--	RTCP (MCPT) Floor ACK	-	-
52	Check: UE1 send periodically floor QUEUE POSITION REQUEST message; following information included: Subtype 9 Floor Queue Position Info Request	-->	RTCP (MCPT) Floor QUEUE POSITION REQUEST	6	P
53	Check: MCx_S1 send floor ACK	<--	RTCP (MCPT) Floor ACK	-	-
54	Check: UE1 receive floor QUEUE POSITION INFO message; following information included: Subtype 8 Floor Queue Position Info	<--	RTCP (MCPT) Floor QUEUE POSITION INFO containing the queue info field with the position (index 2 has changed to index 1) and the ID of the queued user	-	-
55a	UE1 send floor RELEASE message; the following information included: Subtype 4 Floor Release {user ID | track info | floor indicator}	-->	RTCP (MCPT) Floor RELEASE	6	-
56a	Check: MCx_S1 send floor ACK	<--	RTCP (MCPT) Floor ACK	-	-
55b	UE-Con send floor RELEASE message; the following information included: Subtype 4 Floor Release {user ID | track info | floor indicator}	-->	RTCP (MCPT) Floor RELEASE	6	-
56b	Check: MCx_S1 send floor ACK	<--	RTCP (MCPT) Floor ACK	-	-
	EXCEPTION: no action until hang timer (240 s) expires	-		7	-
57a	Check: Hang time expired, MCx_s1 send termination request to all participants; UE 1 receive SIP BYE	<--	SIP BYE	-	-
58a	UE1 responds with STATUS message to the call termination request	-->	SIP STATUS1) 200 (OK)	7	P
57b	Check: Hang time expired, MCx_s1 send termination request to all participants; UE 1 receive SIP BYE	<--	SIP BYE	-	-
58b	UE2 responds with STATUS message to the call termination request	-->	SIP STATUS1) 200 (OK)	7	P
57c	Check: Hang time expired, MCx_s1 send termination request to all participants; UE 1 receive SIP BYE	<--	SIP BYE	-	-
58c	UE-Con responds with STATUS message to the call termination request	-->	SIP STATUS1) 200 (OK)	7	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
59a	Pre-arranged group communication case Check: UE1 initiates pre-arranged group communication call	-->	SIP INVITE	8	P
59b	Ad-hoc Group Communication case Check: UE1 initiates ad-hoc group communication call	-->	SIP INVITE	8	P
60	UE1 receives STATUS information from MCx_S1 regarding the establishment procedure	<--	SIP STATUS 100 (Trying)	-	-
	EXCEPTION: UE2 and UE-Con receive INVITE to the group communication in step 3a/3b provided by MCx_S1	<--	SIP INVITE	-	-
61	UE2 and UE-Con return STATUS information: group communication established	-->	SIP STATUS 200 (OK)	8	P
62	UE1 receives STATUS information: group communication established	<--	SIP STATUS 200 (OK)	-	-
63	Floor activation by one participant to ensure the usage of the call time limit expiry Check: UE send floor REQUEST message	-->	RTCP (MCPT) Floor REQUEST; message containing field {Floor priority=0}	8	P
64	Check: UE-Con receive floor GRANTED message; following information included: Subtype 1 Floor Granted	<--	RTCP (MCPT) Floor GRANTED	-	-
65	Check: UE-Con send floor ACK message to confirm the received Granted message	-->	RTCP (MCPT) Floor ACK	8	P
	None. (Wait until call time limit expires)	--		-	-
66	Call time limit timer expires after 300s. Check: MCx_S1 disconnect UE1 from group communication. UE1 receive DISCONNECT message	<--	RTPC (MTPC) DISCONNECT with reason clause 7 according to ETSI TS 124.380 [12] (3GPP TS 24.380) clauses 8.3.5 and 8.3.3.11	-	-
67	Ceck: UE1 acknowledge the DISCONNECT message	-->	SIP STATUS 200 (OK)	8	P
68a	Pre-arranged group communication case Check: UE1 initiates pre-arranged group communication call.	-->	SIP INVITE	9	P
68b	Ad-hoc Group Communication case Check: UE1 initiates ad-hoc group communication call	-->	SIP INVITE	9	P
69	UE1 receives STATUS information from MCx_S1 regarding the establishment procedure	<--	SIP STATUS 100 (Trying)	-	-
	EXCEPTION: UE2 and UE-Con receive INVITE to the group communication in step 3a/3b provided by MCx_S1	<--	SIP INVITE	-	-
70	UE2 and UE-Con return STATUS information: group communication established	-->	SIP STATUS 200 (OK)	9	P
71	UE1 receives STATUS information: group communication established	<--	SIP STATUS 200 (OK)	-	-
	MCx_S1 removes UE1 due to capacity limitations. Check: UE receive DISCONNECT message	<--	RTCP (MPTC) DISCONNECT with reason clause 10 according to ETSI TS 124.380 [12] (3GPP TS 24.380) clauses 8.3.5 and 8.3.3.11	-	-
	MCx_S1 sends group dynamic notification to all remaining participants Check: UE2 and UE-Con receive group dynamic notification	<--	RTCP (MPTC) Group Dynamic NOTIFY	-	-

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
	MCx_S1 terminates the group communication call using BYE Check: UE2 and UE-Con receive BYE	<--	SIP BYE	-	-
	UE2 confirms the received BYE message	-->	SIP STATUS 200 (OK)	-	-
	UE-Con confirms the received BYE message	-->	SIP STATUS 200 (OK)	-	-
NOTE 1: UE2 and UE-Con are in status floor GRANTED and MaxTalkers exceeds with a new request. These states the precondition to check that the change of the MaxTalker counter will be possible during an ongoing group communication (pre-arranged or ad hoc). NOTE 2: Reject Causes are given in ETSI TS 124 380 [12] (3GPP TS 24.380) clause 8.2.6.2. NOTE 3: Priorities depending on the MCx Service system configuration. For this test consider 4=low priority, 2=mid priority and 0=high priority. NOTE 4: According to ETSI TS 122 179 [1] (3GPP TS 22.179) clause [R-6.2.3.7.2-008] the selection of the participants for revoking the permission to talk is left for implementation.					

5.3.3.3 Specific message contents

Table 5.3.3.3-1: SIP INVITE
(Table 5.3.3.2-1; step 3a, ETSI TS 137 579-4 [7] (3GPP TS 37.579-4) Table 5.5.2.5.1-1)

Derivation Path: Table 5.1.3.3-1: SIP INVITE from the UE

Table 5.3.3.3-2: SIP STATUS 100 from the MCx Service system
(Table 5.3.3.2-1; step 4, ETSI TS 137 579-4 [7] (3GPP TS 37.579-4) Table 5.5.2.16-1)

Derivation Path: Table 5.1.3.3-4: SIP STATUS from the MCx Service system

Table 5.3.3.3-3: SIP STATUS 200
(step 5, Table 5.3.3.2-1; step x, ETSI TS 137 579-4 [7] (3GPP TS 37.579-4) Table 5.5.2.17-1)

Derivation Path: Table 5.1.3.3-6: SIP STATUS

Table 5.3.3.3-4: SDP message
(step 7, Table 5.3.3.2-1; ETSI TS 137 579-4 [7] (3GPP TS 37.579-4) Table 5.5.3.1.0-1)

Derivation Path: Derivation Path: IETF RFC 3556 [18]; condition SDP-ANSWER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Subtype	0 - Floor Request	UE request the floor		SDP
	1 - Floor Granted	MCx Service system grants		
	2 - Floor Taken	Other UE has priority to speak		
	3 - Floor Deny			
	4 - Floor Release	UE or MCx Service system release the floor usage		
	5 - Floor Idle	No party has active floor		
	6 - Floor Revoke			
	10 - Floor Ack			
Media attribute			ETSI TS 124 229 [8] (3GPP TS 24.229)	MCPTT
Mission Critical Push To Talk (MCPTT)	<list of parameter>	Relevant parameter given in the test step		

Table 5.3.3.3-5: RTP DISCONNECT
(step 66, Table 5.3.3.2-1; ETSI TS 124 380 [12] (3GPP TS 24.380) clause 8.3.5

Derivation Path: Derivation Path: ETSI TS 124 380 [12] (3GPP TS 24.380) clause 8.3.5; condition REASON-CAUSE-FIELD				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Stream setup				
Method	DISCONNECT			
Reason Cause	7	Group timer expired	ETSI TS 124 380 [12] (3GPP TS 24.380) clause 8.3.3.11	
	10	Group participant disconnected		

Table 5.3.3.3-6: RTP DENY
(step 29b, Table 5.3.3.2-1; ETSI TS 124 380 [12] (3GPP TS 24.380) clause 8.2.6

Derivation Path: Derivation Path: ETSI TS 124 380 [12] (3GPP TS 24.380) clause 8.2.6; condition REJECT-CAUSE-FIELD				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Stream setup				
Method	DENY			
Reject Cause	1	Another MCPTT client has permission	ETSI TS 124 380 [12] (3GPP TS 24.380) clause 8.3.3.11	
	4	Media Burst pre-empted		
	7	Queue full		

Table 5.3.3.3-7: RTP REVOKE
(step 29b, Table 5.3.3.2-1; ETSI TS 124 380 [12] (3GPP TS 24.380) clause 8.3.5

Derivation Path: Derivation Path: ETSI TS 124 380 [12] (3GPP TS 24.380) clause 8.3.5; condition REJECT-CAUSE-FIELD				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Stream setup				
Method	DENY			
Reject Cause	1	Another MCPTT client has permission	ETSI TS 124 380 [12] (3GPP TS 24.380) clause 8.2.6.2	
	4	Media Burst pre-empted		

Table 5.3.3.3-8: RTP QUEUE POSITION INFO
(step 46b, Table 5.3.3.2-1; ETSI TS 124 380 [12] (3GPP TS 24.380) clause 8.2.12

Derivation Path: Derivation Path: ETSI TS 124 380 [12] (3GPP TS 24.380) clause 8.2.12; condition REJECT-CAUSE-FIELD				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Stream setup				
Method	QUEUE POSITION INFO			
Queue info field	<index>	Position	ETSI TS 124 380 [12] (3GPP TS 24.380) Table 8.2.12-1	
ID of the queued user	<ID>	Requesting user ID		

Table 5.3.3.3-9: RTP (step 9a, Table 5.3.3.2-1; IETF RFC 3550 [19]

Derivation Path: Derivation Path: [ETSI TS 124 380 [12] (3GPP TS 24.380); condition STREAM-INFORMATION				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Stream setup				
Method	SDP			
Payload-type	Used codec e.g. ITU-T G.711 PCMU			

Table 5.3.3.3-10: SIP BYE from the UE
(step 57a, Table 5.3.3.2-1; ETSI TS 137 579-4 [7] (3GPP TS 37.579-4) Table 5.5.2.2.1-1)

Derivation Path: Table 5.1.3.3-12: SIP BYE from the UE

5.4 Arbitration function

5.4.1 Test purpose

(1)

```
with {client arbitration function mobile originated scenario}
ensure that {
    when {MCPTT service user has active call and initiate an additional outgoing call with equal or
different priority and MCx Service system does not pre-empt the connection}
    then {MCPTT service user handle arbitration based on the configurable priority}
}
```

(2)

```
with {client arbitration function mobile terminated scenario}
ensure that {
    when {MCPTT service user has active call and receive an additional incoming call with equal or
different priority and MCx Service system does not pre-empt the connection}
    then {MCPTT service user handle arbitration based on the configurable priority}
}
```

5.4.2 Conformance requirements

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.8.7.2-005]]

"There shall be at least 8 and preferably 30 configurable levels of priority."

5.4.3 Test Description

5.4.3.1 Pre-test conditions

5.4.3.1.1 General

System Configuration:

- MCPTT client is connected to the system in accordance with the previous test cases. According to ETSI TS 122 280 [2] (3GPP TS 22.280) the client shall support four different priorities High > Imminent Peril > normal > low. The levels shall be configured on the MCPTT client.
- MCx Service system is configured to support the following features:
  - According to ETSI TS 122 280 [2] (3GPP TS 22.280) the MCx Service system shall support at least four different priorities High > Imminent Peril > normal > low. The levels shall be configured on the MCPTT client.
  - The IUT behaviour for the arbitration scenarios is given with Table 5.5.3.1.1-1 for mobile originated and Table 5.5.3.1.2-1 for mobile terminated calls.
  - The arbitration function in this chapter will not consider functional aliasing or PTT functionality. It is limited to the call scenario handling of the IUT.
  - For each priority level (high, imminent peril, normal) the specified tuples will be checked:
    - MO {11,21}, {12,22} and {12,24}.
    - MT {11,21}, {12,23} and {12,24}.

IUT:

- IUT (UE1).
- The IUT (UE1) consists of all sub-systems of the MCx user service including the ability for floor control.
- The IUT (UE1) is the acting user.

Preamble:

- The IUT (UE1) is connected to PLMN of the primary 5GS. UE3 is connected to the PLMN of the secondary 5GS.
- The MCx Service system (MCx_S1) is connected to the IUT (UE1, MCx client) and to the supporting devices UE2 (MCx client) and UE-Con (MCx client with controller/dispatcher functionality).
- MCx service SIP registration and affiliation are executed on MCx_S1.
- No resource limitations on the backbone system. Neither MCx_S1 nor the primary 5G system shall handle the arbitration.
- The indications (F1) and (F2) are linked to the calls.

5.4.3.1.2 Mobile Originated (MO)

Table 5.4.3.1.2-1: Arbitration matrix for MO call scenarios

		New Outgoing Call (F2)	Emergency Voice Call (P2P)	Emergency Group Comm. Service	Ad-hoc Emergency Group Comm. Service	Imminent Peril Private Call (P2P)	Imminent Peril Group Comm. Service	Ad-hoc Imminent Peril Group Comm. Service	Point-2-Point Audio Call	Voice Group Comm. Service	Ad-hoc Group Comm. Service	Data service
			MO	MO	MO	MO	MO	MO	MO	MO	MO	MO
	Call	Type	EP2P	EGCS	AEGS	IPP	IGCS	AIGS	P2P	VGCS	AGS	DS
		Prio	High	High	High	Imminent	Imminent	Imminent	Normal	Normal	Normal	High
Ongoing Call (F1)			a	b	C	d	e	f	g	h	i	j
Emergency Voice Call (P2P)	EP2P	High	0	11,21	11,21	12,22	12,22	12,22	12,22	12,22	12,22	12,24
Emergency Group Comm. Service	EGCS	High	12,22	0	11,21	12,22	12,22	12,22	12,22	12,22	12,22	12,24
Ad-hoc Emergency Group Comm. Service	AEGS	High	12,22	12,22	0	12,22	12,22	12,22	12,22	12,22	12,22	12,24
Imminent Peril Private Call (P2P)	IPP	Imminent	11,21	11,21	11,21	0	11,21	11,21	12,22	12,22	12,22	12,24
Imminent Peril Group Comm. Service	IGCS	Imminent	11,21	11,21	11,21	12,22	0	11,21	12,22	12,22	12,22	12,24
Ad-hoc Imminent Peril Group Comm. Service	AIGS	Imminent	11,21	11,21	11,21	12,22	12,22	0	12,22	12,22	12,22	12,24
Point-2-Point Audio Call	P2P	Normal	11,21	11,21	11,21	11,21	11,21	11,21	12,22	11,21	11,21	12,24
Voice Group Comm. Service	VGCS	Normal	11,21	11,21	11,21	11,21	11,21	11,21	11,21	12,22	11,21	12,24
Ad-hoc Group Comm. Service	AGS	Normal	11,21	11,21	11,21	11,21	11,21	11,21	11,21	11,21	12,22	12,24
Data service	DS	High	12,24	12,24	12,24	12,24	12,24	12,24	12,24	12,24	12,24	12,24

NOTE:
- Ongoing call is left/terminated: 11
- Ongoing call is maintained: 12
- Ongoing call is put on hold: 13
- Not possible (UI): x
- Not possible (system): -
- No Change: 0
- New call is joined / answered / established: 21

- New call is not established: 22
- New call is indicated only: 23
- Parallel operation: 24

5.4.3.1.3 Mobile Terminated (MT)

Table 5.4.3.1.3-1: Arbitration matrix for MT call scenarios

		New Incoming Call (F2)	Emergency Voice Call (P2P)	Emergency Group Comm. Service	Ad-hoc Emergency Group Comm. Service	Imminent Peril Private Call (P2P)	Imminent Peril Group Comm. Service	Ad-hoc Imminent Peril Group Comm. Service	Point-2-Point Audio Call	Voice Group Comm. Service	Ad-hoc Group Comm. Service	Data service
			MO	MO	MO	MO	MO	MO	MO	MO	MO	MO
	Call	Type	EP2P	EGCS	AEGS	IPP	IGCS	AIGS	P2P	VGCS	AGS	DS
		Prio	High	High	High	Imminent	Imminent	Imminent	Normal	Normal	Normal	High
Ongoing Call (F1)			a	b	C	d	e	f	g	h	i	j
Emergency Voice Call (P2P)	EP2P	High	12,23	11,21	11,21	12,23	12,23	12,23	12,23	12,23	12,23	12,24
Emergency Group Comm. Service	EGCS	High	12,23	12,23	11,21	12,23	12,23	12,23	12,23	12,23	12,23	12,24
Ad-hoc Emergency Group Comm. Service	AEGS	High	11,21	12,23	12,23	12,23	12,23	12,23	12,23	12,23	12,23	12,24
Imminent Peril Private Call (P2P)	IPP	Imminent	11,21	11,21	11,21	12,23	11,21	11,21	12,23	12,23	12,23	12,24
Imminent Peril Group Comm. Service	IGCS	Imminent	11,21	11,21	11,21	12,23	12,23	11,21	12,23	12,23	12,23	12,24
Ad-hoc Imminent Peril Comm. Call Service	AIGS	Imminent	11,21	11,21	11,21	12,23	12,23	12,23	12,23	12,23	12,23	12,24
Point-2-Point Audio Call	P2P	Normal	11,21	11,21	11,21	11,21	11,21	11,21	12,23	11,21	11,21	12,24
Voice Group Comm. Service	VGCS	Normal	11,21	11,21	11,21	11,21	11,21	11,21	11,21	12,23	11,21	12,24
Ad-hoc Group Comm. Service	AGS	Normal	11,21	11,21	11,21	11,21	11,21	11,21	11,21	11,21	12,23	12,24
Data service	DS	High	12,24	12,24	12,24	12,24	12,24	12,24	12,24	12,24	12,24	12,24

NOTE:
- Ongoing call is left/terminated: 11
- Ongoing call is maintained: 12

• Ongoing call is put on hold:	13
• Not possible (UI):	x
• Not possible (system):	-
• No Change:	0
• New call is joined / answered / established:	21
• New call is not established:	22
• New call is indicated only:	23
• Parallel operation:	24

5.4.3.2 Test procedure sequence

Table 5.4.3.2-1: Main behaviour (MO)

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
	EXCEPTION: Procedure 'MCX CO communication' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1) clause 5.4.3	--		-	-
	EXCEPTION: Procedure 'Initial MCX Authentication, Registration, Configuration and Subscription' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1) clause 5.3.2	--		-	-
1	**Ongoing call: high priority** Group Communication (F1) Check: UE1 initiates emergency group communication call (F1) with priority level "high"	-->	SIP INVITE	1	P
2	UE1 receives STATUS information from MCx_S1 regarding the establishment procedure	<--	SIP STATUS 100 (Trying)	-	-
3	UE1 receives STATUS information: group communication established (see note 1)	<--	SIP STATUS 200 (OK)	-	-
4a	**Test of tuple {12,22}** Check: UE1 user action to initiate an **imminent peril** private call (F2). Emergency group communication will be maintained (12). No change on the user interface (22) indicated (see note 2)	-->	No message	1	P
4b	**Test of tuple {12,22}** Check: UE1 user action to initiate a **normal** private call (F2). Emergency group communication will be maintained (12). No change on the user interface (22) indicated (see note 2)	-->	No message	1	P
5a	**Test of tuple {11,21}** Check: UE1 user action to initiate an Emergency Ad-hoc group communication call (F2). In the first step the emergency group communication (F1) will be terminated (11); UE1 sends BYE	-->	SIP BYE	1	P
5b	MCx_S1 responds with STATUS message to the call termination request	<--	SIP STATUS 200 (OK)	-	-
5c	Check: UE1 initiated the Emergency Ad-hoc group communication (F2)	-->	SIP INVITE	1	P
5d	MCx_S1 responds with STATUS message to the call termination request	<--	SIP STATUS 200 (OK)	-	-
6	Check: UE1 terminates the call (F2) by sending BYE	-->	SIP BYE	1	P
7	MCx_S1 responds with STATUS message to the call termination request	<--	SIP STATUS 200 (OK)	-	-
8	Ongoing call: high priority Group Communication (F1) Check: UE1 initiates emergency group communication call (F1) with priority level "high".	-->	SIP INVITE	1	P
9	UE1 receives STATUS information from MCx_S1 regarding the establishment procedure	<--	SIP STATUS 100 (Trying)	-	-

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
10	UE1 receives STATUS information: group communication established (see note 1)	<--	SIP STATUS 200 (OK)	-	-
11	**Test of tuple {12,24}** Check: UE1 user action to initiate a data service call (F2) with priority "high". The emergency group communication (F1) will be maintained (12); UE1 send INVITE for the additional data service	-->	SIP INVITE	1	P
12	UE1 receives STATUS information from MCx_S1 regarding the establishment procedure	<--	SIP STATUS 100 (Trying)	-	-
13	UE1 receives STATUS information: group communication established (see note 1)	<--	SIP STATUS 200 (OK)	-	-
	Call (F1) and (F2) established	--		-	-
14a	Check: UE1 terminates the call (F1) by sending BYE	-->	SIP BYE	1	P
14b	MCx_S1 responds with STATUS message to the call termination request	<--	SIP STATUS 200 (OK)	-	-
14c	Check: UE1 terminates the call (F2) by sending BYE	-->	SIP BYE	1	P
14d	MCx_S1 responds with STATUS message to the call termination request	<--	SIP STATUS 200 (OK)	-	-
15	**Ongoing call: Imminent Peril** Group Communication (F1) Check: UE1 initiates imminent peril group communication call (F1) with priority level "imminent peril".	-->	SIP INVITE	1	P
16	UE1 receives STATUS information from MCx_S1 regarding the establishment procedure	<--	SIP STATUS 100 (Trying)	-	-
17	UE1 receives STATUS information: group communication established (see note 1)	<--	SIP STATUS 200 (OK)	-	-
18a	**Test of tuple {12,22}** Check: UE1 user action to initiate an **imminent peril** private call (F2). Imminent peril group communication will be maintained (12). No change on the user interface (22) indicated (see note 2)	-->	No message	1	P
18b	**Test of tuple {12,22}** Check: UE1 user action to initiate a **normal** private call (F2). Imminent peril group communication will be maintained (12). No change on the user interface (22) indicated (see note 2)	-->	No message	1	P
19a	**Test of tuple {11,21}** Check: UE1 user action to initiate an Emergency Ad-hoc group communication call (F2). In the first step the imminent peril group communication (F1) will be terminated (11); UE1 sends BYE	-->	SIP BYE	1	P
19b	MCx_S1 responds with STATUS message to the call termination request	<--	SIP STATUS 200 (OK)	-	-
19c	Check: UE1 initiated the Emergency Ad-hoc group communication (F2)	-->	SIP INVITE	1	P
19d	MCx_S1 responds with STATUS message to the call invite request	<--	SIP STATUS 200 (OK)	-	-
	Call (F2) established	--		-	-
20	Check: UE1 terminates the call (F2) by sending BYE	-->	SIP BYE	1	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
21	MCx_S1 responds with STATUS message to the call termination request	<--	SIP STATUS 200 (OK)	-	-
22	**Ongoing call: high priority** Group Communication (F1) Check: UE1 initiates emergency group communication call (F1) with priority level "high"	-->	SIP INVITE	1	P
23	UE1 receives STATUS information from MCx_S1 regarding the establishment procedure	<--	SIP STATUS 100 (Trying)	-	-
24	UE1 receives STATUS information: group communication established (see note 1)	<--	SIP STATUS 200 (OK)	-	-
22	**Test of tuple {12,24}** Check: UE1 user action to initiate a data service call (F2) with priority "high". The imminent peril group communication (F1) will be maintained (12); UE1 send INVITE for the additional data service	-->	SIP INVITE	1	P
23	UE1 receives STATUS information from MCx_S1 regarding the establishment procedure	<--	SIP STATUS 100 (Trying)	-	-
24	UE1 receives STATUS information: group communication established (see note 1)	<--	SIP STATUS 200 (OK)	-	-
25a	Check: UE1 terminates the call (F1) by sending BYE	-->	SIP BYE	1	P
25b	MCx_S1 responds with STATUS message to the call termination request	<--	SIP STATUS 200 (OK)	-	-
	Call (F1) and (F2) established	--		-	-
25c	Check: UE1 terminates the call (F2) by sending BYE	-->	SIP BYE	1	P
25d	MCx_S1 responds with STATUS message to the call termination request	<--	SIP STATUS 200 (OK)	-	-
26	**Ongoing call: Normal** Point-to-Point (P2P) Communication (F1) Check: UE1 initiates the P2P call (F1) with priority level "normal".	-->	SIP INVITE	1	P
27	UE1 receives STATUS information from MCx_S1 regarding the establishment procedure	<--	SIP STATUS 100 (Trying)	-	-
28	UE1 receives STATUS information: communication established (see note 1)	<--	SIP STATUS 200 (OK)	-	-
29	**Test of tuple {12,22}** Check: UE1 user action to initiate a **normal** private call (F2). The ongoing P2P call will be maintained (12). No change on the user interface (22) indicated (see note 2)	-->	No message	1	P
30	**Test of tuple {11,21}** Check: UE1 user action to initiate an Emergency Ad-hoc group communication call (F2). In the first step the normal P2P call (F1) will be terminated (11); UE1 sends BYE	-->	SIP BYE	1	P
31	MCx_S1 responds with STATUS message to the call termination request	<--	SIP STATUS 200 (OK)	-	-
32	Check: UE1 initiated the Emergency Ad-hoc group communication (F2)	-->	SIP INVITE	1	P
33	MCx_S1 responds with STATUS message to the call invite request	<--	SIP STATUS 200 (OK)	-	-
	Call (F2) established	--		-	-
34	Check: UE1 terminates the call (F2) by sending BYE	-->	SIP BYE	1	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
35	MCx_S1 responds with STATUS message to the call termination request	<--	SIP STATUS 200 (OK)	-	-
36	**Ongoing call: Normal** Point-to-Point (P2P) Communication (F1) Check: UE1 initiates the P2P call (F1) with priority level "normal"	-->	SIP INVITE	1	P
37	UE1 receives STATUS information from MCx_S1 regarding the establishment procedure	<--	SIP STATUS 100 (Trying)	-	-
38	UE1 receives STATUS information: communication established (see note 1)	<--	SIP STATUS 200 (OK)	-	-
39a	**Test of tuple {11,21}** Check: UE1 user action to initiate an Imminent peril Ad-hoc group communication call (F2). In the first step the normal P2P call (F1) will be terminated (11); UE1 sends BYE	-->	SIP BYE	1	P
39b	MCx_S1 responds with STATUS message to the call termination request	<--	SIP STATUS 200 (OK)	-	-
39c	Check: UE1 initiated the Emergency Ad-hoc group communication (F2)	-->	SIP INVITE	1	P
39d	MCx_S1 responds with STATUS message to the call invite request	<--	SIP STATUS 200 (OK)	-	-
	Call (F2) established	--		-	-
40	Check: UE1 terminates the call (F2) by sending BYE	-->	SIP BYE	1	P
41	MCx_S1 responds with STATUS message to the call termination request	<--	SIP STATUS 200 (OK)	-	-
42	**Ongoing call: Normal** Point-to-Point (P2P) Communication (F1) Check: UE1 initiates the P2P call (F1) with priority level "normal"	-->	SIP INVITE	1	P
43	UE1 receives STATUS information from MCx_S1 regarding the establishment procedure	<--	SIP STATUS 100 (Trying)	-	-
44	UE1 receives STATUS information: communication established (see note 1)	<--	SIP STATUS 200 (OK)	-	-
45	**Test of tuple {12,24}** Check: UE1 user action to initiate a data service call (F2) with priority "high". The normal P2P call (F1) will be maintained (12); UE1 send INVITE for the additional data service	-->	SIP INVITE	1	P
46	UE1 receives STATUS information from MCx_S1 regarding the establishment procedure	<--	SIP STATUS 100 (Trying)	-	-
47	UE1 receives STATUS information: group communication established (see note 1)	<--	SIP STATUS 200 (OK)	-	-
	Call (F1) and (F2) established	--		-	-
48	Check: UE1 terminates the call (F1) by sending BYE	-->	SIP BYE	1	P
49	MCx_S1 responds with STATUS message to the call termination request	<--	SIP STATUS 200 (OK)	-	-
50	Check: UE1 terminates the call (F2) by sending BYE	-->	SIP BYE	1	P
51	MCx_S1 responds with STATUS message to the call termination request	<--	SIP STATUS 200 (OK)	-	-

NOTE 1: The indication of the call on the user interface shall allow an interpretation of the type of call and its priority.
NOTE 2: Depending on the implementation it is recommended to provide acoustical feedback to the user regarding the action of pushing a button or function not available.

Table 5.4.3.2-2: Main behaviour (MT)

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
	EXCEPTION: Procedure 'MCX CO communication' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1) clause 5.4.3	--		-	-
	EXCEPTION: Procedure 'Initial MCX Authentication, Registration, Configuration and Subscription' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1) clause 5.3.2	--		-	-
1	**Ongoing call: high priority** Emergency Group Communication (F1) Check: UE1 initiates emergency group communication call (F1) with priority level "high"	-->	SIP INVITE	2	P
2	UE1 receives STATUS information from MCx_S1 regarding the establishment procedure	<--	SIP STATUS 100 (Trying)	-	-
3	UE1 receives STATUS information: group communication established (see note 1)	<--	SIP STATUS 200 (OK)	-	-
4	**Test of tuple {12,22}** Incoming INVITE for imminent peril private call (F2)	<--	SIP INVITE	-	-
5	UE1 responds with STATUS message to the invitation request	-->	SIP STATUS 486 (Busy Here)	2	P
6	Check: UE1 user action in respect to the received **imminent peril** private call (F2). Emergency group communication will be maintained (12). The received call will be indicated (22) on the user interface (see note 2)	-->	Visual and audible indication of the received call on the user interface	2	P
	Call (F1) established	--		-	-
7	Check: UE1 terminates the call (F1) by sending BYE	-->	SIP BYE	2	P
8	MCx_S1 responds with STATUS message to the call termination request	<--	SIP STATUS 200 (OK)	-	-
9	Check: UE1 changes to idle state	-->	Idle state according to 'Initial MCX Authentication, Registration, Configuration and Subscription' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1) clause 5.3.2	2	P
10	**Ongoing call: high priority** Emergency Group Communication (F1) Check: UE1 initiates emergency group communication call (F1) with priority level "high".	-->	SIP INVITE	2	P
11	UE1 receives STATUS information from MCx_S1 regarding the establishment procedure	<--	SIP STATUS 100 (Trying)	-	-
12	UE1 receives STATUS information: group communication established (see note 1)	<--	SIP STATUS 200 (OK)	-	-
13	**Test of tuple {11,21}** Incoming INVITE for Ad-hoc Emergency group communication (F2)	<--	SIP INVITE	-	-
14	UE1 responds with STATUS message to the invite request	-->	SIP STATUS 200 (OK)	2	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
15	Check: UE1 user action to initiate an Ad-hoc Emergency group communication call (F2). In the first step the High priority Emergency Group communication call (F1) will be terminated (11); UE1 sends BYE	-->	SIP BYE	2	P
16	MCx_S1 responds with STATUS message to the call termination request	<--	SIP STATUS 200 (OK)	-	-
	Call (F2) established	--		-	-
17	Check: UE1 terminates the call (F2) by sending BYE	-->	SIP BYE	1	P
18	MCx_S1 responds with STATUS message to the call termination request	<--	SIP STATUS 200 (OK)	-	-
19	**Ongoing call: high priority** Emergency Group Communication (F1) Check: UE1 initiates emergency group communication call (F1) with priority level "high".	-->	SIP INVITE	2	P
20	UE1 receives STATUS information from MCx_S1 regarding the establishment procedure	<--	SIP STATUS 100 (Trying)	-	-
21	UE1 receives STATUS information: group communication established (see note 1)	<--	SIP STATUS 200 (OK)	-	-
22	**Test of tuple {12,24}** Incoming INVITE for Data service (F2)	<--	SIP INVITE	-	-
23	Check: UE1 user action to initiate a data service call (F2) with priority "high". The emergency group communication call (F1) will be maintained (12)	-->	SIP STATUS 200 (OK	2	P
	Call (F1) and (F2) established	--		-	-
24	Check: UE1 terminates the call (F1) by sending BYE	-->	SIP BYE	2	P
25	MCx_S1 responds with STATUS message to the call termination request	<--	SIP STATUS 200 (OK)	-	-
26	Check: UE1 terminates the call (F2) by sending BYE	-->	SIP BYE	2	P
27	MCx_S1 responds with STATUS message to the call termination request	<--	SIP STATUS 200 (OK)	-	-
28	**Ongoing call: Imminent peril** private call (F1) Check: UE1 initiates imminent private call (F1) with priority level "high".	-->	SIP INVITE	2	P
29	UE1 receives STATUS information from MCx_S1 regarding the establishment procedure	<--	SIP STATUS 100 (Trying)	-	-
30	UE1 receives STATUS information: communication established (see note 1)	<--	SIP STATUS 200 (OK)	-	-
31	**Test of tuple {12,22}** Incoming INVITE for normal private call (F2)	<--	SIP INVITE	-	-
32	UE1 responds with STATUS message to the (F2) invitation request	-->	SIP STATUS 486 (Busy Here)	2	P
33	Check: UE1 user action in respect to the received **normal** private call (F2). **Imminent peril** private call will be maintained (12). The received call will be indicated (22) on the user interface (see note 2)	-->	Visual and audible indication of the received call on the user interface	2	P
	Call (F1) established	--		-	-
34	Check: UE1 terminates the call (F1) by sending BYE	-->	SIP BYE	2	P
35	MCx_S1 responds with STATUS message to the call termination request	<--	SIP STATUS 200 (OK)	-	-

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
36	Check: UE1 changes to idle state	-->	Idle state according to 'Initial MCX Authentication, Registration, Configuration and Subscription' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1) clause 5.3.2	2	P
37	**Ongoing call: Imminent peril** private call (F1) Check: UE1 initiates imminent peril private call (F1) with priority level "high".	-->	SIP INVITE	2	P
38	UE1 receives STATUS information from MCx_S1 regarding the establishment procedure	<--	SIP STATUS 100 (Trying)	-	-
39	UE1 receives STATUS information: communication established (see note 1)	<--	SIP STATUS 200 (OK)	-	-
40	**Test of tuple {11,21}** Incoming INVITE for Ad-hoc Emergency group communication (F2)	<--	SIP INVITE	-	-
41	UE1 responds with STATUS message to the (F2) invite request	-->	SIP STATUS 200 (OK)	2	P
42	Check: UE1 user action to initiate an Ad-hoc Emergency group communication call (F2). In the first step the imminent peril private call (F1) will be terminated (11); UE1 sends BYE	-->	SIP BYE	2	P
43	MCx_S1 responds with STATUS message to the call termination request	<--	SIP STATUS 200 (OK)	-	-
	Call (F2) established	--		-	-
44	Check: UE1 terminates the call (F2) by sending BYE	-->	SIP BYE	1	P
45	MCx_S1 responds with STATUS message to the call termination request	<--	SIP STATUS 200 (OK)	-	-
46	**Ongoing call: Imminent peril** private call (F1) Check: UE1 initiates imminent private call (F1) with priority level "high".	-->	SIP INVITE	2	P
47	UE1 receives STATUS information from MCx_S1 regarding the establishment procedure	<--	SIP STATUS 100 (Trying)	-	-
48	UE1 receives STATUS information: communication established (see note 1)	<--	SIP STATUS 200 (OK)	-	-
49	**Test of tuple {12,24}** Incoming INVITE for Data service (F2)	<--	SIP INVITE	-	-
50	Check: UE1 user action to initiate a data service call (F2) with priority "high". The imminent peril private call (F1) will be maintained (12);	-->	SIP STATUS 200 (OK	2	P
	Call (F1) and (F2) established	--		-	-
51	Check: UE1 terminates the call (F1) by sending BYE	-->	SIP BYE	2	P
52	MCx_S1 responds with STATUS message to the call termination request	<--	SIP STATUS 200 (OK)	-	-
53	Check: UE1 terminates the call (F2) by sending BYE	-->	SIP BYE	2	P
54	MCx_S1 responds with STATUS message to the call termination request	<--	SIP STATUS 200 (OK)	-	-
55	**Ongoing call: Normal** Point-to-Point call (F1) Check: UE1 initiates normal P2P call (F1) with priority level "high".	-->	SIP INVITE	2	P
56	UE1 receives STATUS information from MCx_S1 regarding the establishment procedure	<--	SIP STATUS 100 (Trying)	-	-

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
57	UE1 receives STATUS information: communication established (see note 1)	<--	SIP STATUS 200 (OK)	-	-
58	**Test of tuple {12,22}** Incoming INVITE for normal private call (F2)	<--	SIP INVITE	-	-
59	UE1 responds with STATUS message to the (F2) invitation request	-->	SIP STATUS 486 (Busy Here)	2	P
60	Check: UE1 user action in respect to the received **normal** private call (F2). **Normal** private call will be maintained (12). The received call will be indicated (22) on the user interface (see note 2)	-->	Visual and audible indication of the received call on the user interface	2	P
	Call (F1) established	--		-	-
61	Check: UE1 terminates the call (F1) by sending BYE	-->	SIP BYE	2	P
62	MCx_S1 responds with STATUS message to the call termination request	<--	SIP STATUS 200 (OK)	-	-
63	Check: UE1 changes to idle state	-->	Idle state according to 'Initial MCX Authentication, Registration, Configuration and Subscription' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1) clause 5.3.2	2	P
64	**Ongoing call: Normal** Point-to-Point call (F1) Check: UE1 initiates normal P2P call (F1) with priority level "high"	-->	SIP INVITE	2	P
65	UE1 receives STATUS information from MCx_S1 regarding the establishment procedure	<--	SIP STATUS 100 (Trying)	-	-
66	UE1 receives STATUS information: communication established (see note 1)	<--	SIP STATUS 200 (OK)	-	-
67	**Test of tuple {11,21}** Incoming INVITE for Ad-hoc Emergency group communication (F2)	<--	SIP INVITE	-	-
68	UE1 responds with STATUS message to the (F2) invite request	-->	SIP STATUS 200 (OK)	2	P
69	Check: UE1 user action to initiate an Ad-hoc Emergency group communication call (F2). In the first step the normal private call (F1) will be terminated (11); UE1 sends BYE	-->	SIP BYE	2	P
70	MCx_S1 responds with STATUS message to the call termination request	<--	SIP STATUS 200 (OK)	-	-
	Call (F2) established	--		-	-
71	Check: UE1 terminates the call (F2) by sending BYE	-->	SIP BYE	1	P
72	MCx_S1 responds with STATUS message to the call termination request	<--	SIP STATUS 200 (OK)	-	-
73	**Ongoing call: normal** private call (F1) Check: UE1 initiates normal private call (F1) with priority level "high"	-->	SIP INVITE	2	P
74	UE1 receives STATUS information from MCx_S1 regarding the establishment procedure	<--	SIP STATUS 100 (Trying)	-	-
75	UE1 receives STATUS information: communication established (see note 1)	<--	SIP STATUS 200 (OK)	-	-
76	**Test of tuple {12,24}** Incoming INVITE for Data service (F2)	<--	SIP INVITE	-	-
77	Check: UE1 user action to initiate a data service call (F2) with priority "high". The normal private call (F1) will be maintained (12)	-->	SIP STATUS 200 (OK	2	P
	Call (F1) and (F2) established	--		-	-

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
78	Check: UE1 terminates the call (F1) by sending BYE	-->	SIP BYE	2	P
79	MCx_S1 responds with STATUS message to the call termination request	<--	SIP STATUS 200 (OK)	-	-
80	Check: UE1 terminates the call (F2) by sending BYE	-->	SIP BYE	2	P
81	MCx_S1 responds with STATUS message to the call termination request	<--	SIP STATUS 200 (OK)	-	-
NOTE 1: The indication of the call on the user interface shall allow an interpretation of the type of call and its priority. NOTE 2: Depending on the implementation it is recommended to provide visual indication (e.g. as missed call) and acoustical feedback to the user regarding the received call.					

5.4.3.3 Specific message contents

Table 5.4.3.3-1: SIP STATUS from UE
(step 5, Table 5.4.3.2-1; ETSI TS 137 579-4 [7] (3GPP TS 37.579-4) Table 5.5.2.18-1)

Derivation Path: Derivation Path: IETF RFC 3261 [16]; condition STATUS-CODE				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Status-Line				
Status-Code	486		ETSI TS 124 229 [8] (3GPP TS 24.229)	
Reason-Phrase	(Busy here)			

6 Mission critical ad hoc group communications

6.1 Interception interface on MCX service system

6.1.1 Test purpose

(1)

```
with {MCX service system providing discrete listen capabilities}
ensure that {
    when {MCX service system provides discrete listening capabilities to a target MCX service user
which requires spec authorizations}
    then {No noticeable impact, information, facility or impact can be detected by non-interception-
target MCX service users and all information from or to the ad hoc group communication MCX service
users are transmitted to the authorized interception-target MCX service user including all session
audit information (e.g. regrouping, leaving of participants, voice durations, position information)}
            }
```

6.1.2 Conformance requirements

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.1-001a]]

"The MCX Service shall provide discreet listening capabilities without noticeable impact on or knowledge of the target MCX User, or the members of the target MCX Service Group including Ad hoc groups, and all other unauthorized MCX Users."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause[R-6.15.1-002a]]

"The MCX Service shall provide a mechanism for an authorized MCX User to receive MCX Service Ad hoc Group Communication transmissions to and from a specific target MCX User that is within the authority of the authorized MCX User."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.4-011]]

"The MCX Service shall provide recording and audit functions for all types of MCX Service group communication (e.g., normal group, regroup group, Ad hoc group)."

6.1.3 Test description

6.1.3.1 Pre-test conditions

System Configuration:

- MCx Service system consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (UE1) also consists of all sub-systems of the MCx user service.
- The configuration for recording, audition and discrete listening is prepared according to Figure 4.1-3.
- For the underlying "transport bearer" over which the UE1 (MCx Service system) and the IUT will communicate Parameters are set to the default parameters specific for the basic E-UTRA single simulated cells cell network scenarios, as defined in ETSI TS 138 508-1 [6] (3GPP TS 38.508-1) clause 6.3. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCx operation in the MCx configuration document).
- MCx Service system is configured to support the following features:
  - Configuration options for discrete listening, recording and session audition.
  - UEa1 shall be configured as discrete listening user.
  - Pre-arranged groups containing minimum UE2 and UE-Con.
  - Ad-hoc Group Communication functionality.
  - multiple user floor control.
  - multi-talker floor control.
  - prioritisation of floor control requests.
  - In case the MCx Service system detects floor inactivity for a configurated period, the call will be terminated to unblock resources. The inactivity timer for any group communication type is configured to 240 seconds.
  - In case floor control will be used by the participants, the MCx service system provide an overall duration for an ongoing communication. This call time limit timer is configured to 500 seconds.
  - Time period for keeping recording and session details shall be configurable between 0 (off) and 30 days.

IUT:

- IUT (UEa1)
- The IUT (UEa1) consists of all sub-systems of the MCx user service including the ability for floor control, pre-arranged group communication functionality and ad-hoc group communication functionality. Ensure the correct configuration.
- The UE1 is the acting user.

Preamble:

- The IUT (UEa1)], UE1 and UE2 are connected to PLMN of the primary 5GS.
- The MCx Service system (MCx_S1) is connected to the IUT (UEa1, MCx client, discrete listener)], UE1, UE2 (MCx client) and UE-Con (MCx client with controller/dispatcher functionality).

- MCx service SIP registration and affiliation are executed on MCx_S1.

6.1.3.2 Test procedure sequence

Table 6.1.3.2-1: Main behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
	EXCEPTION: Procedure 'MCPTT Server - MCPTT Client / Configuration / Authentication / User Authorisation / UE Configuration / User Profile' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1) clause 5.1.3				
	MCx Service system provides a configuration interface for feature activation in respect to the FA	--		-	-
1a	Check: MCx_S1 provides a configuration interface based on HTTPS request. Login required. With correct credentials the access to the discrete listening configuration management will be granted (see note 1)	<--	HTTPS link to configuration interface	-	-
1b	Check MCx_S1 provides a configuration interface based on XML file import. Login for access to the file transfer function required. With correct credentials the access to the discrete listening configuration management will be granted (see note 1)	<--	XML access (e.g. via SCP or SFTP connection)	-	-
	EXCEPTION: Procedure 'MCX CO communication' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1) clause 5.4.3 (see note 2)	--		-	-
	EXCEPTION: Procedure 'Initial MCX Authentication, Registration, Configuration and Subscription' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1) clause 5.3.2 (see note 2)	--		-	-
	EXCEPTION: Procedure 'Functional aliasing client side' as described in clause 5.1.3 (see note 2)	--		-	-
	EXCEPTION: Procedure 'Floor control in general' as described in clause 5.2.3 (see note 2)	--		-	-
	EXCEPTION: Procedure 'MCPTT Groups configured for multi-talker control' as described in clause 5.3.3 (see note 2)	--		-	-
	EXCEPTION: Procedure 'Arbitration function' as described in clause 5.4.3 (see note 2)	--		-	-
2a	Check: Open the MCx Service system GUI and login as "Replay Client" Session audition log will be presented on the GUI	<--	The session audition log contains all information regarding {PTT activities | regrouping | leaving of participants | voice durations | position information | termination causes | optional information}	-	-
2b	Check: Open the MCx Service system GUI and login as "user" or "admin"	<--	The session audition log is not available	-	-
3a	Pre-arranged group communication case Check: UE1 initiates pre-arranged group communication call	-->	SIP INVITE	1	P
3b	Ad-hoc Group Communication case Check: UE1 initiates ad-hoc group communication call	-->	SIP INVITE	1	P
4	UE1 receives STATUS information from MCx_S1 regarding the establishment procedure	<--	SIP STATUS 100 (Trying)	-	-

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
5	EXCEPTION: UEa1, UE2 and UE-Con receive INVITE to the group communication in step 2a/2b provided by MCx_S1	<--	SIP INVITE	-	-
6	UEa1, UE2 and UE-Con return STATUS information: group communication established	-->	SIP STATUS 200 (OK)	-	-
7	UE1 receives STATUS information: group communication established	<--	SIP STATUS 200 (OK)	-	-
8a	Check for pre-arranged group communication: UE1 does not see any information about UE1a in the group communication participants list	-->	RTCP (MCPT) Floor REQUEST	1	P
9a	MCx Service system replies with floor granted	<--	RTCP (MCPT) Floor GRANTED	-	-
10a	Check: UE1 can send RTP data - media stream is transmitted	-->	RTP stream	1	P
11a	RTP data from UE1 is provided to UE1a	<--	RTP stream	-	-
12a	Check: UE1a is not allowed to use the PTT. If floor request is sent by the UE1a, it is denied by the MCx Service system	-->	RTCP (MCPT) Floor REQUEST	1	P
13a	MCx Service system replies with floor deny	<--	RTCP (MCPT) Floor DENY	-	-
8b	Check for ad hoc group communication: UE1 does not see any information about UE1a in the ad hoc group communication participants list	-->	RTCP (MCPT) Floor REQUEST	1	P
9b	MCx Service system replies with floor granted	<--	RTCP (MCPT) Floor GRANTED	-	-
10b	Check: UE1 can send RTP data - media stream is transmitted	-->	RTP stream	1	P
11b	RTP data from UE1 is provided to UE1a	<--	RTP stream	-	-
12b	Check: UE1a is not allowed to use the PTT. If floor request is sent by the UE1a, it is denied by the MCx Service system	-->	RTCP (MCPT) Floor REQUEST	1	P
13b	MCx Service system replies with floor deny	<--	RTCP (MCPT) Floor DENY	-	-
14	MCx_S1 terminates the group communication call using BYE Check: UE2 and UE-Con receive BYE	<--	SIP BYE	-	-
15	UE2 confirms the received BYE message	-->	SIP STATUS 200 (OK)	-	-
16	UE-Con confirms the received BYE message	-->	SIP STATUS 200 (OK)	-	-

NOTE 1: The configuration section on the server shall be in available only with specific user rights to prevent misuse. The discrete listening function can be enabled or disabled. On enabling the settings for recording and user selection can be configured. The recording shall include SIP messaging as well as RTP audio stream data. The discrete listening UE terminal shall be excluded in all lists. PTT function shall be disabled; UE is in listening only mode.

NOTE 2: The exceptional procedures are required as input for the recording and session audition function.

6.2 Ad hoc group communication functionality for MCX service user involving one MCX system

6.2.1 Test purpose

(1)

with {Establishment of an ad hoc group communication by an authorized service user}
ensure that {
    when {MCX service user is allowed to initiate ad hoc communication (initiator) manually or based on automatic criterion (e.g. location) **and** Multiple MCX service users using FA are available for ad hoc group communication and participants can be added or removed by initiator or authorized MCX server user}
    then {MCX service user initiates an ad hoc group communication manually or based on an automatic criterion (e.g. location) **and** MCX service user with multiple FA receives only one ad hoc group communication based on the unique MCX service ID **and** the FA can be determined by the MCX service user **and** the initiator or an authorized MCX service user is able to add or remove participants from the ad hoc group communication **and** the list of participants will be updated by the MCX service system and information will be sent to the initiator of the ad hoc group communication}
    else when {MCX service user has no authorisation to initiate an ad hoc group communication}
    then {MCX service user receive a reason for denial from the MCX service system}}
            }

(2)

with {Ad hoc group communication processing}}
ensure that {
    when {Ad hoc group communication is in preparation **or** ad hoc group communication is active}
    then {MCX service user receive information about the initialisation and transmit request queue and hang time and mode (e.g. broadcast mode)}
            }

(3)

with {Termination of an ad hoc group communication}
ensure that {
    when {MCX ad hoc group communication service is configured, that ad hoc group communication can be released by initiator **or** by timer expiration only **or** termination by MCX service system. Timer is set to 300 seconds}
    then {Initiator of an ad hoc group communication releases the ad hoc communication or timer expires, and ad hoc group communication no longer exists after release}
    else when {MCX ad hoc group communication service is configured, that ad hoc group communication can be released by initiator or by timer expiration after 300 seconds only and ad hoc group call participant which is not initiator release the group communication}
    then {MCX service user leaves the ad hoc group call communication, and the list of participants will be updated by the MCX service system and information will be sent to the initiator of the ad hoc group communication}
            }

(4)

with {Suppression of the participants list}
ensure that {
    when {MCX service system suppress the list of participants for MCX service users which are not the initiator of the ad hoc group communication}
    then {The MCX service user, which is participant in an ad hoc group communication has no option to request the list of participants **or** the MCX service system reject the request}
            }

(5)

with {Generating a permanent group from an ad hoc group communication participants list}
ensure that {
    when {MCS service user is able to initiate an ad hoc group communication by selecting different participants}
    then {The MCX service user application provides options to save the ad hoc group configuration as permanent and the MCX service user application provides options to manipulate the saved list (e.g. add or remove participants, export or delete the list)}
            }

(6)

```
with {Joining an ongoing ad hoc group communication}
ensure that {
    when {Ongoing ad hoc group communication and} MCX Service user added to the ad hoc group
communication list}
    then {MCX service user is able to join an ongoing ad hoc group communication while entering the
location or on request to the initiator or an authorized MCX service user or on invitation from the
initiator or an authorized MCX service user}
            }
```

(7)

```
with {MCX service user provide location information}
ensure that {
    when {MCX service user is initiator or participant of an ad hoc group communication}
    then {MCX service user provides location information}
            }
```

(8)

```
with {Ad hoc group communication parameter selection and usage}
ensure that {
    when {MCX service user prepares an ad hoc group communication}
    then {MCX service user can select additional parameters for hang time and priority (e.g. general
group, emergency, imminent peril) and operational mode (e.g. broadcast or non-broadcast mode)}
```

6.2.2 Conformance requirements

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.5.2-001]]

"The MCX Service shall provide a mechanism for an authorized MCX User to combine an ad hoc multiplicity of MCX Users into a MCX Service Ad hoc Group Communication.

NOTE: Selection of the list of MCX Users can be manual, or automatic based on certain criteria (e.g., location). This is left for implementation."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.5.2-001a]]

"The MCX Service shall provide the reason for denial to an MCX user who was not authorised to initiate an MCX Service Ad hoc Group Communication."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.5.2-001b]]

"The participant of an MCX Ad hoc Group Communication who has activated several Functional Alias(es) shall receive only one communication based on the unique MCX Service ID."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.5.2-001c]]

"The MCX Service shall provide a mechanism for the participant of an Ad hoc Group Communication to determine the Functional Alias, through which this participant is addressed by the communication."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.5.2-002]]

"An MCX Service Ad hoc Group Communication is a type of MCX Service Group communication and shall support MCX Service Group Communication mechanisms for call processing (e.g., transmit request queuing, hang time, broadcast mode)."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.5.2-003]]

"MCX Service Ad hoc Group Communications shall be terminated by an authorised user or using the same mechanisms as MCX Service Group communications (e.g., initiator release, time expiration, server release)."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.5.2-005]]

"When an MCX Service Ad hoc Group Communication is terminated the group shall not persist."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.5.2-007]]

"The MCX Service shall provide a mechanism for an authorized initiator of a MCX Service Ad hoc Group Communication to define the communication parameters for the Ad hoc Group Communication (e.g. priority, hang time, broadcast/non-broadcast)."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.5.2-008]]

"MCX Service Ad hoc Group Communications shall be able to support the same urgency as MCX Service Group communication (e.g., general group, emergency, imminent peril)."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.5.2-010]]

"The MCX Service shall provide a mechanism for the initiator of an MCX Service Ad hoc Group Communication to request that the list of participants is suppressed."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.5.2-011]]

"The MCX Service shall provide a mechanism for authorized MCX Users to create a permanent MCX Service Group from the members of the MCX Service Ad hoc Group communication."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.5.2-012]]

"The MCX Service shall provide a mechanism for the initiator and/or an authorised user to add or remove participants during an in progress MCX Service Ad hoc Group communication."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.5.2-013]]

"The MCX Service shall provide a mechanism for a participant to join an in progress MCX Service Ad hoc Group communication."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.5.2-014]]

"The MCX Service shall provide a mechanism for the initiator of an MCX Service Ad hoc Group Communication to request that the list of participants be determined and updated by the MCX Service system using pre-defined criteria."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.5.2-014a]]

"When the list of participants is determined or updated by the MCX Service system, the MCX Service shall provide a mechanism that monitors and ensures that the participants list is applied for an MCX Service Ad hoc Group communication, performing retries when needed."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.5.2-015]]

"The MCX Service shall provide a mechanism for a participant of an MCX Service Ad hoc Group Communication to provide his location information to an authorised participant of the same group."

6.2.3 Test description

6.2.3.1 Pre-test conditions

System Configuration:

- MCx Service system consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (UE1) also consists of all sub-systems of the MCx user service.
- For the underlying "transport bearer" over which the UE1 (MCx Service system) and the IUT will communicate Parameters are set to the default parameters specific for the basic E-UTRA single simulated cells cell network scenarios, as defined in [ETSI TS 138 508-1 [6] (3GPP TS 38.508-1) clause 6.3]. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCx operation in the MCx configuration document).

- MCx Service system is configured to support the following features:
  - Ad-hoc Group Communication supporting criterion-based functionality:
    - Functional Alias (optional).
    - Location Information (optional).
    - Criteria for determining the participants (mandatory).
  - In case the MCx Service system detects floor inactivity for a configurated period, the call will be terminated to unblock resources. The inactivity timer T4 for any ad hoc group communication is configured to T4=300 seconds.
  - Supporting user UE2 has permission for using ad hoc functionality and has activated multiple FA (minimum two).
  - Supporting user UEa1 user has following configuration:
    - Local parameter set allows the usage of ad hoc functionality.
    - MMCx_S1 has a parameter set with no permission for UEa1 to use ad hoc functionality.

IUT:

- IUT (UE1).
- The IUT (UE1) consists of all sub-systems of the MCx user service including the ability for floor control, pre-arranged group communication functionality and ad-hoc group communication functionality. Ensure the correct configuration.
- The UE1 is the acting user.

Preamble:

- The IUT UE1 is connected to PLMN of the primary 5GS.
- The MCx Service system (MCx_S1) is connected to the IUT, UEa1 (MCx client with restrictions)], UE2 (MCx client with multiple FA) and UE-Con (MCx client with controller/dispatcher functionality).

6.2.3.2 Test procedure sequence

Table 6.2.3.2-1: Main behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
	Check: configuration (as file or user settings entry) with multiple FA is available for the UE1. The configuration contains numerical and organizational arrangements of FA and flags for expiration timer, priority and location. MCx Service user accepts the configuration	--		-	--
	Execute MCX Group Affiliation Status Change as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1) clause 5.3.34. No FA is active at the MCx Service system. Check: UE1 publish the basic POC settings	-->	SIP PUBLISH according ETSI TS 124 229 [8] (3GPP TS 24.229), IETF RFC 3903 [17]	1	P
1	MCx_S1 confirm the POC settings	<--	STATUS 200 OK (PUBLISH) according ETSI TS 124 229 [8] (3GPP TS 24.229)	-	--
2	Check: UE1 announce interest for a specific FA to the MCx_S1 (Interrogation, entrance to a specific list); UE1 is not allowed to use interrogation	-->	SIP SUBSCRIBE according ETSI TS 124 229 [8] (3GPP TS 24.229), IETF RFC 3856 [9] and IETF RFC 6665 [10]	1	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
3	Check: MCx_S1 responds with a STATUS (see note 1) message	<--	SIP STATUS (see note 1) 200 OK	-	--
4	Check: UE1 send PUBLISH message containing an automatic FA information	-->	SIP PUBLISH	1	P
5	Check: MCx_S1 responds with a STATUS (see note 1) message	<--	SIP STATUS (see note 1) 200 OK	-	--
6	Check: UE1 is allowed to participate on ad hoc group communication based on criterion <allow-emergency-adhoc-group-call> in the <ruleset> of the client profile	-->	- none - Client checks the ruleset configuration according to ETSI TS 124 379 [21] (3GPP TS 24.379) clause 6.2.8.1.24 is set to "true"	1	P
7a	Check: UE1 initiates ad hoc group communication based (criterion: manual; list based) {InitiateGroup, ParticipantList}	-->	SIP INVITE according to ETSI TS 124 379 [21] (3GPP TS 24.379) clause 17.2.2.1.1	1	P
8a	System validates authorization of UE1		DB/AAA check → VALID: 200 OK {Group Established, GroupID} → ACK	-	--
9a	UE1 receives confirmation from MCx_S1	<--	SIP ACK		
7b	Check: UE1 initiates ad hoc group communication based (automatic; criterion based: functional alias) {InitiateGroup, ParticipantList, FA ID}	-->	SIP INVITE according to ETSI TS 124 379 [21] (3GPP TS 24.379) clause 17.2.2.1.1	1	P
8b	System validates authorization of UE1		DB/AAA check → VALID: ACK	-	--
9b	UE1 receives confirmation from MCx_S1	<--	SIP ACK		
7c	Check: UE1 initiates ad hoc group communication based (automatic criterion: location information) {InitiateGroup, ParticipantList, location information}	-->	SIP INVITE according to ETSI TS 124 379 [21] (3GPP TS 24.379) clause 17.2.2.1.1	1	P
8c	System validates authorization of UE1		DB/AAA check → VALID: ACK	-	--
9c	UE1 receives confirmation from MCx_S1	<--	SIP ACK	-	--
10	Session preparation phase. MCx_S1 sends NOTIFICATION	<--	RTCP (MPTC) Group Dynamic NOTIFY {InitStatus, QueuePolicy, HangTime, Mode} → All UEs	-	--
11	Check: MCx_S1 invites UE-con into the ongoing ad hoc group communication	-->	SIP INVITE	1	P
12	UE-con responds to the INVITE request (see note 2)	-->	SIP STATUS 100 (TRYING)	1	P
13	OPTIONAL: outgoing response to TRYING if no automatic answering mode is activated	-->	SIP STATUS 180 (RINGING)	1	P
14	Check: UE-con call established confirmation towards MCx_S1	-->	SIP ACK	1	P
15	Session preparation phase. MCx_S1 sends NOTIFICATION		RTCP (MPTC) Group Dynamic NOTIFY {InitStatus, QueuePolicy, HangTime, Mode} → All UEs	-	--
16	Floor activation by one participant to ensure the usage of the call time limit expiry Check: UE1 send floor REQUEST message		RTCP (MCPT) Floor REQUEST; message containing field {Floor priority=0}		
17	Check: MCx_S1 invites UE2 (multiple FA) into the ongoing ad hoc group communication	-->	SIP INVITE (see note 3)	1	P
18	UE2 responds to the INVITE request (see note 2)	-->	SIP STATUS 100 (TRYING)		
19	OPTIONAL: outgoing response to TRYING if no automatic answering mode is activated	-->	SIP STATUS 180 (RINGING)		
20	Check: UE2 call established confirmation towards MCx_S1	-->	SIP STATUS 200 (OK) {Group Established, GroupID} → ACK		
21	Session preparation phase. MCx_S1 sends NOTIFICATION		RTCP (MPTC) Group Dynamic NOTIFY {InitStatus, QueuePolicy, HangTime, Mode} → All UEs	-	--
22	Check: UEa1 initiates ad hoc group communication based (automatic criterion: functional alias) {InitiateGroup, ParticipantList, FA ID}	-->	SIP INVITE according to ETSI TS 124 379 [21] (3GPP TS 24.379) clause 17.2.2.1.1	1	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
23	System validates authorization of UEa1		DB/AAA check → NOT VALID: → 40x (see note 4) FORBIDDEN or UNAUTHORIZED {supressed}	-	--
24	UEa1 receives confirmation from MCx_S1 State: "MEAG 1: no-emergency"	<--	SIP STATUS 40x (see note 4)		
25b	Criterion based on FA UE1 change FA to a value which is not part of the ad hoc group communication		none		
26b	Check: PUBLISH message for subscribe functional alias request to MCx_S1	-->	SIP PUBLISH Message content according to ETSI TS 123 280 [4] 3GPP TS 23.280 clause 10.13.10.2	1	P
27b	Check: MCx Service system responds with a STATUS (see note 4) message	<--	SIP STATUS 200 OK	-	--
25c	Criterion based on location information UE1 receive new location information via internal interface and change to a location which is not part of the ad hoc group communication		none		
26c	Check: PUBLISH message containing Location Information Report to MCx_S1	-->	SIP PUBLISH Message content according to ETSI TS 123 280 [4] 3GPP TS 23.280 clause 10.9.3.2	7	P
27c	Check: MCx Service system responds with a STATUS (see note 4) message	<--	SIP STATUS 200 OK		
28	System validates authorization of UE1.		DB/AAA check → VALID but not part of the current ad hoc: → BYE	-	--
29	Check: MCx_S1 disconnect UE1 from ad hoc group communication. UE1 receive BYE message	<--	SIP BYE		
30	UE1 confirms the received BYE message	-->	SIP STATUS 200 (OK)	1	P
31	Session preparation phase. MCx_S1 sends NOTIFICATION	<--	RTCP (MPTC) Group Dynamic NOTIFY {InitStatus, QueuePolicy, HangTime, Mode} → All UEs	-	--
32	Check: ad hoc group communication activity timer T4 expires. Termination by system timer (300 s)		Internal Timer Expired → SIP NOTIFY {Group Terminated by Timer} → Participants receive BYE		
33a	Check: MCx_S1 disconnect UE2 from ad hoc group communication. UE2 receive BYE message	<--	SIP BYE	-	--
34a	UE2 confirms the received BYE message	-->	SIP STATUS 200 (OK)	3	P
33b	Check: MCx_S1 disconnect UE-con from ad hoc group communication. UE-con receive BYE message	<--	SIP BYE	-	--
34b	UE-con confirms the received BYE message	-->	SIP STATUS 200 (OK)	3	P
	Check: configuration (as file or user settings entry) with multiple FA is available for the UE1. The configuration contains numerical and organizational arrangements of FA and flags for expiration timer, priority and location. MCx Service user accepts the configuration	--		-	--
	Execute MCX Group Affiliation Status Change as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1) clause 5.3.34. No FA is active at the MCx Service system. Check: UE1 publish the basic POC settings	-->	SIP PUBLISH according ETSI TS 124 229 [8] (3GPP TS 24.229), IETF RFC 3903 [17]	1	P
35	MCx_S1 confirm the POC settings	<--	STATUS 200 OK (PUBLISH) according ETSI TS 124 229 [8] (3GPP TS 24.229)	-	--

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
36	Check: UE1 announce interest for a specific FA to the MCx_S1 (Interrogation, entrance to a specific list); UE1 is not allowed to use interrogation	-->	SIP SUBSCRIBE according ETSI TS 124 229 [8] (3GPP TS 24.229), IETF RFC 3856 [9] and IETF RFC 6665 [10]	1	P
37	Check: MCx_S1 responds with a STATUS (see note 4) message	<--	SIP STATUS (see note 1) 200 OK	-	--
38	Check: UE1 send PUBLISH message containing an automatic FA information	-->	SIP PUBLISH	1	P
39	Check: MCx_S1 responds with a STATUS message	<--	SIP STATUS (see note 1) 200 OK	-	--
40	Check: UE1 is allowed to participate on ad hoc group communication based on criterion <allow-emergency-adhoc-group-call> in the <ruleset> of the client profile	-->	- none - Client checks the ruleset configuration according to ETSI TS 124 379 [21] (3GPP TS 24.379) clause 6.2.8.1.24 is set to "true"	1	P
41	Check: UE1 initiates ad hoc group communication based (criterion: manually) {InitiateGroup, ParticipantList}	-->	SIP INVITE according to ETSI TS 124 379 [21] (3GPP TS 24.379) clause 17.2.2.1.1	1	P
42	System validates authorization of UE1.		DB/AAA check → VALID: ACK {Group Established, GroupID} → ACK	-	--
43	UE1 receives confirmation from MCx_S1	<--	SIP ACK		
44	Session preparation phase. MCx_S1 sends NOTIFICATION	<--	RTCP (MPTC) Group Dynamic NOTIFY {InitStatus, QueuePolicy, HangTime, Mode} → All UEs	-	--
45	Check: MCx_S1 invites UE-con into the ongoing ad hoc group communication	-->	SIP INVITE	1	P
46	Session preparation phase. MCx_S1 sends NOTIFICATION		RTCP (MPTC) Group Dynamic NOTIFY {InitStatus, QueuePolicy, HangTime, Mode} → All UEs	-	--
47	Floor activation by one participant to ensure the usage of the call time limit expiry. Check: UE1 send floor REQUEST message		RTCP (MCPT) Floor REQUEST; message containing field {Floor priority=0}		
48	Check: MCx_S1 invites UE2 into the ongoing ad hoc group communication	-->	SIP INVITE	1	P
49	UE2 responds to the INVITE request	-->	SIP STATUS 100 (TRYING)		
50	OPTIONAL: outgoing response to TRYING if no automatic answering mode is activated	-->	SIP STATUS 180 (RINGING)		
51	Check: UE2 call established confirmation towards MCx_S1	-->	SIP ACK		
52	Session preparation phase. MCx_S1 sends NOTIFICATION		RTCP (MPTC) Group Dynamic NOTIFY {InitStatus, QueuePolicy, HangTime, Mode} → All UEs	-	--
53	Check: UE2 (participant) leaves ad hoc group communication sending BYE	-->	SIP BYE	3	P
54	MCx_S1 confirms the received BYE message	<--	SIP STATUS 200 (OK)	-	--
55	Check: Session preparation phase. MCx_S1 sends NOTIFICATION	<--	RTCP (MPTC) Group Dynamic NOTIFY {Updated Participant List} → Remaining UEs	-	--
60	Check: UE1 (initiator) terminates ad hoc group communication sending BYE	-->	SIP BYE	3	P
61	MCx_S1 confirms the received BYE message	<--	SIP STATUS 200 (OK)	-	--
62	Check: MCx_S1 terminates the ad hoc group communication; UE-con receive BYE message	<--	SIP BYE	-	--
63	UE-con confirms the received BYE message	-->	SIP STATUS 200 (OK)	3	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
NOTE 1: May notify the answer state to the user (i.e. "Unconfirmed" or "Confirmed") if received in the P-Answer-State header field. NOTE 2: The application shall indicate the detected status of the floor parameter on the graphical user interface as icon and text: a) Status is IDLE, REVOKE or RELEASED: green; text: e.g. "push PTT to talk!". b) Status REQUESTED, DENY or TAKEN by other user, no multi-talker: red; text: none. c) Status REQUESTED, DENY or TAKEN by other user, multi-talker: blue; text: e.g. "push PTT to talk!". d) Status is GRANTED to own UE: yellow; text: e.g. "please talk!". NOTE 3: User with multiple FA receives only one ad hoc group communication based on the unique MCX service ID and the FA can be determined by the MCX service user and the initiator. NOTE 4: According to ETSI TS 124 229 [8] (3GPP TS 24.229) a request should be rejected by sending a suitable 4xx response. NOTE 5: The application on the UE GUI shall indicate a menu for editing the participant lists via user action. This may include: a) Adding, renaming and remove a list. b) Adding, renaming, repositioning and removing of participants in a list. c) Save an ad hoc participants list during ongoing ad hoc group communication call.				

6.2.3.3 Specific message contents

Table 6.2.3.3-1: SIP INVITE (criteria based) from UE1 (step 7a, Table 6.2.3.2-1; ETSI TS 137 579-4 [7] (3GPP TS 37.579-4) Table 5.5.2.5.1-1)

Derivation Path: ETSI TS 137 579-4 [7] (3GPP TS 37.579-4) clause 5.5.2.5.1; condition MCPTT-INFO-EVENT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Method	**"INVITE"**			
Request-URI	INVITE SIP/2.0	MCPTT originating user public ID		MCPTT
Message-body			IETF RFC 3261 [16]	
MIME Multipart Media Encapsulation		Type: multipart/mixed		
application/sdp		Session description parameter set		
Application/vnd.3gpp.mcptt-info+xml	<mcptt-Params>			
<session-type>	Ad hoc	Type: ad hoc group communication		
<mcptt-request-uri>				
<mcptt-client-id>				
<any-ext>/<call-participants- criterias>	active	Criteria based		

**Table 6.2.3.3-2: SIP INVITE (list based) from UE1
(step 7b, Table 6.2.3.2-1; ETSI TS 137 579-4 [7] (3GPP TS 37.579-4) Table 5.5.2.5.1-1)**

Derivation Path: ETSI TS 137 579-4 [7] (3GPP TS 37.579-4) clause 5.5.2.5.1; condition MCPTT-INFO-EVENT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Method	**"INVITE"**			
Request-URI	INVITE SIP/2.0	MCPTT originating user public ID		MCPTT
Message-body			IETF RFC 3261 [16]	
MIME Multipart Media Encapsulation		Type: multipart/mixed		
application/sdp		Session description parameter set		
application/vnd.3gpp.mcptt-info+xml	<mcptt-Params>			
<session-type>	Ad hoc	Type: ad hoc group communication		
<mcptt-request-uri>				
<mcptt-client-id>				
Application/resource-lists+xml	<resource-lists>			
<lists>	<entry>	List entries		

6.3 Ad hoc group communication security

6.3.1 Test purpose

(1)

```
with {Ad hoc group communication security compliance}
ensure that {
    when {MCX Ad hoc group communication is initiated by a MCX service user}
    then {Ensure that the MCX Service user initiating the session using secure encryption mechanisms
on the application layer for signalling and the encryption is end-to-end for all transmitted media
between all involved MCX UE and traffic and MCX service supports information exchange for regulatory
constraints to support confidentiality, message integrity, and source authentication for some
information exchanges (e.g., MCX Service User Profile management, kill commands)}
}
```

(2)

```
with {Identities for MCX Service Users}
ensure that {
    when {MCX Service users are registered as identities on the MCX Service}
    then {Each MCX User shall register to a unique identity and the MCX Service system shall assign
the MCX Service user to this globally unique identifier and the identifier is independent of the
IMSI, provided by the 3GPP system and the unique identity shall use as basis for the MCX Service
application security layer}
}
```

(3)

```
with {Mechanism to perform a single authentication}
ensure that {
when {MCX Service user request the authentication for a set of unique identifiers}
then {The MCX Service provides a mechanism for a single authentication procedure to the requesting
MCX Service user to access all authorized features and with authentication the MCX Service system
grants full access to all authorized features connected to the requested identity and authorized
features can vary between the different requested identities}
    else when {MCX Service user not performing authentication procedure or not all required
identities are authenticated by the MCX Service user}
    then {MCX Service user has no access or only limited access to the authorized features on the
MCX Service system depending on the requested set of identities}
}
```

(4)

```
with {Limited feature access Prior to authentication}
ensure that {
    when {MCX Service system provides access for selected features for a MCX UE prior the MCX
Service user authentication}
then {An authorized MCX UE shall have access to selected features prior the authentication of the
MCX Service user}
Only predefined low risk features are accessible before the full authentication, and all emulate the
access levels for unauthenticated users}
}
```

(5)

```
with {End-to-End security for media traffic and cryptographic key management}
ensure that {
    when {Media traffic is sent between MCX UEs}
then {The MCX Service system provides a management for cryptographic key services and ensures that
the cryptographic keys are confidentiality protected, integrity protected and authenticated when
delivered over air and media traffic remains end-to-end encrypted throughout transmission and
security mechanisms persist regardless of network connectivity and secure direct peer to peer
communication is established when out of network coverage and the cryptographic key management
service support pre-provisioning and over-the air provisioning and the MCX Service provide end-to-
end confidentiality and integrity protection to the MCX User Profile when transferred to and/or from
and while stored on an MCX Server, an MCX UE or both}
}
```

6.3.2 Conformance requirements

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.5.2-009]]

"MCX Service Ad hoc Group Communications shall support the applicable security requirements as identified in sub-clause 5.12."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.12-001]]

"The MCX Service shall provide a means to support the confidentiality and integrity of all user traffic and signalling at the application layer."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.12-002]]

"The MCX Service shall support MCX User with globally unique identities, independent of the mobile subscriber identity (IMSI) assigned by a 3GPP network operator to UEs."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.12-003]]

"The MCX Service identities shall be part of the MCX Service application service domain."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.12-004]]

"The MCX Service identities shall form the basis of the MCX Service application layer security for the MCX Service."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.12-005]]

"The MCX Service shall provide the MCX User with a mechanism to perform a single authentication for access to all authorized features."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.12-006]]

"The MCX Service shall provide a means for an authorized MCX UE to access selected MCX Service features prior to MCX User authentication."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.12-007]]

"The MCX Service shall require authentication of the MCX User before service access to all authorized MCX Service features is granted.

NOTE: The MCX Service features available are based on the authenticated user identity(s)."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.12-008]]

"Subject to regulatory constraints, the MCX Service shall provide a means to support confidentiality, message integrity, and source authentication for some information exchanges (e.g., MCX Service User Profile management, kill commands) that have the potential to disrupt the operation of the target MCX UE."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.12-009]]

"The MCX Service shall provide a means to support end-to-end security for all media traffic transmitted between MCX UEs."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.12-010]]

"End-to-end security shall be supported both within and without network coverage and regardless of whether the traffic is transmitted directly or via the network infrastructure."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.12-011]]

"Subject to regulatory constraints, the MCX Service shall provide a cryptographic key management service(s)."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.12-012]]

"The cryptographic key management service(s) shall support both pre-provisioning and over-the-air provisioning of cryptographic keys."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.12-013]]

"The cryptographic key management service(s) shall ensure that cryptographic keys are confidentiality protected, integrity protected and authenticated when delivered over-the-air."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.12-014]]

"The MCX Service shall provide end-to-end confidentiality and integrity protection to the MCX User Profile when transferred to and/or from and while stored on an MCX Server, an MCX UE or both."

6.3.3 Test description

6.3.3.1 Pre-test conditions

System Configuration:

- The system is configured as shown in Figure 4.1-4.
- Security policy database (SPD) is defined and all required values for authentication header (AH) and Encapsulating Payload (ESP). Pre Shared Keys (PSK) are agreed between UE (MCx client) and MCx Service system.
- For generation of the keys the Internet Key Exchange (IKE) shall be used. Keys and required passwords are specified.

IUT:

- IUT (UE1).
- The IUT (UE1) consists of all sub-systems of the MCx user service including the ability for floor control, pre-arranged group communication functionality and ad-hoc group communication functionality. Ensure the correct configuration.
- The UE1 is the acting user.

6.3.3.2 Test procedure sequence

Table 6.3.3.2-1: Main behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
	Check: configuration (as file or user settings entry) with multiple FA is available for the UE1. The configuration contains numerical and organizational arrangements of FA and flags for expiration timer, priority and location. MCx Service user accepts the configuration	--		-	--
1	Check: UE1 initiate IPSec Phase 1 for providing policy proposal (start ISAKMP SA negotiation)	-->	UE sends ISAKMP message with proposal; Identity Protection (Main Mode) regarding IETF RFC 2408 [23]		P
2	SIP core on MCx_S1 responds Identity Protection (Main Mode) request from UE with its own policy proposals (respond ISAKMP SA negotiation)	<--	SIP core responds ISAKMP message with proposal; Identity Protection (Main Mode) regarding IETF RFC 2408 [23]		
3	Check: UE1 starts key exchange procedure (Phase 2); UDP message format Identity Protection (Main Mode); init other SA negotiation	-->	UE sends ISAKMP message with proposal; Identity Protection (Main Mode) regarding IETF RFC 2408 [23]		P
4	SIP core on MCx_S1 responds Identity Protection (Main Mode) request from UE (respond other SA negotiation)	<--	SIP core responds ISAKMP message with proposal; Identity Protection (Main Mode)		
5	Check: UE1 starts authentication request; UDP message format Identity Protection (Main Mode); Other (KE, ID, etc.)	-->	UE sends ISAKMP message with authentication request; Identity Protection (Quick Mode) regarding IETF RFC 2408 [23]		P
6	SIP core on MCx_S1 confirms authentication request with message format Identity Protection (Main Mode)	<--	SIP core confirms ISAKMP message with own ID; Identity Protection (Quick Mode)		
7	Check: UE1 start tunnel protected by ESP or AH; establishment via Quick Mode	-->	UE sends ISAKMP message with tunnel protection mechanism; (Main Mode) regarding IETF RFC 2408 [23]		P
8	SIP core on MCx_S1 confirms tunnel establishment request with message format (Quick Mode);	<--	SIP core sends ISAKMP message for tunnel establishment (Quick Mode)		
9	Check: UE1 starts TLS request via TLSv1.3 handshake; sending "Client Hello"	-->	UE sends TLS handshake with Cipher Suite proposal according to IETF RFC 8446 [24] including supported ciphers, random number and session ID		P
10	MCx_S1 responds with TLS message "Server Hello"	<--	Server sends TLS handshake "Server Hello" including certificate and "Hello Done" information including selected ciphers, random number and session ID		
11	Check: UE1 initiates client key exchange procedure	-->	UE sends request for "Cipher Key Exchange" with encrypted premaster secret		P
12	MCx_S1 confirms cipher change	<--	Server confirms the cipher key exchange procedure with Cipher specification exchange finished		

6.3.3.3 Specific message contents

Table 6.1.3.3-1: ISAKMP from UE1 (step 1, Table 6.2.3.2-1; IETF RFC 2408 [23])

Derivation Path: IETF RFC 2408 [23]; condition ISAKMP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Internet Security and Key Management Protocol	**Payload**	**Main Mode**		
Payload				
Security Association (1)	IPSEC			
Proposal (2)	ISAKMP			
Transform (3)/ IKE Attribute	KEY_IKE			
Encryption-Algorithm	AES-CBC			
Key-Length	128			
Hash-Algorithm	SHA2-256			
Group-Description	1024 MODP			
Authentication-Method	Pre-shared key			
Life-Type	Seconds			
Life-Duration	86 400			

Table 6.1.3.3-2: ISAKMP from UE1 (step 3, Table 6.2.3.2-1; IETF RFC 2408 [23])

Derivation Path: IETF RFC 2408 [23]; condition ISAKMP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Internet Security and Key Management Protocol	**Payload**	**Main Mode**		
Payload				
Key Exchange (4)	IPSEC			
Key Exchange Data	<key>			
Payload				
Vendor ID (13)	Dead Peer Detection		IETF RFC 3706 [36]	
Vendor ID (13)	XAUTH			

Table 6.1.3.3-3: ISAKMP from UE1 (step 5, Table 6.2.3.2-1; IETF RFC 2408 [23])

Derivation Path: IETF RFC 2408 [23]; condition ISAKMP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Internet Security and Key Management Protocol	**Payload**	**Main Mode**		
Payload				
Key Exchange (4)	IPSEC			
Key Exchange Data	<key>			
Payload				
Vendor ID (13)	Dead Peer Detection		IETF RFC 3706 [36]	
Vendor ID (13)	XAUTH			

Table 6.1.3.3-4: ISAKMP from UE1 (step 7, Table 6.2.3.2-1; IETF RFC 2408 [23])

Derivation Path: IETF RFC 2408 [23]; condition ISAKMP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Internet Security and Key Management Protocol	**Payload**	**Main Mode**		
Next Payload: Identification (5)	HASH Data			

Table 6.1.3.3-5: TLS from UE1 (step 9, Table 6.2.3.2-1; IETF RFC 8446 [24])

Derivation Path: IETF RFC 8446 [24]; condition TLS				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Secure Socket Layer	**Payload**			
TLSv1.3 Record Layer				
Content Type	Handshake (22)			
Handshake Protocol	Client Hello			
Handshake Type	Client Hello (1)			
Cipher Suites (No. of suites)	TLS_RSA_WITH_ <type>	Supported cipher suites		

Table 6.1.3.3-6: TLS from UE1 (step 11, Table 6.2.3.2-1; IETF RFC 8446 [24])

Derivation Path: IETF RFC 8446 [24]; condition TLS				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Secure Socket Layer	**Payload**			
TLSv1.3 Record Layer	**Handshake Protocol**			
Content Type	Handshake (22)			
Handshake Protocol	Client Key Exchange			
Handshake Type	Client Key Exchange (16)			
Diffie-Hellman Client Params	Pubkey	<key>		
TLSv1.3 Record Layer	**Change Cipher Spec Protocol**			
Content Type	Change Cipher Spec (20)			
Change Cipher Spec Message				
TLSv1.3 Record Layer	**Handshake Protocol**			
Content Type	Handshake (22)			
Handshake Protocol	Encrypted Handshake Message			

6.4 Modifying the participants list of on-going ad hoc group communication

6.4.1 Test purpose

(1)

```
with {MCPTT user clients on different partner MCx Service systems meet the criteria to participate
an ad hoc group communication}
ensure that {
when {MCPTT client meets the criteria for adding to an ongoing ad hoc group communication}
then {MCPTT client will be successfully added to the ongoing ad hoc group communication
independently to which partner MCx Service system it belongs}
else when {MCPTT client is in ongoing ad hoc group communication and does not meet the criteria
successfully}
then {MCPTT client will be removed by the originating MCx Service system}
```

6.4.2 Conformance requirements

[ETSI TS 123 379 [5] (3GPP TS 23.379), clause 10.19.3.2.6]

"MCPTT user modifying the ad hoc group call participants procedure involving multiple MCPTT systems."

6.4.3 Test description

6.4.3.1 Pre-test conditions

System Configuration:

- MCx Service system consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (UE1) also consists of all sub-systems of the MCx user service.
- For the underlying "transport bearer" over which the UE1 (MCx Service system) and the IUT will communicate Parameters are set to the default parameters specific for the basic E-UTRA single simulated cells cell network scenarios, as defined in ETSI TS 138 508-1 [6] (3GPP TS 38.508-1), clause 6.3. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCx operation in the MCx configuration document).
- MCx Service system is configured to support the following features:
  - Ad-hoc Group Communication supporting criterion-based functionality; UE1 is allowed to change the list of participants (mandatory).
  - UE2 is not part of the initial list based ad hoc group communication.

IUT:

- IUT (UE1).
- The IUT (UE1) consists of all sub-systems of the MCx user service including the ability for floor control, pre-arranged group communication functionality and ad-hoc group communication functionality. Ensure the correct configuration.
- The UE1 is the acting user.

Preamble:

- The IUT UE1 is connected to PLMN of the primary 5GS.
- The MCx Service system (MCx_S1) is connected to the IUT, UEa1 (MCx client with restrictions)], UE2 (MCx client) and UE-Con (MCx client with controller/dispatcher functionality).

6.4.3.2 Test procedure sequence

Table 6.4.3.2-1: Main behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
	Check: configuration (as file or user settings entry) with multiple FA is available for the UE1. The configuration contains numerical and organizational arrangements of FA and flags for expiration timer, priority and location. MCx Service user accepts the configuration	--		-	--
	EXCEPTION: Procedure 'MCX CO communication' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1) clause 5.4.3	-		-	--
	EXCEPTION: Procedure 'Initial MCX Authentication, Registration, Configuration and Subscription' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1) clause 5.3.2	-		-	--

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Execute MCX Group Affiliation Status Change as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1) clause 5.3.34. No FA is active at the MCx Service system. Check: UE1 publish the basic POC settings	-->	SIP PUBLISH according ETSI TS 124 229 [8] (3GPP TS 24.229), IETF RFC 3903 [17]	1	P
2	MCx_S1 confirm the POC settings	<--	STATUS 200 OK (PUBLISH) according ETSI TS 124 229 [8] (3GPP TS 24.229)	-	--
3	Check: UE1 announce interest for a specific FA to the MCx_S1 (Interrogation, entrance to a specific list); UE1 is not allowed to use interrogation	-->	SIP SUBSCRIBE according ETSI TS 124 229 [8] (3GPP TS 24.229), IETF RFC 3856 [9] and IETF RFC 6665 [10]	1	P
4	Check: MCx_S1 responds with a STATUS (see note 1) message	<--	SIP STATUS (see note 1) 200 OK	-	--
5	Check: UE1 send PUBLISH message containing an automatic FA information	-->	SIP PUBLISH	1	P
6	Check: MCx_S1 responds with a STATUS message	<--	SIP STATUS (see note 1) 200 OK	-	--
7	Check: UE1 is allowed to participate on ad hoc group communication based on criterion <allow-emergency-adhoc-group-call> in the <ruleset> of the client profile	-->	- none - Client checks the ruleset configuration according to ETSI TS 124 379 [21] (3GPP 24.379) clause 6.2.8.1.24 is set to "true"	1	P
8	Check: UE1 initiates ad hoc group communication based (criterion: manually) {InitiateGroup, ParticipantList}	-->	SIP INVITE according to ETSI TS 124 379 [21] (3GPP TS 24.379) clause 17.2.2.1.1	1	P
9	System validates authorization of UE1	--	DB/AAA check → VALID: ACK {Group Established, GroupID} → ACK	-	--
10	UE1 receives confirmation from MCx_S1	<--	SIP ACK	-	--
11	Session preparation phase. MCx_S1 sends NOTIFICATION	<--	RTCP (MPTC) Group Dynamic NOTIFY {InitStatus, QueuePolicy, HangTime, Mode} → All UEs	-	--
12	Check: MCx_S1 invites UE-con into the ongoing ad hoc group communication	-->	SIP INVITE	1	P
13	UE-con responds to the INVITE request	-->	SIP STATUS 100 (TRYING)	1	P
14	OPTIONAL: outgoing response to TRYING if no automatic answering mode is activated	-->	SIP STATUS 180 (RINGING)	1	P
15	Check: UE-con call established confirmation towards MCx_S1	-->	SIP ACK	1	P
16	{Group Established, GroupID} → ACK Session preparation phase. MCx_S1 sends NOTIFICATION	--	RTCP (MPTC) Group Dynamic NOTIFY {InitStatus, QueuePolicy, HangTime, Mode} → All UEs	-	--
17	Floor activation by one participant to ensure the usage of the call time limit expiry. Check: UE1 send floor REQUEST message	--	RTCP (MCPT) Floor REQUEST; message containing field {Floor priority=0}	-	--
18	Check: UE1 add participant UE2 to the resource list (see note 2). UE send re-INVITE	-->	SIP re-INVITE according to ETSI TS 124 379 [21] (3GPP TS 24.379), clause 17.6.2	1	P
19	UE1 receive ACK response from MCx_S1	<--	SIP ACK	-	--
20	UE2 receive INVITE message from MCx_S1	<--	SIP INVITE	-	--
21	UE2 responds to the INVITE request	-->	SIP STATUS 100 (TRYING)	1	P
22	OPTIONAL: outgoing response to TRYING if no automatic answering mode is activated	-->	SIP STATUS 180 (RINGING)	1	P
23	Check: UE-con call established confirmation towards MCx_S1	-->	SIP ACK	1	P
24	{Group Established, GroupID} → ACK Session preparation phase. MCx_S1 sends NOTIFICATION	--	RTCP (MPTC) Group Dynamic NOTIFY {InitStatus, QueuePolicy, HangTime, Mode} → All UEs	-	--
25	Floor activation by one participant to ensure the usage of the call time limit expiry. Check: UE1 send floor REQUEST message	--	RTCP (MCPT) Floor REQUEST; message containing field {Floor priority=0}	-	--

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
26	Check: UE1 remove participant UE2 from the resource list (see note 2). UE send re-INVITE	-->	SIP re-INVITE	1	P
27	UE1 receive ACK response from MCx_S1	<--	SIP ACK	-	--
28	UE2 receive BYE message from MCx_S1	<--	SIP BYE	-	--
29	UE2 responds to the BYE request	-->	SIP STATUS 200 (OK)	1	P
30	{Group Established, GroupID} → ACK Session preparation phase. MCx_S1 sends NOTIFICATION	--	RTCP (MPTC) Group Dynamic NOTIFY {InitStatus, QueuePolicy, HangTime, Mode} → All UEs	-	--
31	Floor activation by one participant to ensure the usage of the call time limit expiry. Check: UE1 send floor REQUEST message	--	RTCP (MCPT) Floor REQUEST; message containing field {Floor priority=0}	-	--
32	Check: UE1 (initiator) terminates ad hoc group communication sending BYE	-->	SIP BYE	1	P
33	MCx_S1 confirms the received BYE message	<--	SIP STATUS 200 (OK)	-	--
34	Check: MCx_S1 terminates the ad hoc group communication; UE-con receive BYE message	<--	SIP BYE	-	--
35	UE-con confirms the received BYE message	-->	SIP STATUS 200 (OK)	1	P
NOTE 1: May notify the answer state to the user (i.e. "Unconfirmed" or "Confirmed") if received in the P-Answer-State header field. NOTE 2: The application on the UE GUI shall indicate a menu for adding or removing participants via user action.					

6.4.3.3 Specific message contents

Table 6.4.3.3-1: SIP re-INVITE (list based) from UE1
(step 18, Table 6.4.3.2-1; ETSI TS 137 579-4 [7] (3GPP TS 37.579-4) Table 5.4.3.3-6)

Derivation Path: Table 6.4.3.2-1, condition MCPTT-SDS, re-INVITE				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Method	**"re-INVITE"**			
Request-URI	re-INVITE SIP/2.0	MCPTT originating user public ID		MCPTT
Message-body			IETF RFC 3261 [16]	
MIME Multipart Media Encapsulation		Type: multipart/mixed		
application/sdp	Not present			
application/vnd.3gpp.mcptt-info+xml	<mcptt-Params>			
<session-type>	Ad hoc	Type: ad hoc group communication		
<mcptt-request-uri>				
<mcptt-client-id>				
Application/resource-lists+xml	<resource-lists>			
<lists>/<add>	<entry>	List entries		
<lists>/<remove>	<entry>	List entries		

6.5 Ad hoc group emergency alert support involving one MC system

6.5.1 Test purpose

(1)

```
with {Number of parallel allowed MCX Service ad hoc group emergency alerts}
ensure that {
    when {MCX Service administrator configures the number of parallel ad hoc group emergency alerts}
    then {The MCX service system supports the permission settings for MCX Service users
participating ad hoc emergency group alerts: (none or one or maximum number)}
            }
```

(2)

```
with {MCX Service ad hoc group emergency alert termination}
ensure that {
    when {MCX Service emergency ad hoc group communication or ad hoc group emergency alert is
terminated by authorized MCX Service user}
    then {In-progress Emergency condition for ad hoc group communication shall be cancelled or ad
hoc group emergency alert is terminated and if no other ad hoc group call is active the list of
participants is not persistent}
            }
```

(3)

```
with {MCX Service user can initiate an MCX Service ad hoc Emergency Group alert}
ensure that {
    when {MCX Service user is enabled for initiating an ad hoc group emergency alert}
    then {MCX UE send MCX Service ad hoc emergency alert and the responsible user change state to
MCX Service emergency and authorized MCX Service user can set the ad hoc group emergency alert into
an ad hoc group communication and MCX Service ad hoc group emergency alert is distributed to the
determined list of participants and MCX Service emergency alert shall contain location information,
MCX Service user ID, functional alias, criteria for determining list of participants, available MCX
Services, additional information related to the alert, and the user's Mission Critical organization
name}
    else when {MCX Service user is disabled for initiating an ad hoc group emergency alert}
    then {MCX user receive the reason for denial}
            }
```

(4)

```
with {MCX Service system shall provide the list of participants of an ad hoc group emergency alert}
ensure that {
    when {The MCX Service administrator or the MCX Service system changes the list of participants
of an ad hoc group emergency alert (e.g. on request or on location information provided by a MCX
Service user}
    then {The authorized MCX Service user requests the determined list of participants of an ad hoc
group emergency alert and the MCX Service system monitors the changes on the list (e.g. periodically
on timer or on specific trigger) and provide an update on any change and if necessary, the MCX
Service system performs retrials to involve changed list participants into the ad hoc group
emergency alert}
            }
```

6.5.2 Conformance requirements

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.4.2-007a]]

"Depending on meaningful elements of a functional alias the MCX Service shall be able to restrict or allow MCX Users to be involved in more than one MCX Service Emergency Group Communication or more than one MCX Service Ad hoc Group Emergency Alert at a time."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.6.2.4.1-003]]

"The MCX Service Emergency Alert shall contain the following information: Location, MCX Service User ID and MCX Service Group ID (i.e., user's selected group or dedicated MCX Service Emergency Group, as per group configuration) and the user's Mission Critical Organization name."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.6.2-001]]

"The MCX Service shall support an MCX Service Ad hoc Group Emergency Alert capability, which on initiation by an MCX User causes that MCX UE to send an MCX Service Ad hoc Group Emergency Alert and may put that MCX User into the MCX Service Emergency State."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.6.2-002]]

"The MCX Service shall provide a means for an authorized user to be able to activate the MCX Service Ad hoc Group Emergency Alert capability."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.6.2-002a]]

"The MCX Service shall provide the reason for denial to an MCX user who was not authorised to activate the MCX Service Ad hoc Group Emergency Alert."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.6.2-003]]

"The MCX Service Emergency Alert shall contain the following information: Location information, MCX Service User ID, Functional Alias, Criteria for determining list of participants, available MCX Services, additional information related to the alert, and the user's Mission Critical Organization name."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.6.2-004]]

"The MCX Service shall provide a mechanism for the initiator of an MCX Service Ad hoc Group Emergency Alert to request that the list of participants is to be determined and updated by the MCX Service system using pre-defined criteria."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.6.2-004a]]

"The MCX Service shall provide a mechanism for an authorised user to update the pre-defined criteria during an on-going Ad hoc Group Emergency Alert."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.6.2-005]]

"The MCX Service Ad hoc Group Emergency Alert shall be distributed to the list of participants determined by the MCX Service system."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.6.2-005a]]

"When the list of participants is determined and updated by the MCX Service system, the MCX Service shall provide notifications to the relevant participants and authorized users."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.6.2-005b]]

"When the list of participants is determined or updated by the MCX Service system, the MCX Service shall provide a mechanism that monitors and ensures that the participants list is applied for MCX Service Ad hoc Group Emergency Alert, performing retries when needed."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.6.2-006]]

"The MCX Service shall support MCX Service Ad hoc Group Emergency Alert cancellation by authorized MCX Users."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.6.2-007]]

"The MCX Service shall provide a mechanism for an authorised user which is participant of an active MCX Service Ad hoc Group Emergency Alert to set up group communications using the ad hoc group."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.6.2-008]]

"When an MCX Service Ad hoc Group Emergency Alert is cancelled and ongoing calls in the ad hoc group are terminated the ad hoc group shall not persist."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-8.8.8.5-001]]

"The In-progress Emergency condition for a MCX Service Ad hoc Group Communication shall be cancelled when the communication is terminated."

[ETSI TS 123 280 [4] (3GPP TS 23.280)], clause 10.10.3]

"The following sub clauses specify the procedures for ad hoc group emergency alert initiation, ad hoc group emergency alert cancel, entering an ongoing ad hoc group emergency alert and leaving an ad hoc group emergency alert that are utilised by the following MC services:

- *MCPTT*
- *MCVideo*
- *MCData".*

6.5.3 Test description

6.5.3.1 Pre-test conditions

System Configuration:

- MCx Service system consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (UE1) also consists of all sub-systems of the MCx user service.
- For the underlying "transport bearer" over which the UE1 (MCx Service system) and the IUT will communicate Parameters are set to the default parameters specific for the basic E-UTRA single simulated cells cell network scenarios, as defined in ETSI TS 138 508-1 [6] (3GPP TS 38.508-1) clause 6.3. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCx operation in the MCx configuration document).
- MCx Service system is configured to support the following features:
  - Ad-hoc Group Emergency Alert supporting functionality.
- Supporting user UEa1, UEb1, UEc1, UE2 and UE-con.

IUT:

- IUT (UE1).
- The IUT (UE1) consists of all sub-systems of the MCx user service including the ability for floor control, pre-arranged group communication functionality and ad-hoc group communication functionality. Ensure the correct configuration.
- The UE1 is the acting user.

Preamble:

- The IUT UE1 is connected to PLMN of the primary 5GS.
- The MCx Service system (MCx_S1) is connected to the IUT, UEa1 (MCx client with restrictions)], UEa1, UEb1, UEc1, UE2 (MCx client) and UE-Con (MCx client with controller/dispatcher functionality).
- UEa1 and UEb1 are not part of the initial ad hoc emergency alert but can be added.

6.5.3.2 Test procedure sequence

Table 6.5.3.2-1: Main behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
	Check: configuration (as file or user settings entry) with multiple FA is available for the UE1. The configuration contains numerical and organizational arrangements of FA and flags for expiration timer, priority and location. MCx Service user accepts the configuration	--		-	--
1a	Check: MCPTT Administrator configures permissions for the number of parallel ad hoc group emergency alerts for UE1 on the MCx_S1; access via HTML web interface; maximum number of parallel ad hoc group emergency alerts set to "MaxAhGEA"=0	-->	HTTPS link to configuration interface	-	--
2a	Check: MCPTT Administrator configures permissions for the number of parallel ad hoc group emergency alerts for UE1 on the MCx_S1; access via XML configuration file interface; ; maximum number of parallel ad hoc group emergency alerts set to "MaxAhGEA"=0	-->	XML access (e.g. via SCP or SFTP connection)	-	--
1b	Check: MCPTT Administrator configures permissions for the number of parallel ad hoc group emergency alerts for UE1 on the MCx_S1; access via HTML web interface; maximum number of parallel ad hoc group emergency alerts set to "MaxAhGEA"=3 (see note 1)	-->	HTTPS link to configuration interface	-	--
2b	Check: MCPTT Administrator configures permissions for the number of parallel ad hoc group emergency alerts for UE1 on the MCx_S1; access via XML configuration file interface; maximum number of parallel ad hoc group emergency alerts set to "MaxAhGEA"=3 (see note 1)	-->	XML access (e.g. via SCP or SFTP connection)	-	--
	EXCEPTION: Procedure 'MCX CO communication' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1) clause 5.4.3	-		-	--
	EXCEPTION: Procedure 'Initial MCX Authentication, Registration, Configuration and Subscription' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1) clause 5.3.2	-		-	--
3	Execute MCX Group Affiliation Status Change as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1) clause 5.3.34. No FA is active at the MCx Service system. Check: UE1 publish the basic POC settings	-->	SIP PUBLISH according ETSI TS 124 229 [8] (3GPP TS 24.229), IETF RFC 3903 [17]	1	P
4	MCx_S1 confirm the POC settings	<--	STATUS 200 OK (PUBLISH) according ETSI TS 124 229 [8] (3GPP TS 24.229)	-	--
5	Check: UE1 announce interest for a specific FA to the MCx_S1 (Interrogation, entrance to a specific list); UE1 is not allowed to use interrogation	-->	SIP SUBSCRIBE according ETSI TS 124 229 [8] (3GPP TS 24.229), IETF RFC 3856 [9] and IETF RFC 6665 [10]	1	P
6	Check: MCx_S1 responds with a STATUS message	<--	SIP STATUS 200 (OK)	-	--
7	Check: UE1 send PUBLISH message containing an automatic FA information	-->	SIP PUBLISH	1	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
8	Check: MCx_S1 responds with a STATUS message	<--	SIP STATUS 200 (OK)	-	--
9	Check: UE1 is allowed to participate on ad hoc group emergency alert based on criterion <allow-adhoc-emergency-alert> in the <ruleset> of the client profile	-->	- none - Client checks the ruleset configuration according to ETSI TS 124 379 [21] (3GPP TS 24.379) clause 6.2.8.1.24 is set to "true"	1	P
10	Check: UE1 activates ad hoc group emergency alert (see note 2) #1	-->	SIP MESSAGE according to ETSI TS 124 379 [21] (3GPP TS 24.379) clause 17.2.2.1.1	1	P
11	System validates authorization of UE1.	-	check → VALID: 200	-	--
12	UE1 receives confirmation from MCx_S1	<--	SIP STATUS 200 (OK)	-	--
13	Check: For the initiator the ad hoc group emergency alert automatically is confirmed. Other participants need to confirm automatically or by manually action		- none -		
14a	Check: UEa1 activates ad hoc group emergency alert (see note 2) #2	-->	SIP MESSAGE according to ETSI TS 124 379 [21] (3GPP TS 24.379) clause 17.2.2.1.1	1	P
15a	System validates authorization of UEa1.	-	check → VALID: NOK; MaxAhGEA=0 (no parallel ad hoc group emergency alert)	-	--
16a	UE1 receives confirmation from MCx_S1	<--	SIP STATUS 403 (FORBIDDEN)	-	--
14b	Check: UEa1 activates ad hoc group emergency alert (see note 2) #2	-->	SIP MESSAGE according to ETSI TS 124 379 [21] (3GPP TS 24.379) clause 17.2.2.1.1	1	P
15b	System validates authorization of UEa1.	-	check → VALID: 200	-	--
16b	UEa1 receives confirmation from MCx_S1	<--	SIP STATUS 200 (OK)	-	--
17b	Check: UEb1 activates ad hoc group emergency alert (see note 2) #3	-->	SIP MESSAGE according to ETSI TS 124 379 [21] (3GPP TS 24.379) clause 17.2.2.1.1	1	P
18b	System validates authorization of UEb1.	-	check → VALID: 200	-	--
19b	UEb1 receives confirmation from MCx_S1	<--	SIP STATUS 200 (OK)	-	--
17b	Check: UEb1 activates ad hoc group emergency alert (see note 2) #4	-->	SIP MESSAGE according to ETSI TS 124 379 [21] (3GPP TS 24.379) clause 17.2.2.1.1	1	P
18b	System validates authorization of UEc1.	-	check → VALID: NOK; MaxAhGEA=3 (no parallel ad hoc group emergency alert)	-	--
19b	UEc1 receives confirmation from MCx_S1	<--	SIP STATUS 403 (FORBIDDEN)	-	--
20b	Check: UE1 shows three different ad hoc group emergency alerts	-->	Visual check on the UE1 GUI	1	P
21	Check: UE1 deactivates ad hoc group emergency alert (see note 2) #1	-->	SIP MESSAGE according to ETSI TS 124 379 [21] (3GPP TS 24.379) clause 17.2.2.1.1	2	P
22	UE1 receives confirmation from MCx_S1	<--	SIP STATUS 200 (OK)	-	--
23	Check: UE-con deactivates ad hoc group emergency alerts #2 and #3			-	--
24	UE-con receives confirmation from MCx_S1 for each, ad hoc group emergency alert #2 and #3	<--	SIP STATUS 200 (OK)	-	--
25	Check: no list of participants is available on UE1 after ad hoc group emergency alert was terminated.	-->	Visual check on the UE1 GUI	2	P
26	Check: UE-con activates ad hoc group emergency alert (see note 2)	-->	SIP MESSAGE	2	P
27	UE-con receives confirmation from MCx_S1	<--	SIP STATUS 200 (OK)	-	--
28	UE1 receives ad hoc group emergency alert activation from MCx_S1	<--	SIP MESSAGE	2	

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
29	Check: UE1 confirms the receipt of the SIP MESSAGE for activation of the ad hoc group emergency alert (no alert confirmation)	-->	SIP STATUS 200 (OK)	2	P
30	Check: UE1 confirm ad hoc group emergency alert activation (see note 3)	-->	SIP MESSAGE	2	P
31	UE1 receives confirmation from MCx_S1	<--	SIP STATUS 200 (OK)	-	--
32	UE2 receives ad hoc group emergency alert activation from MCx_S1	<--	SIP MESSAGE	-	--
33	Check: UE2 confirms the receipt of the SIP MESSAGE for activation of the ad hoc group emergency alert (no alert confirmation)	-->	SIP STATUS 200 (OK)	-	--
34	Check: UE2 confirm ad hoc group emergency alert activation	-->	SIP MESSAGE	-	--
35	UE2 receives confirmation from MCx_S1	<--	SIP STATUS 200 (OK)	-	--
36	UE-con deactivates the ad hoc group emergency alert				

NOTE 1: MaxAhGEA=3 simulates the maximum value for parallel ad hoc emergency alerts.
NOTE 2: The application on the UE GUI shall indicate a menu for editing the participant lists via user action. This may include:
- Adding, renaming and remove a list.
- Adding, renaming, repositioning and removing of participants in a list.
- Save an ad hoc participants list during ongoing ad hoc group communication call.

NOTE 3: The user GUI shall change the visualization between active, unconfirmed alert and confirmed alert e.g. like:
- Unconfirmed: alert tone + blinking indication on main screen.
- Confirmed: static information in the status field.

6.5.3.3 Specific message contents

Table 6.5.3.3-1: SIP MESSAGE from UE1
(step 10, Table 6.5.3.2-1; ETSI TS 137 579-4 [7] (3GPP TS 37.579-4) Table 5.5.2.7.2-1)

Derivation Path: ETSI TS 137 579-4 [7] (3GPP TS 37.579-4) clause 5.5.2.7.2; condition MCPTT-INFO-EVENT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Method	**"MESSAGE"**			
Message-body			IETF RFC 3261 [16]	
application/mcptt-info+xml		Type: multipart/mixed		
<mcptt-request-uri>				
<mcptt-calling-user-id>				
<mcptt-calling-group-id>				
<any-ext>				
<adhoc-emergency-alert-ind>	true			
<location-info>	<report>			
<call-participants-criterias>	active			
<adhoc-alert-participant-list>	<entry>			
<adhoc-alert-participant-not-ack-list>	<entry>			

Table 6.5.3.3-2: SIP MESSAGE from UE1
(step 14, Table 6.5.3.2-1; ETSI TS 137 579-4 [7] (3GPP TS 37.579-4) Table 5.5.2.7.2-1)

Derivation Path: ETSI TS 137 579-4 [7] (3GPP TS 37.579-4) clause 5.5.2.7.2; condition MCPTT-INFO-EVENT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Method	**"MESSAGE"**			
Message-body			IETF RFC 3261 [16]	
application/mcptt-info+xml		Type: multipart/mixed		
<any-ext>				
<adhoc-emergency-alert-ind>	false			
<location-info>	<report>			

Table 6.5.3.3-3: SIP MESSAGE from UE1
(step 20, Table 6.5.3.2-1; ETSI TS 137 579-4 [7] (3GPP TS 37.579-4) Table 5.5.2.7.2-1)

Derivation Path: ETSI TS 137 579-4 [7] (3GPP TS 37.579-4) clause 5.5.2.7.2; condition MCPTT-INFO-EVENT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Method	**"MESSAGE"**			
Message-body			IETF RFC 3261 [16]	
application/mcptt-info+xml		Type: multipart/mixed		
<any-ext>				
<adhoc-emergency-alert-ind>	true			
<location-info>	<report>			

6.6 Ad hoc group call and ad hoc group emergency alert involving multiple MC systems

6.6.1 Test purpose

(1)

```
with {Information provided by the initiator of an Ad hoc group communication}
ensure that {
    when {Initiation of an ad hoc group communication call or ad hoc group emergency alert by a MCX
Service user}
    then {The MCX Service user provide participants list for an Ad hoc group communication call or
ad hoc group emergency alert on procedures for call setup and call release and MCPTT server can
determine the participants list during Ad hoc group call setup and MCPTT server or an authorised
user can modify the participants list and authorized user can modify the ad hoc group communication
criteria}
            }
```

(2)

```
with {Releasing an Ad hoc group call by partner MCPTT system}

ensure that {

   when {The ad hoc group communication is active with participants involving different MCPTT
systems and the ad hoc group call shall be released by partner MCPTT server}

   then {Ad hoc group communication shall be terminated and the partner MCPTT system terminates the
designation of the participant, and a termination notification is received from all active
participants and MCPTT server determines the participants list}
```

6.6.2 Conformance requirements

[ETSI TS 123 379 [5] (3GPP TS 23.379)], clause 10.19.3.2.1]

"Procedure for ad hoc group call setup - Participants list provided by the Initiator."

[ETSI TS 123 379 [5] (3GPP TS 23.379)], clause 10.19.3.2.2]

"Procedure for ad hoc group call release by MCPTT server - Participants list provided by the Initiator."

[ETSI TS 123 379 [5] (3GPP TS 23.379)], clause 10.19.3.2.3]

"Ad hoc group call setup - Participants list determined by the MCPTT server."

[ETSI TS 123 379 [5] (3GPP TS 23.379)], clause [10.19.3.2.4]

"Modification of ad hoc group call participants by the MCPTT server."

[ETSI TS 123 379 [5] (3GPP TS 23.379)], clause 10.19.3.2.5]

"Release ad hoc group call and stop determining the ad hoc group call participants by partner MCPTT system - Participants list determined by the MCPTT server."

[ETSI TS 123 379 [5] (3GPP TS 23.379)], clause 10.19.3.2.6]

"Modification of ad hoc group call participants by an authorized user."

[ETSI TS 123 379 [5] (3GPP TS 23.379)], clause 10.19.3.2.7]

"Modification of ad hoc group call criteria by an authorized user."

6.6.3 Test description

6.6.3.1 Pre-test conditions

System Configuration:

- MCx Service system consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (UE1) also consists of all sub-systems of the MCx user service.
- For the underlying "transport bearer" over which the UE1 (MCx Service system) and the IUT will communicate Parameters are set to the default parameters specific for the basic E-UTRA single simulated cells cell network scenarios, as defined in ETSI TS 138 508-1 [6] (3GPP TS 38.508-1) clause 6.3. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCx operation in the MCx configuration document).
- MCx Service system is configured to support the following features:
  - Ad hoc Group Emergency Alert supporting functionality.
  - Ad hoc group communication call supporting functionality.
  - MCx Service system 1 (MCx_S1) and MCx Service system 2 (MCx_S2) are connected according to Figure 4.1-2.
- Supporting UE2, UE3 and UE-con:
  - UE2 is connected to MCx_S1.
  - UE3 is connected to MCx_S2 with permission to cancel or terminate ad hoc group emergency alerts or ad hoc group communication calls. UE3 can request list of participants.

IUT:

- IUT (UE1).
- The IUT (UE1) consists of all sub-systems of the MCx user service including the ability for floor control, pre-arranged group communication functionality and ad-hoc group communication functionality. Ensure the correct configuration.
- The IUT has the permission to cancel or terminate ad hoc group communication requests.

- UE1 and UE3 are the acting user.

Preamble:

- The IUT UE1 is connected to PLMN of the primary 5GS.
- The MCx Service system 1 (MCx_S1) is connected to the IUT, UE2 (MCx client) and UE-Con (MCx client with controller/dispatcher functionality).
- The MCx Service system 2 (MCx_S2) is connected to UE3.

6.6.3.2 Test procedure sequence

Table 6.6.3.2-1: Main behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
	EXCEPTION: Procedure 'MCX CO communication' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1) clause 5.4.3	-		-	--
	EXCEPTION: Procedure 'Initial MCX Authentication, Registration, Configuration and Subscription' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1) clause 5.3.2	-		-	--
	Execute MCX Group Affiliation Status Change as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1) clause 5.3.34. No FA is active at the MCx Service system. Check: UE1 publish the basic POC settings	-->	SIP PUBLISH according ETSI TS 124 229 [8] (3GPP TS 24.229), IETF RFC 3903 [17]	1	P
1	MCx_S1 confirm the POC settings	<--	STATUS 200 OK (PUBLISH) according ETSI TS 124 229 [8] (3GPP TS 24.229)	-	--
2	Check: UE1 announce interest for a specific FA to the MCx_S1 (Interrogation, entrance to a specific list); UE1 is not allowed to use interrogation	-->	SIP SUBSCRIBE according ETSI TS 124 229 [8] (3GPP TS 24.229), IETF RFC 3856 [9] and IETF RFC 6665 [10]	1	P
3	Check: MCx_S1 responds with a STATUS message	<--	SIP STATUS200 OK	-	--
4	Check: UE1 send PUBLISH message containing an automatic FA information	-->	SIP PUBLISH	1	P
5	Check: MCx_S1 responds with a STATUS message	<--	SIP STATUS 200 OK	-	--
6	Check: UE1 activates ad hoc group emergency alert involving UE2, UE3 and UE-con as list-based group initiation	-->	SIP MESSAGE according to ETSI TS 124 379 [21] (3GPP TS 24.379) clause 17.2.2.1.1	1	P
7	System validates authorization of UE1.	-	check → VALID: 200	-	--
8	UE1 receives confirmation from MCx_S1	<--	SIP STATUS 200 (OK)	-	--
9	UE-con receives ad hoc group emergency alert activation from MCx_S1	<--	SIP MESSAGE	-	--
10	UE-con confirms the receipt of the SIP MESSAGE for activation of the ad hoc group emergency alert (no alert confirmation)	-->	SIP STATUS 200 (OK)	-	--
11	MCx_S1 provides SIP MESSAGE for activating ad hoc group emergency alert to MCx_S2, MCx_S2 provide MESSAGE to UE3	-->	SIP MESSAGE	-	--
12	MCx_S2 confirms the receipt of the SIP MESSAGE for activation of the ad hoc group emergency alert (no alert confirmation)	-->	SIP STATUS 200 (OK)	-	--
13	UE3 receives ad hoc group emergency alert activation from MCx_S1	<--	SIP MESSAGE	-	--

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
14	UE3 confirms the receipt of the SIP MESSAGE for activation of the ad hoc group emergency alert (no alert confirmation)	-->	SIP STATUS 200 (OK)	1	P
16	Check: UE3 change list of participants and includes UE2 into the ad hoc group emergency alert	-->	SIP MESSAGE	1	P
17	MCx_S2 provides SIP MESSAGE for activating ad hoc group emergency alert to MCx_S1, MCx_S1 provide MESSAGE to UE2	-->	SIP MESSAGE	-	--
18	MCx_S1 confirms the receipt of the SIP MESSAGE for activation of the ad hoc group emergency alert (no alert confirmation)	-->	SIP STATUS 200 (OK)	-	--
19	UE2 receives ad hoc group emergency alert activation from MCx_S1	<--	SIP MESSAGE	-	--
20	UE2 confirms the receipt of the SIP MESSAGE for activation of the ad hoc group emergency alert (no alert confirmation)	-->	SIP STATUS 200 (OK)	-	--
21	Check: UE1 confirm ad hoc group emergency alert activation	-->	SIP MESSAGE	1	P
22	UE1 receives confirmation from MCx_S1	<--	SIP STATUS 200 (OK)	-	--
23	Check: UE1 deactivates ad hoc group emergency alert	-->	SIP MESSAGE according to ETSI TS 124 379 [21] (3GPP TS 24.379) clause 17.2.2.1.1	1	P
24	UE1 receives confirmation from MCx_S1	<--	SIP STATUS 200 (OK)	-	--
25	MCx_S2 provides SIP MESSAGE for deactivating ad hoc group emergency alert to MCx_S1, MCx_S1 provide MESSAGE to UE2	-->	SIP MESSAGE	-	--
26	MCx_S1 confirms the receipt of the SIP MESSAGE for deactivation of the ad hoc group emergency alert (no alert confirmation)	-->	SIP STATUS 200 (OK)	-	--
27	UE3 receives ad hoc group emergency alert deactivation from MCx_S1	<--	SIP MESSAGE	-	--
28	UE3 confirms the receipt of the SIP MESSAGE for deactivation of the ad hoc group emergency alert (no alert confirmation)	-->	SIP STATUS 200 (OK)	1	P
29	Check: all involved participants have changed back to idle mode	--		-	--
30	Check: UE1 initiates ad hoc group communication based (criterion: manual; list based) {InitiateGroup, ParticipantList}	-->	SIP INVITE according to ETSI TS 124 379 [21] (3GPP TS 24.379) clause 17.2.2.1.1	2	P
31	System validates authorization of UE1		DB/AAA check → VALID: 200 OK {Group Established, GroupID} → ACK	-	--
32	UE1 receives confirmation from MCx_S1	<--	SIP ACK	-	--
33	Check: MCx_S1 invites UE-con into the ongoing ad hoc group communication	-->	SIP INVITE	2	P
34 35	UE-con responds to the INVITE request	-->	SIP STATUS 100 (TRYING)	2	P
36	OPTIONAL: outgoing response to TRYING if no automatic answering mode is activated	-->	SIP STATUS 180 (RINGING)	2	P
37	Check: UE-con call established confirmation towards MCx_S1	-->	SIP ACK	2	P
38	Session preparation phase. MCx_S1 sends NOTIFICATION		RTCP (MPTC) Group Dynamic NOTIFY {InitStatus, QueuePolicy, HangTime, Mode} → All UEs	-	--
39	Floor activation by one participant to ensure the usage of the call time limit expiry. Check: UE1 send floor REQUEST message		RTCP (MCPT) Floor REQUEST; message containing field {Floor priority=0}		
40	Check: MCx_S1 forwards the INVITE for the ad hoc group communication to MCx_S2	-->	SIP INVITE	-	--

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
41	Check: MCx_S2 responds to the INVITE request	-->	SIP STATUS 100 (TRYING)	-	--
42	Check: UE3 receive INVITE from MCx_S2 for the ongoing ad hoc group communication	<--	SIP INVITE	2	P
43	UE3 responds to the INVITE request	-->	SIP STATUS 100 (TRYING)	2	P
44	Check: MCx_S2 updates the INVITE request from MCx_S1	-->	SIP STATUS 200 (OK)	-	--
45	OPTIONAL: UE3 sends outgoing response to TRYING if no automatic answering mode is activated	-->	SIP STATUS 180 (RINGING)	2	P
46	Check: UE3 call established confirmation towards MCx_S2	-->	SIP STATUS 200 (OK) {Group Established, GroupID} → ACK	2	--
47	Check: UE3 call established confirmation towards MCx_S2	-->	SIP ACK	2	P
48	Session preparation phase. MCx_S2 sends NOTIFICATION to MCx_S1 and UE3	-->	RTCP (MPTC) Group Dynamic NOTIFY {InitStatus, QueuePolicy, HangTime, Mode} → All UEs	-	--
49	Session preparation phase. MCx_S2 sends NOTIFICATION	-->	RTCP (MPTC) Group Dynamic NOTIFY {InitStatus, QueuePolicy, HangTime, Mode} → All UEs	-	--
50a	Check: UE3 terminates ad hoc group communication; sending BYE	-->	SIP BYE	2	P
51a	UE3 receive termination request confirmation from MCx_S2	<--	SIP STATUS 200 (OK)	-	--
52a	Session termination phase. MCx_S2 sends BYE to MCx_S1	-->	SIP BYE	-	--
53a	Check: MCx_S1 confirms the receipt of the termination request to MCx_S2	-->	SIP STATUS 200 (OK)	-	--
54a	Check: MCx_S1 sends BYE to UE1 and UE-con	-->	SIP BYE	-	--
55a	Check: UE1 and UE-con confirm the receipt of the termination request to MCx_S1	-->	SIP STATUS 200 (OK)	-	--
50b	Check: UE1 terminates ad hoc group communication; sending BYE	-->	SIP BYE	2	P
51b	UE1 receive termination request confirmation from MCx_S1	<--	SIP STATUS 200 (OK)	-	--
52b	Session termination phase. MCx_S1 sends BYE to MCx_S2	-->	SIP BYE	-	--
53b	Check: MCx_S2 confirms the receipt of the termination request to MCx_S1	-->	SIP STATUS 200 (OK)	-	--
54b	Check: MCx_S2 sends BYE to UE3	-->	SIP BYE	-	--
55b	Check: MCx_S1 sends BYE to UE-con	-->	SIP BYE	-	--
56b	Check: UE3 and UE-con confirm the receipt of the termination request	-->	SIP STATUS 200 (OK)	-	--

6.7 MCData subservices SDS and FD support involving one MC system

6.7.1 Test purpose

(1)

with {MCX Service system supporting MCData functionality SDS}
ensure that {
    when {SDS is initiated by an MCX Service user}
    then {The MCX Service user can enter the content of the message and size for the message is limited to a content payload of at least [1000] characters of 8 bit text or [500] characters of 16 bit text or [250] characters of 32 bit text and the necessary character encoding information **or** at least [1000] characters of hyperlink or interleaved text and hyperlink(s) to allow subsequent access to linked content (which may be a large file) **or** at least [1000] bytes of binary data to be used by a local running application and the necessary addressing detail to identify the intended application **and** SDS is provided to a MCX service user with policy assertion capabilities (e.g. policy to limit certain types of message or content to certain users due, for example, to location or user privilege)}
}

(2)

with {MCX Service system supporting MCData functionality SDS to a group of service affiliated members}
ensure that {
    when {SDS is initiated by an MCX Service user to a group}
    then {SDS is provided to a group service to affiliated members with policy assertion capabilities (e.g. certain types of messages or content may only be relevant to certain members of a group due, for example, to location)}
}

(3)

with {MCX Service system supporting MCData message thread indication functionality}
ensure that {
    when {MCData SDS functionality}
    then {The MCX Service user provides a message thread indication function so that multiple message flows can be managed independently **and** on reply to a message or any message coupled with a previous message (sent or received) the SDS message thread indication function use the same indication mechanism as for previous messages}
}

(4)

with {Additionally MCData SDS functionality configurable for the sender of an SDS}
ensure that {
    when {MCData SDS is configured for read confirmation **active**}
    then {The MCX Service user initiating a SDS can select a read confirmation tag and the activation includes automatically the presentation of the delivery status for the specific SDS add a delivery history can requested by an authorized user **and** the MCX Service user initiating a SDS containing no information or payload according (1) can select a location tag and system add the current location of the MCX UE and send the SDS}
else when {MCData SDS is configured for read confirmation **not active**}
then {The MCX Service user initiating a SDS is **not able** to select a read confirmation tag and the presentation of the delivery status for the specific SDS will not be delivered **and** the MCX Service user initiating a SDS containing no information or payload according (1) can select a location tag and system add the current location of the MCX UE and send the SDS}
}

(5)

with {MCX Service system provides file distribution functionality}
ensure that {
    when {MCX Service user distributes files to the MCData System}
    then {MCData file distribution capability provide a service to allow a user to send a file to any combination of individual users and/or affiliated groups **and** it provides an option for each recipient to reject **or** choose to receive the file or not (e.g. by storing the file and sending a link (URL) to all relevant members **and** the user can choose when to receive the file}
}

(6)

```
with {MCX Service system provides file distribution additionally functionality}
ensure that {
    when {MCX Service user distributes files to the MCData System}
    then {MCData file distribution capability provide a service to allow a user to send a file to
any combination of individual users and/or affiliated groups and it provides an option for each
recipient to reject or choose to receive the file or not (e.g. by storing the file and sending a
link (URL) to all relevant members and the user can choose when to receive the file and the sending
MCX Service user is able to indicate the file as immediately to send or mandatory for download and a
system level capability on the receiving UE invoke the download automatically by need or by
selection of a user and on download complete the MCData file distribution capability provides an
indication to each recipient successfully downloading the file and on availability of system
delivery efficiencies the MCData file distribution capability is able to distribute common
information to users within a specific geographical area at the same time and the user (sender) or
an authorized user is able to cancel the distribution of files which waiting to be sent or have been
sent but not have been delivered}
}
```

6.7.2 Conformance requirements

[ETSI TS 122 282 [3] (3GPP TS 22.282)], clause [R-5.2.2-001]]

"The MCData Service shall provide an SDS feature for conveyance of limited size, variable content, messages."

[ETSI TS 122 282 [3] (3GPP TS 22.282)], clause [R-5.2.2-002]]

"*The MCData SDS shall provide a group service to affiliated members with policy assertion capabilities (e.g. certain types of messages or content may only be relevant to certain members of a group due, for example, to location)."*

[ETSI TS 122 282 [3] (3GPP TS 22.282)], clause [R-5.2.2-003]]

"The MCData SDS shall provide a one-to-one service with policy assertion capabilities (e.g. policy to limit certain types of messages or content to certain users due, for example, to location or user privilege)."

[ETSI TS 122 282 [3] (3GPP TS 22.282)], clause [R-5.2.2-004]]

"The MCData SDS shall provide the option to include a content payload of at least [1000] characters of 8-bit text or [500] characters of 16-bit text or [250] characte32-bit text and the necessary character encoding information (for example to identify alphabet used)."

[ETSI TS 122 282 [3] (3GPP TS 22.282)], clause [R-5.2.2-005]]

"The MCData SDS shall provide the option to include a content payload of at least [1000] characters of hyperlink or interleaved text and hyperlink(s) to allow subsequent access to linked content (which may be a large file)."

[ETSI TS 122 282 [3] (3GPP TS 22.282)], clause [R-5.2.2-006]]

"The MCData SDS shall provide the option to include a content payload of at least [1000] bytes of binary data to be used by a local running application and the necessary addressing detail to identify the intended application."

[ETSI TS 122 282 [3] (3GPP TS 22.282)], clause [R-5.2.2-007]]

"The MCData SDS shall provide a message thread indication so that multiple message flows can be managed independently."

[ETSI TS 122 282 [3] (3GPP TS 22.282)], clause [R-5.2.2-008]]

"When replying to a message on the MCData SDS or sending any message which should be coupled with previously sent or received messages or message flows; the message thread indication shall use the same indication as was used for those previous messages."

[ETSI TS 122 282 [3] (3GPP TS 22.282)], clause [R-5.2.2-009]]

"The MCData SDS shall provide a selectable read receipt indication. When requested, the receiving entity shall provide receipt indication for delivered and read messages as appropriate.

NOTE 1: The read indication will implicitly indicate delivered as well."

[ETSI TS 122 282 [3] (3GPP TS 22.282)], clause [R-5.2.2-010]]

"The MCData SDS shall provide a configurable read receipt indication. When configured, the receiving entity shall provide receipt indication addressed to the application for delivered and read messages as appropriate.

NOTE 2: The read indication will implicitly indicate delivered as well."

[ETSI TS 122 282 [3] (3GPP TS 22.282)], clause [R-5.2.2-011]]

"The MCData SDS shall permit delivery history interrogation for suitably authorized users."

[ETSI TS 122 282 [3] (3GPP TS 22.282)], clause [R-5.2.2-012]]

"The MCData SDS shall provide the option to add a field indicating location of the sending user/UE."

[ETSI TS 122 282 [3] (3GPP TS 22.282)], clause [R-5.2.2-013]]

"The MCData SDS shall allow empty messages including only a field indicating location of the sending user/UE."

[ETSI TS 122 282 [3] (3GPP TS 22.282)], clause [R-5.3.2-001]]

"The MCData Service shall provide a file distribution capability."

[ETSI TS 122 282 [3] (3GPP TS 22.282)], clause [R-5.3.2-002]]

"The MCData file distribution capability shall provide a service to allow a user to send a file to any combination of individual users and/or affiliated groups."

[ETSI TS 122 282 [3] (3GPP TS 22.282)], clause [R-5.3.2-003]]

"The MCData file distribution capability shall provide an option for each recipient to choose to receive the file or not (e.g. by storing the file and sending a link (URL) to all relevant members)."

NOTE 1: By using this capability, a user can choose when to receive the file."

[ETSI TS 122 282 [3] (3GPP TS 22.282)], clause [R-5.3.2-004]]

"The MCData file distribution capability shall allow a user to reject to receive the file where appropriate."

[ETSI TS 122 282 [3] (3GPP TS 22.282)], clause [R-5.3.2-005]]

"The MCData file distribution capability shall provide a sending user selectable indication for mandatory download so that the UE, for all relevant receiving members, will automatically download the file.

NOTE 2: The mandatory indication provides a system level capability and could be invoked automatically by need of the application or might be provided for selection by the user."

[ETSI TS 122 282 [3] (3GPP TS 22.282)], clause [R-5.3.2-006]]

"The MCData file distribution capability shall provide download complete indications for each recipient successfully downloading the file."

[ETSI TS 122 282 [3] (3GPP TS 22.282)], clause [R-5.3.2-007]]

"The MCData file distribution capability shall allow the sender to select to send the file immediately to all chosen users."

[ETSI TS 122 282 [3] (3GPP TS 22.282)], clause [R-5.3.2-008]]

"The MCData file distribution capability shall make use of available system delivery efficiencies for distribution of common information to users within a specific geographic area and able to receive at the same time."

[ETSI TS 122 282 [3] (3GPP TS 22.282)], clause [R-5.3.2-009]]

"The MCData file distribution capability shall allow a user to cancel distribution of files they have sent but have not been delivered."

[ETSI TS 122 282 [3] (3GPP TS 22.282)], clause [R-5.3.2-010]]

"The MCData file distribution capability shall allow an authorised user to cancel distribution of files being sent or waiting to be sent."

6.7.3 Test description

6.7.3.1 Pre-test conditions

System Configuration:

- MCx Service system consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (UE1) also consists of all sub-systems of the MCx user service.
- For the underlying "transport bearer" over which the UE1 (MCx Service system) and the IUT will communicate Parameters are set to the default parameters specific for the basic E-UTRA single simulated cells cell network scenarios, as defined in ETSI TS 138 508-1 [6] (3GPP TS 38.508-1) clause 6.3. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCx operation in the MCx configuration document).
- MCx Service system is configured to support the following features:
  - MCData services SDS and File Distribution.
- Supporting MCx Service user UE2 and UE-con:
  - UE2 is connected to MCx_S1.
  - UE-con is able to receive and display SDS and FD.

IUT:

- IUT (UE1).
- The IUT (UE1) consists of all sub-systems of the MCx user service including the ability for floor control, pre-arranged group communication functionality and ad-hoc group communication functionality. Ensure the correct configuration.
- IUT supports MCData client functionality.
- UE1 is the acting user.

Preamble:

- The IUT UE1 is connected to PLMN of the primary 5GS.
- The MCx Service system 1 (MCx_S1) is connected to the IUT (MCData client) and UE-Con (MCData client with controller/dispatcher functionality).

6.7.3.2 Test procedure sequence

Table 6.7.3.2-1: Main behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
	EXCEPTION: Procedure 'MCX CO communication' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1) clause 5.4.3	-		-	--
	EXCEPTION: Procedure 'Initial MCX Authentication, Registration, Configuration and Subscription' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1) clause 5.3.37	-		-	--
1	Check: Execute MCX Affiliation Status Change as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1) clause 5.3.34. No FA is active at the MCx Service system. Check: UE1 publish the basic settings	-->	SIP PUBLISH according ETSI TS 137 579-1 [28] (3GPP TS 37.579-1), IETF RFC 3903 [17]	1	P
2	MCx_S1 confirm the basic settings	<--	STATUS 200 OK (PUBLISH) according ETSI TS 124 229 [8] (3GPP TS 24.229)	-	--
3	Check: local configuration on the UE provides options for setting the character format used for SDS	-->	Support of format types: 8bit (max. 1000 characters) 16bit (max. 500 characters) 32bit (max. 250 characters)	1	P
4a	Check: UE1 provides a user GUI for entering a message in 8bit character format (see note 1)	-->	The input of 1000 characters is possible	1	P
4b	Check: UE1 provides a user GUI for entering a message in 16bit character format (see note 1)	-->	The input of 500 characters is possible	1	P
4c	Check: UE1 provides a user GUI for entering a message in 32bit character format (see note 1)	-->	The input of 250 characters is possible	1	P
4d	Check: UE1 provides a user GUI for adding a hyperlink to connected content (see note 1)	-->	The input of the hyperlink is possible.	1	P
4e	Check: UE1 provides a user GUI for providing abstract binary data (see note 1)	-->	The input of the abstract binary data up to 1000 bytes is possible.	1	P
5a1	Check: UE1 (MCData client) send a SIP MESSAGE request with the content generated in test steps 4a, 4b, 4c, 4d and 4e to a unique user UE-con	-->	SIP MESSAGE according to ETSI TS 137 579-1 [28] (3GPP TS 37.579-1), clause 5.3C.1.3	1	P
5a2	UE1 receive (Accepted) response	<--	SIP STATUS 202 (ACCEPTED)	-	--
5a3	MCx_S1 checks the policy rules for the targeting user	-		-	--
5a4	UE-con receive the SDS message	<--	SIP MESSAGE	-	--
5a5	UE-con confirms reception with (Accepted) response	-->	SIP STATUS 202 (ACCEPTED)	-	--
5b1	Check: UE1 (MCData client) send a SIP MESSAGE request with the content generated in test steps 4a, 4b, 4c, 4d and 4e to a user group	-->	SIP MESSAGE according to ETSI TS 137 579-1 [28] (3GPP TS 37.579-1), clause 5.3C.1.3	2	P
5b2	UE1 receive (Accepted) response	<--	SIP STATUS 202 (ACCEPTED)	-	--
5b3	MCx_S1 checks the policy rules for the targeting user group	-		-	--
6	Check: UE1 have received multiple SDS with {InReplyTo message ID}; UE1 organize the messages as thread indication function for managing multiple message flows (see note 3)	-->	Visualisation on GUI	3	P
7	Check: UE1 provides an option for activating a read confirmation (see note 4)	-->	Read confirmation can be activated on the GUI	4	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
8	Check: UE1 (MCData client) send a SIP MESSAGE with SDS NOTIFICATION request to a unique user UE-con	-->	SIP MESSAGE	4	P
9	UE1 receive (Accepted) response	<--	SIP STATUS 202 (ACCEPTED)	-	--
10	UE-con receive the SDS message	<--	SIP MESSAGE	-	--
11	UE-con confirms reception with (Accepted) response	-->	SIP STATUS 202 (ACCEPTED)	-	--
12	SDS read confirmation: UE-con read the SDS and transmit the confirmation reply	-->	SIP MESSAGE	-	--
13	UE1 confirms reception with (Accepted) response	-->	SIP STATUS 202 (ACCEPTED)	-	--
14	Repeat test steps 7, 8, 9, 10 and 11 with UE-con as acting user (UE1 receives SDS with read confirmation request)			-	--
15	Check: SDS read confirmation; UE1 read the SDS and transmit a SIP MESSAGE containing SDS NOTIFICATION with the confirmation reply including {date|time information}	-->	SIP MESSAGE	4	P
16	UE-con confirms reception with (Accepted) response	<--	SIP STATUS 202 (ACCEPTED)	-	--
17	Check: UE1 provides an option for activating a read confirmation (see note 4)	-->	Read confirmation can be activated on the GUI	4	P
	EXCEPTION: UE1 supports FD according to the procedure described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1) clause 5.3C.10.3			5	P

NOTE 1: The GUI shall provide an information for the user of characters left based on the configuration setting for the format (8 bit, 16 bit or 32 bit).
NOTE 2: Local application provides an option for providing binary information (e.g. abstract control data).
NOTE 3: The GUI shall provide functions for organising and filtering to optimise the presentation of the message threads on the GUI.
NOTE 4: GUI shall provide the following options regarding to the creation of an SDS or FD:
- [SDS] Boolean option active/inactive for read confirmation.
- [FD] Create an URL instead of adding a file on upload.
- [FD] Provide an option for the user to receive the file or not.

6.7.3.3 Specific message contents

Table 6.7.3.3-1: SIP MESSAGE from the UE
(Table 6.7.3.2-1; step 5a1, ETSI TS 137 579-4 [7] (3GPP TS 37.579-4) Table 5.5.2.7.1-1)

Derivation Path: ETSI TS 137 579-4 [7] (3GPP TS 37.579-4) clause 5.5.2.7.1; SIP MESSAGE				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Method	**"MESSAGE"**			
Request-URI	MCPTT_PublicServiceID	MCPTT originating user public ID		MCPTT
Message-body			IETF RFC 3261 [16] IETF RFC 3909 [i.3]	
MIME -part-body	MCData-Info as described in Table 6.7.3.3-2			
Application/				
vnd3.3gpp.mcdata-info+xml	<mcdata-Params>	one-to-one-sds group-sds		
resource-list+xml	<list>			
vnd3.3gpp.mcdata-signalling				
vnd3.3gpp.mcdata-payload		Payload		

Table 6.7.3.3-2: SIP MESSAGE from the UE
(Table 6.7.3.2-1; step 15 ETSI TS 124 282 [29] (3GPP TS 24.282) Table 15.1.5.1-1)

Derivation Path: ETSI TS 124 282 [29] (3GPP TS 24.282) clause 15.1.5.-1; condition SDS NOTIFICATION message content

7 Enhanced Mission Critical Push-to-talk architecture phase 4

7.1 Mission critical location enhancements for on-network and off-network

7.1.1 Test purpose

(1)

```
with {MCx Service obtaining and conveying high accuracy location information}
ensure that {
    when {MCx Service user detects its position}
    then {MCx Service system can describe exact position of the MCx Service user and MCx Service has access to multiple external sources (e.g. magnetometers, orientation sensors, GNSS) and }
            }
```

(2)

```
with {Manage the privacy of Location information for MCx Users}
ensure that {
    when {MCx Service user can provide location information}
    then {The MCx Service system provides the Location information only to authorised participants (e.g. dispatcher) and the Location information is provided by the MCx Service user at the time of conveying }
            }
```

(3)

```
with {Manage the Location information update trigger}
ensure that {
    when {MCx Service system can configure several Location information update triggers}
    then {Based on identified conditions the MCx Service system can provide several Location information update triggers (e.g. initial registration, distance travelled, elapsed time, cell change, tracking area change, PLMN change or MCX communication initiation) and the MCx Service system is able to modify the triggers on the MCx Service user while connected to the network and the MCx Service administrator can connect the triggers with a geographical area}
            }
```

(4)

```
with {Location information on ad hoc group emergency communication or alert}
ensure that {
    when {MCx Service user initiates ad hoc group emergency alert or ad hoc group communication}
    then {The MCx Service user is providing Location information independent from the duration of the ongoing alert or communication and in any case, a defined trigger condition will be satisfied}
            }
```

7.1.2 Conformance requirements

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.11-001]]

"The MCX Service shall support obtaining and conveying Location information describing the position of the MCX UE."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.11-002]]

"The MCX Service should support obtaining and conveying high accuracy Location information describing the position of the MCX UE."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.11-002a]]

"The MCX Service shall be able to provide a mechanism for obtaining high accuracy Location information by integrating position information from multiple external sources (e.g. magnetometers, orientation sensors, GNSS)."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.11-003]]

"The MCX Service shall provide for the flexibility to convey future formats of Location information."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.11-004]]

"The MCX Service shall provide a means for MCX Service Administrators to manage the privacy of Location information for MCX Users within their authority."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.11-005]]

"An authorized MCX User shall be able to control the supplying of Location information by the MCX UE for MCX Service communications."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.11-006]]

"The conveyed Location information shall be the most recently obtained information about the position of the MCX UE at the time of the Location information conveyance."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.11-007]]

"The MCX Service shall be capable of configuring and re-configuring one or more Location information update triggers (i.e., identified conditions that, when satisfied, cause the MCX UE to report its current Location information)."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.11-008]]

"The MCX Service shall be able to modify Location information update triggers of an MCX User while the MCX User is on the network."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.11-009]]

"The MCX Service shall provide a means for an MCX UE to send a Location information update whenever a trigger condition is satisfied (e.g., initial registration, distance travelled, elapsed time, cell change, tracking area change, PLMN change, MCX Service communication initiation)."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.11-010]]

"The MCX Service shall provide a means for an MCX UE to send a Location information update whenever the MCX User initiates an MCX Service Emergency Alert."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.11-011]]

"The MCX Service shall provide a means for an MCX UE to send a Location information update whenever the MCX User initiates an MCX Service Emergency Group Communication."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.11-012]]

"The MCX Service shall provide a means for an MCX UE to send a Location information update if the MCX User is in an MCX Service Emergency State and a configured amount of time has passed since the previous location information update."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.11-013]]

"The MCX Service shall provide a means for an MCX UE to send a Location information update whenever a trigger condition is satisfied while the MCX User is in MCX Service Emergency State (e.g., initial registration, distance travelled, elapsed time, cell change, tracking area change, PLMN change, MCX Service communication initiation).

NOTE 1: The Location information update triggers for an MCX User in an MCX Service Emergency State might be different than the Location information update triggers used when the MCX User is not in an MCX Service Emergency State."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.11-014]]

"The MCX Service shall provide a means for an MCX Service Administrator to define geographical areas to be used for Location information update triggers for MCX Users within their authority."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.11-015]]

"The MCX Service shall provide a means for an MCX UE in a predefined area to send a Location information update whenever a trigger condition configured in an MCX User's active MCX Service User Profile is satisfied (e.g., initial registration, distance travelled, elapsed time, cell change, tracking area change, PLMN change, MCX Service communication initiation).

NOTE 2: The Location information update triggers for an MCX User in a predefined area might be different than the Location information update triggers used when the MCX User is not in a predefined area."

7.1.3 Test description

7.1.3.1 Pre-test conditions

System Configuration:

- MCx Service system consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (UE1) also consists of all sub-systems of the MCx user service.
- For the underlying "transport bearer" over which the UE1 (MCx Service system) and the IUT will communicate Parameters are set to the default parameters specific for the basic E-UTRA single simulated cells cell network scenarios, as defined in ETSI TS 138 508-1 [6] (3GPP TS 38.508-1) clause 6.3. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCx operation in the MCx configuration document).
- MCx Service system is configured to support the following features:
  - Handling of geographical information data.
  - Configuration of different triggers is possible. The following triggers shall be supported:
    - initial registration;
    - distance travelled;
    - elapsed time;
    - cell change;
    - tracking area change;
    - PLMN change;
    - MCX communication initiation.
  - Supporting MCx Service user UE2, UE-con:
    - UE2 and UE-con are connected to MCx_S1.

IUT:

- IUT (UE1).
- The IUT (UE1) consists of all sub-systems of the MCx user service including the ability for floor control, pre-arranged group communication functionality and ad-hoc group communication functionality. Ensure the correct configuration.
- IUT supports the detection of the geographical position and authorized to provide the Location information to the MCx Service.
- UE1 is the acting user.

Preamble:

- The IUT UE1 is connected to PLMN of the primary 5GS.
- The MCx Service system 1 (MCx_S1) is connected to the IUT, UE2 (MCx Service client) and UE-Con (MCx Service client with controller/dispatcher functionality).
- UE2 has no authorisation to receive Location information from UE1.
- UE-con is authorised to fetch Location information from UE1. UE-con provides a graphical interface to visualize the position of UE1 on a map.

7.1.3.2 Test procedure sequence

Table 7.1.3.2-1: Main behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
	EXCEPTION: Procedure 'MCX CO communication' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1) clause 5.4.3	-		-	--
	Check: MCx_S1 provides a configuration interface based on HTTPS request. Login required. Configure the different Location information trigger for UE1	<--	HTTPS link to configuration interface	-	--
1	Check MCx_S1 provides a configuration interface based on XML file import. Login for access to the file transfer function required. Configure the different Location information trigger for UE1	<--	XML access (e.g. via SCP or SFTP connection)	-	--
2	EXCEPTION: Procedure 'Initial MCX Authentication, Registration, Configuration and Subscription' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1) clause 5.3.37	-		-	--
3	Check: Execute MCX group Affiliation Status Change as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1) clause 5.3.34. No FA is active at the MCx Service system. Check: UE1 publish the basic settings	-->	SIP PUBLISH according ETSI TS 137 579-1 [28] (3GPP TS 37.579-1), IETF RFC 3903 [17]	1	P
4	MCx_S1 confirm the basic settings	<--	STATUS 200 OK (PUBLISH) according ETSI TS 124 229 [8] (3GPP TS 24.229)	-	--
5	MCx_S1 provides location information configuration to UE1; receive SIP MESSAGE	<--	SIP MESSAGE	-	--
6	Check: UE1 respond with SIP STATUS 200 (OK)	-->	SIP STATUS 200 (OK)	1	P
7	Check: UE1 provides location information (see notes 1 and 2) on MCPTT signalling event: INVITE UE-con	-->	SIP INVITE	2	P
8	MCx_S1 confirm INVITE	<--	SIP STATUS 100 (TRYING)		

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
9	UE-con receive incoming call	<--	SIP INVITE		
10	UE-con responds with SIP STATUS 200	-->	SIP STATUS 200 (OK)		
11	MCx_S1 confirms call setup towards UE1	-->	SIP STATUS 200 (OK)		
12a	Check: UE1 provides location information (see notes 1 and 2) on cell change event: MESSAGE	-->	SIP MESSAGE	2	P
13a	MCx_S1 responds with SIP STATUS 200 (OK)	<--	SIP STATUS 200 (OK)	-	--
12b	Check: UE1 provides location information (see notes 1 and 2) on tracking area change event: MESSAGE	-->	SIP MESSAGE	2	P
13b	MCx_S1 responds with SIP STATUS 200 (OK)	<--	SIP STATUS 200 (OK)	-	--
12c	Check: UE1 provides location information (see notes 1 and 2) on PLMN change event: MESSAGE	-->	SIP MESSAGE	2	P
13c	MCx_S1 responds with SIP STATUS 200 (OK)	<--	SIP STATUS 200 (OK)	-	--
12d	Check: UE1 provides location information (see notes 1 and 2) on periodic event: MESSAGE	-->	SIP MESSAGE	2	P
13d	MCx_S1 responds with SIP STATUS 200 (OK)	<--	SIP STATUS 200 (OK)	-	--
12e	Check: UE1 provides location information (see notes 1 and 2) on travel distance event: MESSAGE	-->	SIP MESSAGE	2	P
13e	MCx_S1 responds with SIP STATUS 200 (OK)	<--	SIP STATUS 200 (OK)	-	--
12f	Check: UE1 provides location information (see notes 1 and 2) on geographical area change event: MESSAGE	-->	SIP MESSAGE	2	P
13f	MCx_S1 responds with SIP STATUS 200 (OK)	<--	SIP STATUS 200 (OK)	-	--
14	MCx_S1 provides updated location information configuration to UE1; receive SIP MESSAGE	<--	SIP MESSAGE	-	--
15	Check: UE1 respond with SIP STATUS 200 (OK)	-->	SIP STATUS 200 (OK)	3	P
16	MCx_S1 terminates the call; UE1 receive BYE		SIP BYE	-	--
17	Check: UE1 respond with SIP STATUS 200 (OK)	-->	SIP STATUS 200 (OK)	-	--
18a	Check: UE1 initiates ad hoc group communication based (criterion: manual; list based) {InitiateGroup | ParticipantList | Location Information}	-->	SIP INVITE	4	P
19a	System validates authorization of UE1.	-	DB/AAA check → VALID: 200 OK {Group Established, GroupID} → ACK	-	--
20a	UE1 receives confirmation from MCx_S1	<--	SIP ACK	-	--
18b	Check: UE1 activates ad hoc group emergency alert {Location Information}	-->	SIP MESSAGE	4	P
19b	System validates authorization of UE1.	-	check → VALID: 200	-	--
20b	UE1 receives confirmation from MCx_S1	<--	SIP STATUS 200 (OK)	-	--

NOTE 1: MCx application of the user shall determinate the geographical position, velocity, time, travel distance, etc. based on external sensors or supporting applications, which are not on focus for this conformance test. The MCx application further shall authorised to merge all the available information into the MCx stack.

NOTE 2: MCx_S1 forwards the SIP MESSAGE with the Location information to the UE-con. UE-con updates the position of UE1 on the underlaying map on its graphical user interface.

7.1.3.3 Specific message contents

Table 7.1.3.3-1: SIP MESSAGE from UE1
(step 5, Table 7.1.3.2-1; ETSI TS 137 579-4 [7] (3GPP TS 37.579-4) Table 5.5.2.7.2-1)

Derivation Path: ETSI TS 137 579-4 [7] (3GPP TS 37.579-4) clause 5.5.2.7.2; condition MCPTT-INFO-EVENT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Method	**"MESSAGE"**			
Message-body			IETF RFC 3261 [16]	
application/vnd.3gpp.mcptt-location-info+xml		Type: multipart/mixed		
<location-info>				
<mcpttloc:configuration>	<mcpttlog:Triggering Criteria>	CellChange TrackingAreaChange PlmnChange MbmsSaChange MbsfnAreaChange PeriodicReport TravelDistance McpttSignalingEvent GeogrphicalAreaChange		

Table 7.1.3.3-2: SIP INVITE from the UE
(Table 5.1.3.2-1; step 7, ETSI TS 137 579-4 [7] (3GPP TS 37.579-4) Table 5.5.2.5.1-1)

Derivation Path: ETSI TS 137 579-4 [7] (3GPP TS 37.579-4) clause 5.5.2.5.1; condition MCPTT-INFO-EVENT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Method	**"INVITE"**			
Message-body				
application/vnd.3gpp.mcptt-location-info+xml		Type: multipart/mixed		
<location-info>				
<mcpttloc:configuration>	<mcpttlog:NonEmergency LocationInformation>	Geographical Coordinates, minimum interval length		
	<mcpttlog:Emergency LocationInformation>	Geographical Coordinates, minimum interval length		
	<mcpttlog:TriggeringCriteria>	Triggering criterion		

Table 7.1.3.3-3: SIP MESSAGE from UE1
(step 12, Table 7.1.3.2-1; ETSI TS 137 579-4 [7] (3GPP TS 37.579-4) Table 5.5.2.7.2-1)

Derivation Path: ETSI TS 137 579-4 [7] (3GPP TS 37.579-4) clause 5.5.2.7.2; condition MCPTT-INFO-EVENT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Method	**"MESSAGE"**			
Message-body			IETF RFC 3261 [16]	
application/vnd.3gpp.mcptt-location-info+xml		Type: multipart/mixed		
<location-info>				
<mcpttloc:configuration>	<mcpttlog:NonEmergency LocationInformation>	Geographical Coordinates, minimum interval length		
	<mcpttlog:Emergency LocationInformation>	Geographical Coordinates, minimum interval length		
	<mcpttlog:TriggeringCriteria>	Triggering criterion		

Table 7.1.3.3-4: SIP MESSAGE from UE1
(step 18b, Table 6.5.3.2-1; ETSI TS 137 579-4 [7] (3GPP TS 37.579-4) Table 5.5.2.7.2-1)

Derivation Path: Table 6.5.3.3-2

7.2 Preconfigured group enhancements

7.2.1 Test purpose

(1)

```
with {MCx Service system handles preconfigured user regroup procedures}
ensure that {
    when {Authorised MCx Service user request regroup of preconfigured user groups}
    then {MCx Service system can handle the procedures request, remove request and remove response}
        }
```

7.2.2 Conformance requirements

[ETSI TS 123 280 [4] (3GPP TS 23.280)], clause 10.15.2.15]

"Preconfigured user regroup remove request (MC service server - MC service client)."

[ETSI TS 123 280 [4] (3GPP TS 23.280)], clause 10.15.2.16]

"Preconfigured user regroup remove response (MC service client - MC service server)."

[ETSI TS 123 280 [4] (3GPP TS 23.280)], clause 10.15.2.17]

"Preconfigured regroup request return (MC service server - MC service client)."

7.2.3 Test description

7.2.3.1 Pre-test conditions

System Configuration:

- MCx Service system consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (UE1) also consists of all sub-systems of the MCx user service.
- For the underlying "transport bearer" over which the UE1 (MCx Service system) and the IUT will communicate Parameters are set to the default parameters specific for the basic E-UTRA single simulated cells cell network scenarios, as defined in ETSI TS 138 508-1 [6] (3GPP TS 38.508-1) clause 6.3. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCx operation in the MCx configuration document).
- MCx Service system is configured to support the following features:
  - Re-grouping of predefined user groups.
- Supporting MCx Service user UE2, UE-con:
  - UE2 and UE-con are connected to MCx_S1.

IUT:

- IUT (UE1).
- The IUT (UE1) consists of all sub-systems of the MCx user service including the ability for floor control, pre-arranged group communication functionality and ad-hoc group communication functionality. Ensure the correct configuration.

- UE1 is the acting user:
  - Can request a removal from the preconfigured list.

Preamble:

- The IUT UE1 is connected to PLMN of the primary 5GS.
- The MCx Service system 1 (MCx_S1) is connected to the IUT, UE2 (MCx Service client) and UE-Con (MCx Service client with controller/dispatcher functionality).

7.2.3.2 Test procedure sequence

Table 7.2.3.2-1: Main behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
	EXCEPTION: Procedure 'MCX CO communication' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1) clause 5.4.3	-			
	MCx_S1 provides a configuration interface based on HTTPS request. Login required. Authorise UE1 for re-grouping functionality	<--	HTTPS link to configuration interface		
	MCx_S1 provides a configuration interface based on XML file import. Login for access to the file transfer function required. Authorise UE1 for re-grouping functionality	<--	XML access (e.g. via SCP or SFTP connection)		
	EXCEPTION: Procedure 'Initial MCX Authentication, Registration, Configuration and Subscription' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1) clause 5.3.37	-			
1	Check: UE1 send a SIP MESSAGE request for removal from the preconfigured list (see note 1)	-->	SIP MESSAGE with group regroup notification request according to ETSI TS 123 280 [4] (3GPP TS 23.280)	1	P
2	MCx_S1 confirms the reception with SIP STATUS	<--	SIP STATUS 200 (OK) (see note 2)		
3	The group management part of the MCx_S1 changes the preconfigured groups and confirm the change towards UE1	<--	MCx_S1 responds with group regroup notification response	-	--
4	Check: UE1 confirm with SIP STATUS	-->	SIP STATUS 200 (OK) (see note 2)	1	P
NOTE 1: UE transmit a changed MC service group ID list. NOTE 2: According to ETSI TS 124 229 [8] (3GPP TS 24.229) a request should be rejected by sending a suitable 4xx response.					

7.2.3.3 Specific message contents

Table 7.2.3.3-1: SIP MESSAGE from UE1
(step 1, Table 7.2.3.2-1; ETSI TS 124 379 [21] (3GPP TS 24.379) Table 5.5.2.7.2-1)

Derivation Path: ETSI TS 124 379 [21] (3GPP TS 24.379) clause 16.2.1; condition client procedures				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Method	**"MESSAGE"**			
Message-body			IETF RFC 3261 [16]	
Extensible Markup Language				
application/vnd.3gpp.mcptt-regroup+xml	<mcptt-regroup-URI>	MCPTT user identity		MCPTT
<regroup-action>	<create>			
	<remove>			

7.3 Remotely initiated call request enhancements to support additional call setup parameters and implicit affiliation to the groups

7.3.1 Test purpose

(1)

```
with {Remotely initiated and requested MCx services}
ensure that {
    when {MCx Service user is authorised to initiate remote controlled communications}
    then {Authorised MCx Service user (e.g. Dispatcher) can request or initiate remotely a private
communication or a group communication from another MCx Service user UE and MCx Service user can
accept or deny the request and Authorised MCx Service user (e.g. Dispatcher) can terminate remotely
the connection}
            }
```

(2)

```
with {MCx Service user can pull MCx communication data towards an authorised MCx Service user (e.g.
dispatcher)}
ensure that {
    when {MCX User to request an authorized MCX User (e.g., a dispatcher) to send an MCX
Communication (e.g., video or data) to the MCX UE (downlink pull)}
    then {Data received by the authorised MCx Service user}
            }
```

7.3.2 Conformance requirements

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.21.1.2-001]]

"The MCX Service shall provide a mechanism for an authorized MCX User (e.g. dispatch operator) to remotely initiate an MCX Service Private Communication or MCX Service Group Communication from another user's MCX UE."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.21.1.2-002]]

"The MCX Service shall provide a mechanism for an authorized MCX User (e.g. dispatch operator) to request remote activation of an MCX Service Group Communication from another user's (e.g. officer in the field with wearable camera) MCX UE."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.21.1.2-003]]

"The MCX Service shall provide a mechanism for an authorized MCX User (e.g. officer in the field with wearable camera) to accept or deny access to an MCX Service Group Communication from their MCX UE."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.21.1.2-004]]

"The MCX Service shall provide a mechanism for an MCX User to request an authorized MCX User (e.g., a dispatcher) to send an MCX Communication (e.g., video or data) to the MCX UE (downlink pull). NOTE: A video could be stored on the MCX UE or taken online by the camera."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.21.2.1-001]]

"The MCX Service shall provide a mechanism for an authorized MCX User (e.g. dispatch operator) to remotely terminate an MCX Service Private Communication or MCX Service Group Communication from another user's MCX UE."

7.3.3 Test description

7.3.3.1 Pre-test conditions

System Configuration:

- MCx Service system consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (UE1) also consists of all sub-systems of the MCx user service.
- For the underlying "transport bearer" over which the UE1 (MCx Service system) and the IUT will communicate Parameters are set to the default parameters specific for the basic E-UTRA single simulated cells cell network scenarios, as defined in ETSI TS 138 508-1 [6] (3GPP TS 38.508-1) clause 6.3. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCx operation in the MCx configuration document).
- MCx Service system is configured to support the following features:
  - Remote initiation of association and call activation.
- Supporting MCx Service user UE2, UE-con:
  - UE2 and UE-con are connected to MCx_S1.

IUT:

- IUT (UE1).
- The IUT (UE1) consists of all sub-systems of the MCx user service including the ability for floor control, pre-arranged group communication functionality and ad-hoc group communication functionality. Ensure the correct configuration.
- UE1 is the acting user:
  - Remote part: receives request and initiates call.
  - Acting part: requests the remotely initiated actions.

Preamble:

- The IUT UE1 is connected to PLMN of the primary 5GS.
- The MCx Service system 1 (MCx_S1) is connected to the IUT, UE2 (MCx Service client) and UE-Con (MCx Service client with controller/dispatcher functionality).

7.3.3.2 Test procedure sequence

Table 7.3.3.2-1: Main behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
	EXCEPTION: Procedure 'MCX CO communication' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1) clause 5.4.3	-		-	--
1	Check: MCx_S1 provides a configuration interface based on HTTPS request. Login required. Authorise UE1 for acting as remotely controlled user entity	<--	HTTPS link to configuration interface	-	--
2	Check MCx_S1 provides a configuration interface based on XML file import. Login for access to the file transfer function required. Authorise UE1 for acting as remotely controlled user entity	<--	XML access (e.g. via SCP or SFTP connection)	-	--

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
3	EXCEPTION: Procedure 'Initial MCX Authentication, Registration, Configuration and Subscription' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1) clause 5.3.37	-		-	--
4	Check: Execute MCX group Affiliation Status Change as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1) clause 5.3.34. No FA is active at the MCx Service system. Check: UE1 publish the basic settings	-->	SIP PUBLISH according ETSI TS 137 579-1 [28] (3GPP TS 37.579-1), IETF RFC 3903 [17]	1	P
5	MCx_S1 confirm the basic settings	<--	STATUS 200 OK (PUBLISH) according ETSI TS 124 229 [8] (3GPP TS 24.229)	-	--
6a	UE-con initiates the request for remote initiated private communication by SIP MESSAGE	-->	SIP MESSAGE containing <remotely-initiated-private-call-request>	-	--
7a	MCx_S1 checks the authorisation for UE-con to initiate remote connections and confirms with SIP STATUS 200 (OK)	<--	SIP STATUS 200 (OK)	-	--
8a	MCx_S1 forwards the request for remote initiated private communication by SIP MESSAGE to UE1	<--		-	--
9a	Check: UE1 confirm the receipt with SIP STATUS 200 (OK)	-->	SIP STATUS 200 (OK)	1	P
10a	Check: UE1 initiates group communication; sending SIP INVITE	-->	SIP INVITE <remotely-initiated-private-call-response>	1	P
6b	UE-con initiates the request for remote initiated group communication by SIP MESSAGE	-->	SIP MESSAGE containing <remotely-initiated-group-call-request>	-	--
7b	MCx_S1 checks the authorisation for UE-con to initiate remote connections and confirms with SIP STATUS 200 (OK)	<--	SIP STATUS 200 (OK)	-	--
8b	MCx_S1 forwards the request for remote initiated group communication by SIP MESSAGE to UE1	<--		-	--
9b	Check: UE1 confirm the receipt with SIP STATUS 200 (OK)	-->	SIP STATUS 200 (OK)	1	P
10b	Check: UE1 initiates group communication; sending SIP INVITE	-->	SIP INVITE <remotely-initiated-group-call-response>	1	P
6c	Check: UE1 initiates the request for remote initiated private communication by SIP MESSAGE	-->	SIP MESSAGE containing <remotely-initiated-private-call-request>	1	P
7c	MCx_S1 checks the authorisation for UE1 to initiate remote connections and confirms with SIP STATUS 200 (OK)	<--	SIP STATUS 200 (OK)	-	--
8c	MCx_S1 forwards the request for remote initiated private communication by SIP MESSAGE to UE2	<--		-	--
9c	Check: UE2 confirm the receipt with SIP STATUS 200 (OK)	-->	SIP STATUS 200 (OK)	-	--
10c	Check: UE2 initiates group communication; sending SIP INVITE		SIP INVITE <remotely-initiated-private-call-response>	-	--
6d	UE1 initiates the request for remote initiated group communication by SIP MESSAGE	-->	SIP MESSAGE containing <remotely-initiated-group-call-request>	1	P
7d	MCx_S1 checks the authorisation for UE1 to initiate remote connections and confirms with SIP STATUS 200 (OK)	<--	SIP STATUS 200 (OK)	-	--
8d	MCx_S1 forwards the request for remote initiated group communication by SIP MESSAGE to UE2	<--		-	--

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
9d	UE2 confirm the receipt with SIP STATUS 200 (OK)	-->	SIP STATUS 200 (OK)	-	--
10d	UE2 initiates group communication, sending SIP INVITE	-->	SIP INVITE <remotely-initiated-group-call-response>	-	--

7.3.3.3 Specific message contents

Table 7.3.3.3-1: SIP MESSAGE from UE1
(step 6c, Table 7.3.3.2-1; ETSI TS 137 579-4 [7] (3GPP TS 37.579-4) Table 5.5.2.7.2-1)

Derivation Path: ETSI TS 137 579-4 [7] (3GPP TS 37.579-4) clause 5.5.2.7.2; condition MCPTT-INFO-EVENT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Method	**"MESSAGE"**			
Message-body			IETF RFC 3261 [16]	
Extensible Markup Language				
application/vnd.3gpp.mcptt-info+xml	<mcpttURI>	MCPTT user identity		MCPTT
<anyExt>	<request-type>	<remotely-initiated-group-call-request>		
		<remotely-initiated-private-call-request>		
	<response-type>	<remotely-initiated-group-call-response>		
		<remotely-initiated-private-call-response>		

Table 7.3.3.3-2: SIP INVITE from UE1
(step 10a, Table 7.3.3.2-1; ETSI TS 137 579-4 [7] (3GPP TS 37.579-4) Table 5.5.2.5.1-1)

Derivation Path: ETSI TS 137 579-4 [7] (3GPP TS 37.579-4) clause 5.5.2.5.1; condition MCPTT-INFO-EVENT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Method	**"INVITE"**			
Message-body			IETF RFC 3261 [16]	
Extensible Markup Language				
application/vnd.3gpp.mcptt-info+xml	<mcpttURI>	MCPTT user identity		MCPTT
<anyExt>	<request-type>	<remotely-initiated-group-call-request>		
		<remotely-initiated-private-call-request>		
	<response-type>	<remotely-initiated-group-call-response>		
		<remotely-initiated-private-call-response>		

7.4 Geographical affiliation/de-affiliation

7.4.1 Test purpose

(1)

```
with {Notification to an MCx Service user on entering a geographical area}
ensure that {
    when {MCx Service system informs MCx Service user about the need of association regarding
entering or leaving a geographical area}
    then {On entering a geographical area which is connected to a pre-defined group the MCx Service
system triggers the MCx Service user to affiliate or de-affiliate on leaving}
```

7.4.2 Conformance requirements

[ETSI TS 124 379 [21] (3GPP TS 24.379), clause 6.3.2.4.2]

"Generating a SIP MESSAGE request for notification of entry into or exit from a group geographic area. This clause describes the procedures for generating a SIP MESSAGE request to notify an MCPTT client that it has entered a pre-defined group geographic area or exited from a pre-defined group geographic area requiring affiliation to or de-affiliation from a group. The procedure is initiated by the participating MCPTT function when the participating MCPTT function determines that the MCPTT client has entered a pre-defined group geographic area or exited from a pre-defined group geographic area."

[ETSI TS 124 379 [21] (3GPP TS 24.379), clause 12.1.1.4]

"MCPTT client receives notification of entry into or exit from a group geographic area. Upon receipt of a "SIP MESSAGE request for notification of entry into or exit from a group geographic area", the MCPTT client."

7.4.3 Test description

7.4.3.1 Pre-test conditions

System Configuration:

- MCx Service system consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (UE1) also consists of all sub-systems of the MCx user service.
- For the underlying "transport bearer" over which the UE1 (MCx Service system) and the IUT will communicate Parameters are set to the default parameters specific for the basic E-UTRA single simulated cells cell network scenarios, as defined in ETSI TS 138 508-1 [6] (3GPP TS 38.508-1) clause 6.3. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCx operation in the MCx configuration document).
- MCx Service system is configured to support the following features:
  - Support Location information.
  - Location information is connected to a pre-defined group communication.

IUT:

- IUT (UE1).
- The IUT (UE1) consists of all sub-systems of the MCx user service including the ability for floor control, pre-arranged group communication functionality and ad-hoc group communication functionality. Ensure the correct configuration.
- The IUT can measure its geographical position based on e.g. GNSS or GPS and provide Location information to the MCx Service system.
- UE1 is the acting user.

Preamble:

- The IUT UE1 is connected to PLMN of the primary 5GS.
- The MCx Service system 1 (MCx_S1) is connected to the IUT and UE-Con (MCx client with controller/dispatcher functionality).

7.4.3.2 Test procedure sequence

Table 7.4.3.2-1: Main behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
	EXCEPTION: Procedure 'MCX CO communication' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1) clause 5.4.3	-		-	--
	EXCEPTION: Procedure 'Initial MCX Authentication, Registration, Configuration and Subscription' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1) clause 5.3.37	-		-	--
1	Check: UE1 provides location information on cell change event: MESSAGE.	-->	SIP MESSAGE according to test case 7.1	1	P
2	MCx_S1 responds with STATUS	<--	SIP STATUS 200 (OK)	-	--
3	MCx_S1 indicates to UE1 that de-affiliation is required; event: MESSAGE	<--	SIP MESSAGE	-	--
4	Check: UE1 confirms with STATUS	-->	SIP STATUS 200 (OK)	1	P
5	Check: UE1 starts de-affiliation with PUBLISH	-->	SIP PUBLISH	1	P
6	MCx_S1 responds with STATUS	<--	SIP STATUS 200 (OK)	-	--
7	Check: UE1 de-affiliated (see note)	--	- none -	1	P
8	Check: UE1 starts re-affiliation with PUBLISH	-->	SIP PUBLISH	-	--
9	MCx_S1 responds with STATUS	<--	SIP STATUS 200 (OK)	-	--
10	Check: UE1 affiliated (see note)	--	- none -	1	P
NOTE:	Affiliation status can be checked via graphical user interface.				

7.5 Geographical sending of an emergency alert feature

7.5.1 Test purpose

(1)

```
with {MCx Service user providing a mechanism to provide geographical location information}
ensure that {
    when {MCx Service user is entering or leaving a specific geographical area}
    then {An ad hoc group emergency alert can be activated on entering or deactivated on leaving for
the MCx Service user depending on a specific geographical area and MCx Service system can authorise
an MCx Service user to activate a preconfigured group emergency alert for this specific area}
            }
```

7.5.2 Conformance requirements

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.1.9-001]]

"The MCX Service shall provide a mechanism for an MCX Service Emergency Alert to be triggered when an MCX UE moves into a predefined area."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.1.9-002]]

"The MCX Service shall provide a mechanism for an MCX Service Emergency Alert to be cancelled when the MCX UE moves out of a predefined area or to remain active until cancelled by the MCX User."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.6.2.4.1-005]]

"The MCX Service shall provide a mechanism for an authorized MCX User to configure an MCX Service Emergency Alert to send a notification to MCX Users within a configurable geographic area of the MCX User entering the MCX Service Emergency State, independent of the MCX Service Group Membership."

7.5.3 Test description

7.5.3.1 Pre-test conditions

System Configuration:

- MCx Service system consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (UE1) also consists of all sub-systems of the MCx user service.
- For the underlying "transport bearer" over which the UE1 (MCx Service system) and the IUT will communicate Parameters are set to the default parameters specific for the basic E-UTRA single simulated cells cell network scenarios, as defined in ETSI TS 138 508-1 [6] (3GPP TS 38.508-1) clause 6.3. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCx operation in the MCx configuration document).
- MCx Service system is configured to support the following features:
  - Support Location information.
  - Location information is connected to ad-hoc group emergency alert.

IUT:

- IUT (UE1).
- The IUT (UE1) consists of all sub-systems of the MCx user service including the ability for floor control, pre-arranged group communication functionality and ad-hoc group communication functionality. Ensure the correct configuration.
- The IUT can measure its geographical position based on e.g. GNSS or GPS and provide Location information to the MCx Service system.
- UE1 is the acting user.

Preamble:

- The IUT UE1 is connected to PLMN of the primary 5GS.
- The MCx Service system 1 (MCx_S1) is connected to the IUT and UE-Con (MCx client with controller/dispatcher functionality).

7.5.3.2 Test procedure sequence

Table 7.5.3.2-1: Main behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
	EXCEPTION: Procedure 'MCX CO communication' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1) clause 5.4.3	-		-	
	EXCEPTION: Procedure 'Initial MCX Authentication, Registration, Configuration and Subscription' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1) clause 5.3.37	-		-	
1	Check: UE1 provides location information on cell change event: MESSAGE.	-->	SIP MESSAGE according to test case 7.1	1	
2	MCx_S1 responds with STATUS	<--	SIP STATUS 200 (OK)	-	--
3	MCx_S1 indicates to UE1 that ad-hoc emergency group alert is required; event: MESSAGE	<--	SIP MESSAGE	-	--
4	Check: UE1 confirms with STATUS	-->	SIP STATUS 200 (OK)	1	P
5	Check: UE1 activates ad hoc group emergency alert	-->	SIP MESSAGE according to ETSI TS 124 379 [21] (3GPP TS 24.379) clause 17.2.2.1.1	1	P
6	System validates authorization of UE1.	-	check → VALID: 200	-	--
7	UE1 receives confirmation from MCx_S1	<--	SIP STATUS 200 (OK)	-	--
8	Check: For the initiator the ad hoc group emergency alert automatically is confirmed. Other participants need to confirm automatically or by manually action		- none -		

8 MCData using IP connectivity

8.1 IP Connectivity

8.1.1 Test purpose

(1)

with {MCData service capability for transmission of IP data towards a receiver in the network}
ensure that {
    when {MCData user initiates the IP data transport}
    then {The MCData service capability enables a MCData user for initiating IP data transport towards a server in the network or another MCData user **and** the MCData Service enables the incoming transport of IP data to another MCData user which is not the initiator **and** the MCData Service supports different per packet priorities for incoming and outgoing IP data transport}
            }

(2)

with {MCData Service authorizations}
ensure that {
    when {The MCData Service capability provides different authorisation options}
    then {The MCData Service system provides preconfigured or authorized MCData user or personnel access to authorize the initiation of incoming or outgoing transport of IP data for the MCData user to specific destinations and authorization can be requested by initiating transport IP data to new destination or can be revoked on demand by authorized access and the MCData system allows an authorized remote access to allow the use of higher per packet priority for a particular MCData user}
            }

(3)

```
with {Authorized control over outgoing IP data transport}
ensure that {
    when {MCData user tries to send IP data to new destination and authorized person has set
predefined transport rules or needs dynamically to approve the request}
    then {MCData service checking if preconfigured authorization exists for the destination. And in
case is not preconfigured, a real-time approval request is sent to the authorization person.}
            then {IP data transport is initiated upon approval, and the authorized person can revoke
the authorization at any time, preventing further transmissions.}
            }
```

(4)

```
with {Control over receiving IP data from specific MCData user}
ensure that {
    when {MCData user attempts to receive IP data from another MCData user. And the sender is either
preauthorized or needs real - time approval}
    then {The MCData service verify if the sender is or not on the preauthorized list. When the
sender is not preauthorized, an approval request is sent to authorized person}
            then {IP data transport is initiated upon approval, and the authorized person can revoke
the authorization at any time, blocking further transmissions.}
            }
```

```
(5)
with {Priority based IP data transport for MCData user}
ensure that {
    when {MCData user request transmission of IP data with higher pre-packet priority and the IP
data transport requires includes priority information}
    then {The priority request in form the MCData service processed and recognized. The IP data is
transmitted with preferential handling over lower - priority data and the receiving MCData user
correctly interprets the priority level}
            then {The network congestion does not interface with priority-based transport.}
            }
```

(6)

```
with {Remote control over higher packet priority assignment}
ensure that {
    when {Authorized person attempts to remotely enable higher per packet priority for an MCData
user}
    then {MCData service is validating the authorization credentials of requesting entity, Higher
priority setting is applied to the designated MCData user, and the priority ca be revoked from the
authorized person at any time. The system logs record all authorization actions precisely}
                        }
```

8.1.2 Conformance requirements

[ETSI TS 122 282 [3] (3GPP TS 22.282)], clause [R-5.5.2-001]]

"The MCData Service shall enable a MCData user to initiate transport of IP data towards a server in the network or another MCData user."

[ETSI TS 122 282 [3] (3GPP TS 22.282)], clause [R-5.5.2-002]]

"The MCData Service shall enable incoming transport of IP data towards an MCData user, initiated by another MCData user."

[ETSI TS 122 282 [3] (3GPP TS 22.282)], clause [R-5.5.2-003]]

"The MCData Service shall enable an authorized person for an MCData user to authorize initiation of transport of IP data from that MCData user to specific destinations. This authorization may be preconfigured or may be provided by the authorized person when IP data transport to a new destination is initiated. Authorization may be revoked on demand by the authorized person."

[ETSI TS 122 282 [3] (3GPP TS 22.282)], clause [R-5.5.2-004]]

"The MCData Service shall enable a MCData user or authorized person for this MCData user to authorize initiation of incoming transport of IP data from specific other MCData users. This authorization may be preconfigured or may be provided by the authorized person when IP data transport from a new destination is initiated. Authorization may be revoked on demand by the authorized person."

[ETSI TS 122 282 [3] (3GPP TS 22.282)], clause [R-5.5.2-005]]

"The MCData Service shall support incoming and outgoing IP data transport for a MCData user with a higher per packet priority."

[ETSI TS 122 282 [3] (3GPP TS 22.282)], clause [R-5.5.2-006]]

"The MCData Service shall enable an authorized person to remotely authorize the use of higher per packet priority for a particular MCData user."

8.1.3 Test description

8.1.3.1 Pre-test conditions

System Configuration:

- MCx Service system consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (UE1) also consists of all sub-systems of the MCx user service.
- For the underlying "transport bearer" over which the UE1 (MCx Service system) and the IUT will communicate Parameters are set to the default parameters specific for the basic E-UTRA single simulated cells cell network scenarios, as defined in ETSI TS 138 508-1 [6] (3GPP TS 38.508-1) clause 6.3. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCx operation in the MCx configuration document).
- MCx Service system is configured to support the following features:
  - IPConn authorisation for several users.
- Supporting MCx Service user UE2:
  - UE2 is connected to MCx_S1.
  - UE2 is authorised to use IPConn functionality.
- Figure 4.1-5 shows the principal setup.

IUT:

- IUT (UE1).
- The IUT (UE1) consists of all sub-systems of the MCx user service including the ability for floor control, pre-arranged group communication functionality and ad-hoc group communication functionality. Ensure the correct configuration.
- IUT supports IPConn client functionality.
- UE1 is the acting user.

Preamble:

- The IUT UE1 is connected to PLMN of the primary 5GS.
- The MCx Service system 1 (MCx_S1) is connected to the IUT (MCData client) and UE-Con (MCData client with controller/dispatcher functionality).

8.1.3.2 Test procedure sequence

Table 8.1.3.2-1: Main behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
	EXCEPTION: Procedure 'MCX CO communication' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1) clause 5.4.3	-		-	--
1	Check: MCx_S1 provides a configuration interface based on HTTPS request. Login required. Activate the IPConn functionality for UE1 and UE2 including prioritisation	<--	HTTPS link to configuration interface	-	--
2	Check MCx_S1 provides a configuration interface based on XML file import. Login for access to the file transfer function required. Activate the IPConn functionality for UE1 and UE2 including prioritisation	<--	XML access (e.g. via SCP or SFTP connection)	-	--
3	Check: UE1 provides local configuration for using IPConn functionality	-->		1	P
	EXCEPTION: Procedure 'Initial MCX Authentication, Registration, Configuration and Subscription' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1) clause 5.3.37	-	<mcdata-client-id> <mcdata-access-token>	-	--
	Check: Execute MCData group Affiliation Status Change as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1) clause 5.3.34. No FA is active at the MCx Service system. Check: UE1 publish the basic settings	-->	SIP PUBLISH according [ETSI TS 137 579-1 [28] (3GPP TS 37.579-1)], IETF RFC 3903 [17]	1	P
	MCx_S1 confirm the basic settings	<--	STATUS 200 OK (PUBLISH) according ETSI TS 124 229 [8] (3GPP TS 24.229)	-	--
4	Check: UE1 starts initiation of IPConn using SIP INVITE message	-->	SIP INVITE	2	
	MCx_S1 checks the policy rules for the targeting users UE1 and UE2		-none- (both users have authorisation for using IPConn)	-	--
5	MCx_S1 provides proceeding information to UE1	<--	SIP STATUS 100 (TRYING)	-	--
	MCx_S1 invites UE2 as second IPConn user	-->	SIP INVITE	-	--
	UE2 confirms invitation request, replying SIP STATUS 200 (OK)	-->	SIP STATUS 200 (OK)	-	--
6	UE1 receive confirmation from MCx_S1	<--	SIP STATUS 200 (OK)	-	--
7	Check: UE1 start establishment of the GRE tunnel sending IP protocol with generic routing encapsulation (GRE) using IPv6	-->	IETF RFC 2784 [30]	3	P
8	UE2 initiates GRE tunnel like UE1; MCx_S1 connects the tunnel	-->	IETF RFC 2784 [30]	-	-
9	Check: UE1 starts Ping test through the GRE tunnel to check the availability of UE2	-->	ICMP request message according to IETF RFC 792 [31]	4	P

8.1.3.3 Specific message contents

Table 8.1.3.3-3: SIP INVITE from the UE
(Table 8.1.3.2-1; step 4, ETSI TS 137 579-4 [7] (3GPP TS 37.579-4) Table 5.5.2.5.1-1)

Derivation Path: ETSI TS 137 579-4 [7] (3GPP TS 37.579-4) clause 5.5.2.5.1; condition MCPTT-INFO-EVENT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Method	**"INVITE"**			
Request-URI	MCPTT_PublicServiceID	MCPTT originating user public ID		MCPTT
Message-header				
P-asserted identity/SIP PAI Address	<SIP PAI User part>			
	<SIP PAI Host part>			
Route				
addr-spec[1]	SIP URI			
user-info and host	P-CSCF address of the SS	P-CSCF address as assigned to the UE via NAS signalling or P-CSCF discovery		
port	protected server port of the SS	as assigned during registration		
uri-parameters	List of parameters			
Message-body				
Application//vnd.3gpp.mcdata-info+xml				
<mcdata-Params>	<request-type>	one-to-one-ipcon		IPCon

8.2 MCData with different QoS requirements

8.2.1 Test purpose

(1)

```
with {FRMCS UE attach to FRMCS network including 5QI activation}
ensure that {
    when {UE is powered on or returns from flight mode}
    then {RRC attaching procedure including context activation and IP address assignment in the
network}
            }
```

8.2.2 Conformance requirements

[ETSI TS 103 765-3 [25**], clause 6.2]**

"The SMF receives policy rules from the PCF via N7 and sends instructions for QoS enforcement, traffic steering and charging control via N4 to the UPF. The UPF enforces the rules via Packet Detection Rules (PDRs) and forwarding actions."

[ETSI TS 103 765-3 [25**], clause 8.1]**

"FRMCS Quality of Service Principles. The FRMCS Transport Stratum QoS framework is based on the following principles:

a) QoS is applied via PDU sessions (up to 15 per UE) and QoS flows (up to 63 per PDU session).

b) QoS is enforced at UE, gNodeB and UPF based on rules propagated from the UDM/PCF or pre-configured.

c) QoS policy is controlled from the FRMCS service stratum via messaging over reference points Rx or N5.

d) The gNodeB can map multiple QoS flows with similar characteristics to a single Data Radio Bearer (DRB).

e) QoS flows can be of type GBR (Guaranteed Bit Rate), non-GBR and delay critical GBR.

f) Each QoS flow is associated with a QoS profile, including:

i) A QFI (QoS Flow identifiers) based on 5QI.

ii) An ARP (Allocation & Retention Priority).

iii) Type specific & optional parameters.

IP data is mapped (filtered) to QoS flows at UE (based on QoS rules) and UPF utilizing packet filter sets.

The FRMCS Application Function (AF), defined in ETSI TR 103 791 [i.1], shall perform mapping from a Session Description Information (SDI), received within the AF session signalling, to service information passed to the PCF over the N5 or Rx reference points as per ETSI TS 129 513 [27] (3GPP TS 29.513). The FRMCS AF shall signal the PCF to request resources for media bearers using one of the following options:

1) as per [ETSI TS 123 289 [14] (3GPP TS 23.289)], clause 7.2.2 (see step 3 in Figure 8.1-1) if the FRMCS SIP Core represents the FRMCS AF; or

2) as per [ETSI TS 123 289 [14] (3GPP TS 23.289)], clause 7.2.3.2 if the MC Service Server represents the FRMCS AF (see step 6 in Figure 8.2-2).

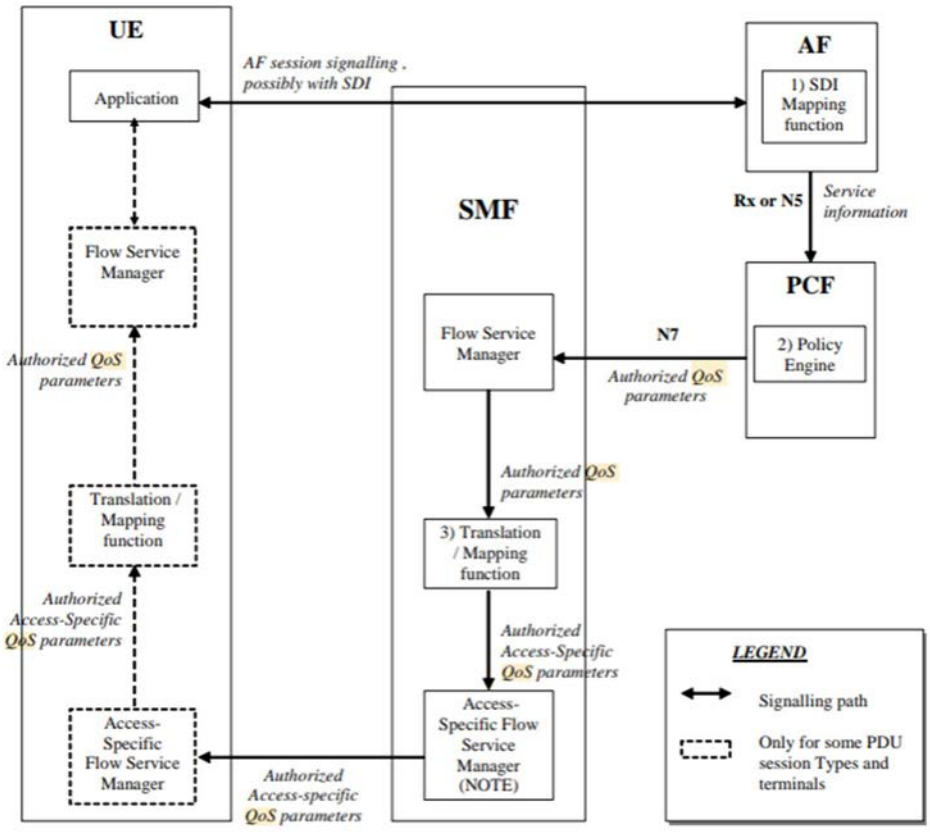


Figure 8.1-1: Communication Security

The MC Service Server shall utilize the Npcf_PolicyAuthorization Service for N5 or Rx procedures to request policies for the PDU session which the FRMCS MC session is bound to. The PCF shall perform mapping from the service information received over the Rx or N5 reference points to the Authorized QoS parameters that shall be passed to the SMF via the N7 reference point.

The Session binding function determines relevant PDU session and associates IP data flows to it. PCF shall consider any 5GC specific restrictions, the AF service information and other information available to the PCF. PCF associates a set of QoS parameters (5QI, ARP, etc.) to each PCC rule as per [ETSI TS 123 289 [14] (3GPP TS 23.289)].

The SMF shall perform mapping from the Authorized QoS parameters received from PCF to the access specific QoS parameters as per [ETSI TS 123 289 [14] (3GPP TS 23.289)].

FRMCS AF can use Npcf_EventExposure service to get notifications about the outcomes of UE policies delivery. The network sets up QoS flow(s) with a suitable QoS and indicates to the UE the QoS characteristics of those QoS flow(s) as per Figure 8.2-1 (reference to [ETSI TS 123 289 [14] (3GPP TS 23.289)], clause 7.1)."

[ETSI TS 103 765-3 [25**], clause 8.3]**

"5QI and ARP Application per FRMCS Service. The FRMCS Transport Domain shall support at least the 5QI identified in Table 8.2-1.

Table 8.3-1: List of 5QI in FRMCS

5QI
4
5
8
65
69

"

[ETSI TS 103 765-3 [25], clause 8.4]

"The QoS of FRMCS Transport Stratum traffic is classified using 5QI. Since the traffic flow to some endpoints (Trackside Gateway, VAS controller, etc.) traverses an IP/MPLS backbone the 5QI shall be mapped to a Traffic Class. For each QoS Flow characterized by its 5QI, the mapping of 5QI to DSCP values are done on different nodes depending on the traffic direction:

- *For UL traffic, the gNodeB maps the 5QI values to transport DSCP values. For DL traffic, the mapping is done by local configuration on the SMF, which provides UPF with the transport DSCP values. The UPF then marks DL traffic on a per QoS Flow basis.*

The DSCP mapping for mandatory FRMCS 5QIs from Table 8.2-1 is identified in Table 8.4-1.

Table 8.4-2: 5QI to DSCP Mapping

5QI	*DSCP*
4	*AF43 (DSCP 38)*
5	*CS5 (DSCP 40)*
8	*AF12 (DSCP 12)*
65	*EF (DSCP 46)*
69	*CS5 (DSCP 40)*

"

8.2.3 Test description

8.2.3.1 Pre-test conditions

System Configuration:

- MCx Service system not required.
- For the underlying "transport bearer" over which the UE1 (MCx Service system) and the IUT will communicate Parameters are set to the default parameters specific for the basic E-UTRA single simulated cells cell network scenarios, as defined in ETSI TS 138 508-1 [6] (3GPP TS 38.508-1) clause 6.3. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCx operation in the MCx configuration document).
- Different bearers are available according to Table 8.2-2.

IUT:

- IUT (UE1).
- The IUT (UE1) consists of all sub-systems of the MCx user service including the ability for floor control, pre-arranged group communication functionality and ad-hoc group communication functionality. Ensure the correct configuration.
- UE1 is the acting user.

Preamble:

- The IUT UE1 is connected to PLMN of the primary 5GS.
- The MCx Service system 1 (MCx_S1) is connected to the IUT (MCData client) and UE-Con (MCData client with controller/dispatcher functionality).

8.2.3.2 Test procedure sequence

Table 8.2.3.2-1: Main behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Switch on the UE1. UE1 starts radio resource allocation procedure (RRC). 5QI contexts are assigned to the SIM application permanently in the network and UE1 assign via PDU session initialisation	-->	RRC activation	1	P
2	Network confirms the radio resource allocation and UE1 is attached to the network	<--			
3	Check: PDU contexts in the UE. For each 5QI there shall be a bearer available (e.g. check for IP address allocation)	-->	5QI contexts are available	1	P
4	Check: application A on UE1 request transmission 5QI=4. Bearer is accessible for UE1	-->	5QI contexts are available	1	P
5	Check: application B on UE1 request transmission 5QI=5. Bearer is accessible for UE1	-->	5QI contexts are available	1	P
6	Check: application C on UE1 request transmission 5QI=8. Bearer is accessible for UE1	-->	5QI contexts are available	1	P
7	Check: application D on UE1 request transmission 5QI=65. Bearer is accessible for UE1	-->	5QI contexts are available	1	P
8	Check: application E on UE1 request transmission 5QI=69. Bearer is accessible for UE1	-->	5QI contexts are available	1	P

8.3 MCData streaming and datagram mode

8.3.1 Test purpose

(1)

```
with {MCData Service system provides a data streaming capability}
ensure that {
    when {MCData service user is authorized to use MCData streaming capability}
    then {Each recipient can decide to select the receipt of a data stream by using a provided link
or MCData service user can reject the receipt of a data stream}
            }
```

(2)

```
with {Automatic reception of streamed data for tracking delivery status per recipient}
ensure that {
    when {MCData user initiates a data stream and marks it for automatic reception by UEs}
    then {Recipient UEs automatically accept and start receiving the stream without user
intervention. MCData system generates and returns a start record for each recipient that has
successfully receiving the stream and when stream ends, MCData systems returns stop record to the
sender identifying, which users completed or terminated reception}
                        }
```

(3)

```
with {Optimize delivery of data streams to users in the same area}
ensure that {
    when {MCData is initiated for a group of users}
    then {MCData service detects the common delivery context, and the systems uses available network
and system efficiencies to deliver the stream using the least amount of transmission resources and
all the recipients within the area receive the stream in parallel with minimized delay and bandwidth
consumption.}
                        }
```

(4)

```
with {User or authorized initiated cancellation of active or buffered data streams}
ensure that {
    when {MCData user decides to terminate streaming that is currently active or cancel pending
initiated by another user}
    then {MCData service holds the transmission of ongoing data to all recipients. All buffered and
undelivered streams segments are purged from system and successful cancelation is returned to the
initiating entity. No further delivery attempts of the cancelled stream data are made}
                        }
```

(5)

```
with {Measurement of data communication setup times under different conditions}
ensure that {
    when {MCData user made request to initiate a data stream under normal and urgent conditions}
    then {MCData systems completes the setup and confirms establishment within less than 1 second
for immediate and for normal data the setup time does not exceed 3 seconds, and the measurement
includes the time from the initial request to successful establishment of communication as perceived
by the user and failures of those timing requirements are logged for systems analyses}
                        }
```

8.3.2 Conformance requirements

[ETSI TS 122 282 [3] (3GPP TS 22.282)], clause [R-5.4.2-001]]

"The MCData Service shall provide a data streaming capability."

[ETSI TS 122 282 [3] (3GPP TS 22.282)], clause [R-5.4.2-002]]

"The MCData data streaming capability shall provide an option that allows each recipient to choose to receive the data stream or not (e.g. by sending a link (URL) to all relevant members)."

[ETSI TS 122 282 [3] (3GPP TS 22.282)], clause [R-5.4.2-003]]

"The MCData data streaming capability shall allow a user to reject to receive the data stream."

[ETSI TS 122 282 [3] (3GPP TS 22.282)], clause [R-5.4.2-004]]

"The MCData data streaming capability shall provide a sending user selectable indication for automatic reception by the UE."

[ETSI TS 122 282 [3] (3GPP TS 22.282)], clause [R-5.4.2-005]]

"The MCData data streaming capability shall provide start and stop records to the sender for each recipient successfully receiving the data stream."

[ETSI TS 122 282 [3] (3GPP TS 22.282)], clause [R-5.4.2-006]]

"The MCData data streaming capability shall make use of available system delivery efficiencies for streaming of common information to users within the same relevant area and able to receive at the same time."

[ETSI TS 122 282 [3] (3GPP TS 22.282)], clause [R-5.4.2-007]]

"The MCData data streaming capability shall allow a user to cancel streaming of data they have initiated including data remaining buffered in the system waiting to be streamed."

[ETSI TS 122 282 [3] (3GPP TS 22.282)], clause [R-5.4.2-008]]

"The MCData data streaming capability shall allow an authorised user to terminate streaming of data being sent and cancel streaming of data remaining buffered in the system waiting to be streamed."

[ETSI TS 122 282 [3] (3GPP TS 22.282)], clause [R-5.4.2-009]]

"The MCData data streaming capability shall provide setup time of:

- *Less than 1 second for immediate setup of a data communication.*
- *Less than 3 second for normal setup of a data communication.*

NOTE 1: Setup time of a communication is essential because of their safety related character.

NOTE 2: Communication setup encompasses the value of the elapsed time between the communication establishment request and the indication of successful communication establishment."

8.3.3 Test description

8.3.3.1 Pre-test conditions

System Configuration:

- MCx Service system consists of all sub-systems of the Common Services Core, according to Figure 4.1-6.
- For the underlying "transport bearer" over which the UE1 (MCx Service system) and the IUT will communicate Parameters are set to the default parameters specific for the basic E-UTRA single simulated cells cell network scenarios, as defined in ETSI TS 138 508-1 [6] (3GPP TS 38.508-1) clause 6.3. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCx operation in the MCx configuration document).
- MCx Service system is configured to support the following features:
  - Support QUIC according to IETF RFC 9000 [33].

IUT:

- IUT (UE1).
- The IUT (UE1) consists of all sub-systems of the MCx user service including the ability for floor control, pre-arranged group communication functionality and ad-hoc group communication functionality. Ensure the correct configuration.
- UE1 is the acting user.

Preamble:

- The IUT UE1 is connected to PLMN of the primary 5GS.
- The MCx Service system 1 (MCx_S1) is connected to the IUT.

8.3.3.2 Test procedure sequence

Table 8.3.3.2-1: Main behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: UE1 initiates UDP Data connection with datagram mode according to IETF RFC 9897 [32] and IETF RFC 9000 [33]	-->	Connection on UDP established		P
2	MCx_S1 confirms transmission after all datagrams are sent	<--	none		-
4	Check: UE1 initiates UDP Data connection with streaming mode according to IETF RFC 9897 [32] and IETF RFC 9000 [33]	-->	Connection on UDP established		P
5	MCx_S1 confirms transmission after all datagrams are sent	<--	none		-

8.3.3.3 Specific message contents

Table 8.3.3.3-1: UDP from UE1 (step 1, Table 8.3.3.2-1; IETF RFC 9897 [32])

Derivation Path IETF RFC 9897 [32]; condition: Stream Information				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
User Datagram Protocol	**Payload**			
Stream Index	Index	In stream mode stream index will be changed for each transmission	IETF RFC 9897 [32]	
Stream Packet Number	Number of packages			
QUIC IETF		Could be more than one QUIC IETF connection	IETF RFC 9000 [33]	

Annex A (normative): Additional functionalities

A.1 Support of MP-QUIC and MP-TCP

A.1.1 Test purpose

(1)

```
with { Support of MP-QUIC }
ensure that {
    when { FRMCS UE open a MP-QUIC stream }
    then {FRMCS UE can establish a data connection using QUIC IETF based on UDP }
            }
```

(2)

```
with { Support of MP-QUIC }
ensure that {
    when { FRMCS UE open a MP-TCP stream }
    then {FRMCS UE can establish a data connection using QUIC IETF based on TCP }
            }
```

A.1.2 Conformance requirements

[ETSI TS 103 765-1 [27], clause 5.3.3]

"Architecture. It is constituted of two functions, the Multipath Client and the Multipath Gateway. The Multipath Client and the Multipath Gateway are connected via two reference points:

- *FSMPM which enables the FRMCS Multipath control plane function, including Multipath Gateway discovery, information exchange on available Multipath Datapaths.*
- *FSMPU which enables the user plane transport of a Multipath Dataflow based on a Multipath Protocol over one or more Multipath Datapaths."*

A.1.3 Test description

A.1.3.1 Pre-test conditions

System Configuration:

- MCx Service system consists of all sub-systems of the Common Services Core according to Figure 4.1-6.
- For the underlying "transport bearer" over which the UE1 (MCx Service system) and the IUT will communicate Parameters are set to the default parameters specific for the basic E-UTRA single simulated cells cell network scenarios, as defined in [ETSI TS 138 508-1 [6] (3GPP TS 38.508-1) clause 6.3]. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCx operation in the MCx configuration document).
- MCx Service system is configured to support the following features:
  - Support of MP-QUIC according IETF RFC 9000 [33].
  - Support of MP-TCP according to IETF RFC 9897 [32].
  - Multipath Control Function (MPF) to control the parameter for the switching of the uniflows.

IUT:

- IUT (UE1).
- The IUT (UE1) consists of all sub-systems of the MCx user service including the ability for floor control, pre-arranged group communication functionality and ad-hoc group communication functionality. Ensure the correct configuration.
- UE1 is the acting user.

Preamble:

- The IUT UE1 is connected to PLMN of the primary 5GS.
- The MCx Service system 1 (MCx_S1) is connected to the IUT.

A.1.3.2 Test procedure sequence

Table A.1.3.2-1: Main behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: UE1 establish UDP based session introducing QUIC content	-->	Connection is accepted by MCx_S1 (see note)	1	P
2	Check: UE1 establish TCP based session introducing QUIC content	-->	Connection is accepted by MCx_S1 (see note)	2	P
NOTE:	If more than one uniflow will be used, then also the destination needs to be checked. A destination can have multiple connection numbers and one connection number can have multiple destinations regarding to the MPF settings.				

A.1.3.3 Specific message contents

Table A.1.3.3-1: MP-QUIC connection from UE1 (step 1, Table A1.1.3.2-1; IETF RFC 9000 [33])

Derivation Path IETF RFC 9000 [33]; condition MP-QUIC				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
User Datagram Protocol	UDP	Bearing protocol type		
QUIC IETF				
QUIC Connection Information	Connection Number	Uniflow index	IETF RFC 9000 [33]	
Destination Connection ID				
Source Connection ID				
Token				
Payload		Payload		
CRYPTO		Cryptography part		

Table A.1.3.3-2: MP-QUIC connection from UE1 (step 2, Table A1.1.3.2-1; IETF RFC 9897 [32])

Derivation Path IETF RFC 9897 [32]; condition MP-TCP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Transmission Control Protocol	TCP	Bearing protocol type		
Options				
Multipath Transmission Control Protocol	Multipath Capable			
Multipath TCP				
Multipath TCP flags				
Senders Key				
Subflow Token	And expected IDSN			
MPTCP analysis			IETF RFC 9897 [32]	
Master flow	Master TCP stream			
Stream index				
TCP subflow stream id	Sub-stream identification			

A.2 Support of IPv6

A.2.1 Test purpose

(1)

```
with { Support of IPv6 }
ensure that {
    when { FRMCS UE is attached to the network }
    then {FRMCS UE is receiving an IP address with IPv6 and the routing function can handle the
entry }
            }
```

A.2.2 Conformance requirements

[ETSI TS 103 765-1 [27], clause 5.3.6]

"IP Address assignment is required:

a) for public service layer IP address for each Multipath Dataflow Handler; and

b) for each transport domain Multipath Datapath for each Multipath Dataflow Handler.

The 5G Core provides IPv6 prefix during the initial default data connection establishment to each UE. The procedure is IPv6 Prefix Delegation. The IPv6 prefix is assigned via DHCPv6.

As illustrated in Figure 5.3.5-1 (Multipath Client), the transport-related IP addresses on the OB side (IP OB 1.1, IP OB 2.1, etc.) are appended with this prefix. The IPv6 prefix is delegated from the 5G Core directly, thus the IP addresses, that are part of the prefix, are reachable and routable from the 5G Core to the correct UE.

The Multipath Client uses the given IPv6 prefix from the 5GC/UE to define the full transport related IP address on the Multipath Client's side (IP OB 1.1, IP OB 2.1, etc.) by IPv6 stateless autoconfiguration:

- *The transport-related IP addresses on the Multipath Client's side (IP OB 1.1, IP OB 2.1, etc.) are defined locally on the Multipath Client. These IP addresses shall be propagated to the Multipath Gateway so that the Multipath Gateway can populate the mapping table for each Multipath Dataflow Handler to map the correct trackside transport Multipath Datapath IP address to the On-Board transport IP Multipath Datapath address.*
- *The MPF-CTRL of Multipath Client shall assign the IPv6 addresses to the TAP Service Clients interfaces. They may be public addresses (e.g. Global Unicast Addresses, GUA) or Link-Local addresses, based on the specific address plan in use."*

A.2.3 Test description

A.2.3.1 Pre-test conditions

System Configuration:

- MCx Service system consists of all sub-systems of the Common Services Core.
- For the underlying "transport bearer" over which the UE1 (MCx Service system) and the IUT will communicate Parameters are set to the default parameters specific for the basic E-UTRA single simulated cells cell network scenarios, as defined in ETSI TS 138 508-1 [6] (3GPP TS 38.508-1) clause 6.3. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCx operation in the MCx configuration document).
- 5GS Common Service Core is supporting IPv6 according to IETF RFC 8200 [34].
- MCx Service system is configured to support the following features:
  - Support of IPv6.

IUT:

- IUT (UE1).
- The IUT (UE1) consists of all sub-systems of the MCx user service including the ability for floor control, pre-arranged group communication functionality and ad-hoc group communication functionality. Ensure the correct configuration.
- UE1 is the acting user.

Preamble:

- The IUT UE1 is connected to PLMN of the primary 5GS.
- The MCx Service system 1 (MCx_S1) is connected to the IUT.

A.2.3.2 Test procedure sequence

Table A.2.3.2-1: Main behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
	EXCEPTION: Procedure 'MCX CO communication' as described in ETSI TS 137 579-1 [28] (3GPP TS 37.579-1) clause 5.4.3	--		-	-
1	Check: request IP settings on UE1	--	UE1 has allocated an IPv6 address	1	P
2	Check: UE1 send ping request to MCx_S1 using IPv6	-->	UE1 is using ICMPv6 according IETF RFC 4443 [35]	1	P
3	MCx_S1 reply to ICMPv6 request (see note)	<--	MCx_S1 is using IPv6 format	-	--
NOTE: The reply on the ICMPv6 request indicates, that the routing between UE1 and MCx_S1 is correctly. This means, that following communication regarding FRMCS or MCx communication is also routed correctly in Ipv6.					

A.2.3.3 Specific message contents

Table A.2.3.3-1: IPv6 from UE1 (step 1, Table A1.2.3.2-1; IETF RFC 8200 [34])

Derivation Path IETF RFC 8200 [34]; condition Internet Protocol part				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Internet Protocol Version 6				
Traffic Class				
Source Address				
Destination Address				
Source SLAAC MAC				
Destination SLAAC MAC				
Stream Index				

History

Version	Date	Status
V1.1.1	September 2026	Publication